POWERWIZE: AI-ENHANCED TRAINING AND CONSULTATION SYSTEM WITH BLOCKCHAIN PAYMENT VERIFICATION AND TOKENIZED CERTIFICATES FOR ENERGYWIZE SOLUTIONS INC.

A Capstone Project
Presented to the Faculty
of the College of Computer Studies
Laguna State Polytechnic University

Jan Mczeal A. Bandao
Chelsea P. Cosio
Leinard Kurt V. Esguerra

In Partial Fulfilment
of the Requirements for the
Degree Bachelor of Science in Information Technology

LAGUNA STATE POLYTECHNIC UNIVERSITY
COLLEGE OF COMPUTER STUDIES
Santa Cruz, Laguna
Philippines

2026

VISION

An ASEAN Polytechnic University by 2030.

MISSION

A premier ASEAN polytechnic university that LEADS multidisciplinary education and applied research, equipping future-ready servant LEADers to drive innovation for sustainable regional development.

QUALITY POLICY

LSPU delivers quality education through responsive instruction, distinctive research, sustainable extension and production service. Thus, we are committed with continual improvement to meet applicable requirements to provide quality, efficient and effective services to the university stakeholder's highest level of satisfaction through an excellent management system imbued with utmost integrity, professionalism and innovation.

I. COLLEGE OF COMPUTER STUDIES GOALS

The College of Computer Studies graduates are exp/ected to become globally competitive and innovative computing professionals imbued with utmost integrity, contributing to the country's national development.

I. PROGRAM EDUCATION OBJECTIVES

The Bachelor of Science in Information Technology graduates are professionals that can adapt with fast-paced computing trends responsive to global IT demands.

It is designed to enable students to achieve the following by the time they graduate:

- Apply knowledge for solving computing problems employing designs and development solutions for business-driven application, installation, processes, operation, maintenance and administration of IT hardware and software.
- Utilize modern computing tools and techniques in research and development projects.
- Communicate effectively with a member or leader of the computing society with social, moral, and legal responsibilities to achieve a common goal
- Engage in life-long learning as a foundation for continuing professional advancement.

APPROVAL SHEET

The thesis entitled **"POWERWIZE: AI-ENHANCED TRAINING AND CONSULTATION SYSTEM WITH BLOCKCHAIN PAYMENT VERIFICATION AND TOKENIZED CERTIFICATES FOR ENERGYWIZE SOLUTIONS INC."** prepared and submitted by **BANDAO, JAN MCZEAL A., COSIO, CHELSEA P.,** and **ESGUERRA, LEINARD KURT V.** in partial fulfilment of the requirements for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY,** major in **WEB AND MOBILE APPLICATION DEVELOPMENT PROGRAM** is hereby recommended for approval and acceptance.

_____
MIA V. VILLARICA, DIT
Thesis Adviser

Approved by the Committee on Oral Examination with a grade of _____.

MARK P. BERNARDINO
Member

GENER F. MOSICO
Member

EDWARD S. FLORES
Member

MA. CEZANNE D. DIMACULANGAN
Member

MARK P. BERNARDINO
Research Coordinator

Accepted and approved in partial fulfilment of the requirement for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY,** major in **WEB AND MOBILE APPLICATION DEVELOPMENT PROGRAM**

MIA V. VILLARICA, DIT
Associate Dean, CCS

ROSARIO G. CATAPANG, PhD, Ffp
Research and Development Chairperson

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGEMENT

The researchers would like to express their sincere gratitude to everyone who contributed to the successful completion of this capstone project.

First and foremost, the researchers thank **Almighty God** for providing them with the wisdom, strength, patience, and guidance throughout the development of this study.

The researchers would like to extend their heartfelt appreciation to their capstone adviser, **Dr. Mia V. Villarica, DIT**, for her valuable guidance, constructive feedback, and continuous support throughout the completion of this manuscript. Her knowledge and recommendations greatly helped the researchers improve their work.

Sincere appreciation is also extended to the faculty members of the **College of Computer Studies at Laguna State Polytechnic University** for the knowledge and learning experiences they provided throughout the researchers' academic journey.

The researchers are also grateful to **EnergyWize Solutions Inc.** for their cooperation, time, and assistance in providing the necessary information and insights that helped in the development of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc.

Special thanks are given to the IT professionals and respondents who participated in the testing and evaluation of the system. Their feedback, suggestions, and time contributed significantly to the improvement of the developed system.

The researchers also wish to thank their families for their unwavering love, understanding, encouragement, and support throughout the completion of this study. Their sacrifices and motivation served as an inspiration for the researchers to persevere despite the challenges encountered.

Lastly, the researchers acknowledge the cooperation, dedication, and teamwork of each member of Team EW06. Through their combined efforts, patience, and determination, they were able to overcome challenges and complete this capstone project.

To everyone who contributed directly or indirectly to the completion of this study, the researchers extend their deepest gratitude and appreciation.

ABSTRACT

The study focused on the development of **PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc.** The system was developed to address the challenges of managing training programs, conducting Solar Photovoltaic (PV) consultations, verifying payments, and issuing training completion certificates through manual and email-based processes. PowerWize is a web-based platform that integrates artificial intelligence (AI), blockchain technology, and automated computation to improve these processes. Its main features include an AI-supported learning assistant for Certified Energy Manager (CEM), Certified Energy Auditor (CEA), and Government Energy Management Program (GEMP) training; an AI-assisted Solar PV consultation computation engine; blockchain-based payment verification; tokenized digital certificates; and an administrative dashboard for managing users, monitoring training progress, reviewing consultation requests, and generating reports. The system was developed using an iterative software development approach and evaluated through Black-Box Testing and User Acceptance Testing (UAT). The evaluation involved 48 respondents composed of EnergyWize Solutions Inc. administrators, trainers, and CEM, CEA, and GEMP training participants. The system was assessed using selected characteristics of ISO/IEC 25010:2023 and criteria from the Technology Acceptance Model (TAM). The study presents an integrated platform designed to streamline training and consultation operations, support more efficient Solar PV computations, provide traceable payment records, and enable the issuance and verification of digital training certificates. The findings provide a basis for assessing the system's functionality, usability, and overall quality within the operational context of EnergyWize Solutions Inc.

Keywords: Artificial Intelligence, Blockchain, Solar PV Consultation, Payment Verification, Tokenized Digital Certificates, Online Training System

TABLE OF CONTENTS

Preliminaries **Page**

TITLE PAGE ..... **I**

VISION, MISSION, QUALITY POLICY, GOALS and OBJECTIVES ..... **II**

APPROVAL SHEET ..... **III**

ACKNOWLEDGEMENT ..... **IV**

ABSTRACT ..... **V**

TABLE OF CONTENTS ..... **VI**

LIST OF TABLES ..... **VIII**

LIST OF FIGURES .....

LIST OF EQUATION .....

DEFINITION OF TERMS .....

OPERATIONAL TERMS .....

TECHNICAL TERMS .....

CHAPTER I **1**

INTRODUCTION 1

Project Context 3

Research Objectives 7

General Objective 7

Specific Objectives 8

Conceptual Framework 10

Project Purpose 13

Scope And Limitation of the Study 16

CHAPTER II **19**

REVIEW OF RELATED LITERATURE 19

Artificial Intelligence in Training and Learning Systems 19

AI-Assisted Solar PV System Design and Computation 24

Blockchain Technology for Payment Verification and Security 28

Blockchain-Based Digital Certificates and Credentialing 33

Agile Software Development Methodology 37

Software Quality Evaluation ISO/IEC 25010 40

Technology Gap Analysis 44

Synthesis 47

CHAPTER III **50**

RESEARCH METHODOLOGY 50

Research Design....50
Population of the Study....51
Sampling Design....53
Data Collection Instrument....54
Preliminary Interview....55
Online Research and Technology Gap Analysis....55
Observation....56
Survey Using Questionnaire....56
Project Design....57
Software Development Model....57
System Testing and Evaluation Procedure....60
System Testing....61
Testing Tool....62
Testing Method....67
System User Evaluation....69
Evaluation Tool....70
Evaluation Method....75
Validation of Questionnaire....76
Statistical Treatment....77
Frequency and Percentage Distribution....77
Weighted Mean....77
Likert Scale....78
Standard Deviation....79
REFERENCE....80

LIST OF TABLES

Tables

Page

Table 1. Respondents for the User Evaluation **52**

Table 2. System Testers **54**

Table 3. Summary of Agile Development Phases for PowerWize **59**

Table 4. Testing for User Login and Authentication **64**

Table 5. Testing for AI-Driven Training Module and AI Learning Assistant **65**

Table 6. Testing for AI-Assisted Solar PV Computation Engine **65**

Table 7. Testing for Blockchain Payment Verification Module **66**

Table 8. Testing for Tokenized Digital Certificate Issuance **66**

Table 9. Testing for Administrative Dashboard **67**

Table 10. Quality Factors – Functional Suitability and Reliability **72**

Table 11. Performance Efficiency **72**

Table 12. Perceived Ease of Use – Interaction Capability **73**

Table 13. Perceived Usefulness **73**

Table 14. User Experience **74**

Table 15. Attitude Toward Using **74**

Table 16. Behavioral Intention to Use **75**

Table 17. Likert Scale for UAT Questionnaire Interpretation **78**

LIST OF FIGURES

Figures **Page**

Figure 1. Conceptual Framework of PowerWize: Input-Process-Output (IPO) Model. **12**

Figure 2. Agile Model **58**

Figure 3. Black Box Testing Model **63**

Figure 4. Dichotomous Testing Format for PowerWize System Evaluation **68**

Figure 5. User Acceptance Testing (UAT) Process adapted from Lanars (2021) **71**

Figure 6. ISO/IEC 25010 Software Quality Metric **71**

LIST OF EQUATION

DEFINITION OF TERMS

The researchers explain the terminologies utilized in the development of the PowerWize system, which can be beneficial for thoroughly understanding the key terms used in the study. The terms are organized into Operational Terms which express how each term is specifically used within the context of this study and Technical Terms which express the universal or scholarly meaning of each term as defined in existing literature and standards.

OPERATIONAL TERMS

Administrative Dashboard	The centralized management interface within PowerWize that allows authorized EnergyWize Solutions Inc. personnel to manage user accounts, monitor training progress across all CEM, CEA, and GEMP programs, review Solar PV consultation requests, and generate operational reports.
Agile Methodology	The iterative and sprint-based software development process used to design, develop, test, and refine all modules of the PowerWize system through seven cyclical phases: Requirement and Analysis, Design, Implementation, Testing, Development, Maintenance, and Launch.
AI Learning Assistant	The artificial intelligence feature embedded within PowerWize's training module that guides CEM, CEA, and GEMP trainees in real-time by providing adaptive quiz feedback, personalized content recommendations, and performance-based hints to improve trainee comprehension and assessment outcomes.

AI-Assisted Computation Engine	The automated Solar PV system design tool within PowerWize that analyzes client-supplied data — including energy load requirements, site location, available area, and equipment preferences — and intelligently generates system sizing, panel configuration, energy yield estimates, and cost projections.
Black Box Testing	The testing method applied by IT professionals during system testing of PowerWize wherein the system's functions are evaluated based on inputs and expected outputs, without examining the internal source code or architecture.
Blockchain Payment Verification	The module within PowerWize that records and validates all enrollment fee payments for CEM, CEA, and GEMP training programs and Solar PV consultation fees on a distributed blockchain ledger, producing a tamper-proof and transparent payment audit trail.
Certified Energy Auditor (CEA)	One of the three DOE-accredited training programs offered by EnergyWize Solutions Inc. and managed through PowerWize, covering systematic energy auditing skills for practitioners in the commercial and industrial sectors across Level 1, Level 2, and Level 3.
Certified Energy Manager (CEM)	One of the three DOE-accredited training programs offered by EnergyWize Solutions Inc. and managed through PowerWize, covering Strategic Energy Management, Cost-Benefit Analysis, and compliance with the Energy Efficiency and Conservation Act (RA 11285).

EnergyWize Solutions Inc.	The client organization and primary stakeholder for whom PowerWize is developed; a DOE-accredited Recognized Training Institution (RTI) authorized to deliver CEM, CEA, and GEMP training programs and Solar PV consultation services.
Government Energy Management Program (GEMP)	One of the three DOE-accredited training programs offered by EnergyWize Solutions Inc. and managed through PowerWize, designed for government-designated Energy Efficiency Officers, Focal Persons, and Coordinators under DOE Department Circular 2023-05-0009.
Input-Process-Output (IPO) Model	The systems analysis framework used as the organizing structure of PowerWize's conceptual framework, representing inputs such as training materials and Solar PV data, processes such as AI training delivery and blockchain payment verification, and outputs such as tokenized certificates and PV design reports.
PowerWize	The proposed AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates developed for EnergyWize Solutions Inc. to digitalize and secure its CEM, CEA, and GEMP training programs and Solar PV consultation workflows.
Purposive Sampling	The sampling technique used in this study wherein respondents are deliberately selected based on their direct experience with EnergyWize Solutions Inc. 's training and consultation operations, ensuring that only the most contextually relevant individuals evaluate the PowerWize system.

Recognized Training Institution (RTI)	The formal accreditation status is held by EnergyWize Solutions Inc. under the Department of Energy of the Philippines, authorizing the company to deliver DOE-mandated energy certification training programs and issue verified certificates of completion.
Sprint	A short, fixed-length development cycle within the Agile Methodology used to build and review specific modules of PowerWize, allowing the research team to deliver functional increments of the system and incorporate stakeholder feedback before proceeding to the next development cycle.
Tokenized Digital Certificate	The blockchain-anchored, NFT-based digital credential issued by PowerWize to participants who successfully complete the CEM, CEA, or GEMP training programs, enabling tamper-proof authentication by the DOE, employers, and other authorized third parties.
User Acceptance Testing (UAT)	The structured evaluation of PowerWize conducted by 48 respondents — EnergyWize Solutions Inc. administrators, trainers, CEM/CEA/GEMP training participants — using an ISO/IEC 25010:2023-based questionnaire and the System Usability Scale (SUS).
Weighted Mean	The statistical tool used to compute the average evaluation scores per ISO/IEC 25010:2023 quality characteristic from the PowerWize UAT questionnaire, determining the overall level of user satisfaction and system acceptability for each evaluated module and quality dimension.

TECHNICAL TERMS

Adaptive Learning	An educational technology approach that automatically modifies the sequence, format, and difficulty of learning content based on the learner's performance, behavior, and knowledge state, enabling personalized instruction without requiring direct instructor intervention (Mavrikis et al., 2025).
Artificial Intelligence (AI)	The branch of computer science that enables machines to simulate human cognitive functions — including learning, reasoning, natural language understanding, and problem-solving — through algorithms such as machine learning, neural networks, and natural language processing (Yu et al., 2025).
Blockchain	A decentralized and distributed digital ledger that records transactions across a peer-to-peer network in immutable, cryptographically linked blocks, ensuring data integrity, transparency, and security without the need for a central authority (Nakamoto, 2008).
Cryptographic Hashing	A mathematical function that converts input data of any size into a fixed-length string of characters called a hash, used in blockchain systems to uniquely identify and protect the integrity of each block of transaction records (Zhao et al., 2024).
Data Integrity	The accuracy, consistency, and trustworthiness of data across its entire lifecycle, ensuring that stored records are complete, uncorrupted, and protected from unauthorized alteration during storage, transmission, and processing (Adanigbo et al., 2024).

Decentralized Application (dApp)	A software application that operates on a blockchain or peer-to-peer network rather than a centralized server, using smart contracts to execute its logic in a transparent, trustless, and censorship-resistant manner (Buterin, 2014).
Energy Efficiency and Conservation Act (RA 11285)	Republic Act No. 11285, a Philippine law that institutionalizes energy efficiency and conservation as a national policy, mandating government agencies and private firms to designate energy managers, conduct audits, and submit energy management plans to the Department of Energy (DOE, 2019).
Immutable Ledger	A record-keeping system in which data, once written and validated, cannot be altered, overwritten, or deleted, providing a permanent and verifiable history of all recorded transactions — a fundamental property of blockchain-based systems (Zhao et al., 2024).
Intelligent Tutoring System (ITS)	A computer-based instructional system that uses artificial intelligence to provide personalized, adaptive learning experiences and immediate feedback to individual learners without requiring direct human instructor involvement (Lin et al., 2023).
ISO/IEC 25010:2023	The current version of the International Organization for Standardization's SQuaRE software product quality model, defining nine quality characteristics — functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety — for evaluating software systems (ISO, 2023).

Learning Management System (LMS)	A software application designed to create, manage, deliver, and track educational courses and training programs in digital environments, supporting features such as course authoring, quiz administration, gradebook management, and learner progress monitoring (Alotaibi, 2024).
Machine Learning (ML)	A subset of artificial intelligence in which algorithms learn patterns from data and improve their performance on specific tasks — such as classification, prediction, and recommendation — without being explicitly programmed for each task (Wang et al., 2024).
Non-Fungible Token (NFT)	A unique, indivisible cryptographic token stored on a blockchain that represents ownership of a specific digital asset and cannot be replicated or exchanged for an identical equivalent, commonly used for digital art, credentials, and other unique digital items (Wu & Liu, 2023).
Photovoltaic (PV) System	A renewable energy system that converts sunlight directly into electricity using semiconductor-based solar cells, configurable as standalone, grid-tied, or hybrid installations for residential, commercial, or industrial energy applications (Lemence & Tamayao, 2021).
Smart Contract	A self-executing program stored on a blockchain that automatically enforces and executes the terms of an agreement when predefined conditions are met, without requiring human intermediaries or manual processing (Buterin, 2014; Hewa et al., 2025).

Solar PV Computation	The engineering process of calculating optimal Solar PV system specifications — including system sizing, panel configuration, energy yield, and cost projections — based on site-specific parameters such as geographic location, energy load requirements, and available installation area (Jobayer et al., 2023).
System Usability Scale (SUS)	A standardized 10-item questionnaire developed by Brooke (1996) that measures perceived system usability. Scores above 68 indicate above-average usability; scores of 80.3 and above correspond to an 'Excellent' usability rating, providing a widely accepted benchmark for software evaluation (Lewis, 2018).
Technology Acceptance Model (TAM)	A theoretical model proposed by Davis (1989) that explains user adoption of information technology systems based on two primary constructs: perceived usefulness — the belief that the system improves performance — and perceived ease of use — the belief that using the system requires minimal effort (Kolog et al., 2022).
Tokenization	The process of converting a physical or digital asset — such as a certificate, document, or credential — into a unique cryptographic token stored on a blockchain, enabling secure ownership, transfer, and third-party verification of the asset without relying on a central authority (Khati et al., 2023).

Verifiable Credential	A cryptographically secured, machine-verifiable digital statement about a subject — such as a certification of achievement or professional qualification — issued by an authorized organization and structured in accordance with the W3C Verifiable Credentials Data Model standard (Mazzocca et al., 2025).

CHAPTER I

INTRODUCTION

1.1 Project Context

Digital technologies have advanced rapidly, reshaping how organisations operate, provide services and engage with their clients. As industries become more interconnected, artificial intelligence (AI), blockchain technology, and smart digital platforms are increasingly adopted by organizations seeking to remain competitive and efficient. This digital transformation is influencing the delivery of services and solutions across the global economy, from health services to finance, from education to energy, and beyond.

In the renewable energy industry, recent years have seen a growing demand for intelligent, scalable, and transparent systems. As the transition to a more sustainable and alternative energy generation system, particularly solar PV begins to take root around the world, so has the demand for professional services, such as professional consultation, technical training and certification. Energy resources are expanding rapidly in both the local and international market and solar PV solutions providers are on the rise, which forces these providers to manage complex workflows such as training programs, client consultations, payments and transactions, certifications, and issuance.

The government of the Philippines has been promoting the use of renewable energy sources and energy efficiency, which led to the passage of the Renewable Energy Act of 2008 (RA 9513) and the Energy Efficiency and Conservation Act (RA 11285). All these legislations are aimed at fostering energy and investment in renewable energy sources like solar energy and to create a skilled and trained workforce to drive the energy industry. The Department of Energy (DOE) of the Philippines has

established a structured certification program for professionalization of the energy industry in order to do so. Amongst these are Certified Energy Manager (CEM) Training Program for training energy managers in energy efficiency principles, project development, project financing and the application of best practices in energy auditing of energy systems, the Certified Energy Auditor (CEA) Training Program for energy managers to learn how to conduct systematic energy audits, develop energy-saving recommendations, and apply industry best practices and energy auditing tools to conduct energy audits of energy systems, and the Government Energy Management Program (GEMP) Training and Workshop, aimed at training designated government personnel (focal Persons, Coordinators, and Energy Efficiency and Conservation (EEC) Officers) to implement energy efficiency initiatives in their respective offices under the framework of DOE DC 2023-05-0009. These projects align with the national energy policy to build a competent energy manpower by acquiring qualifications and will highlight the significance of certificate training and introduction in the energy field.

Expert consultation and training services to private entities and individuals who wish to implement or 'make the switch' to solar energy solutions has become a crucial function that these companies perform to the successful adoption of solar implementation and the cultivation of these DOE-based training programs locally. As a Recognized Training Institution (RTI), EnergyWize Solutions Inc. is responsible for delivering consistent training content, producing high-quality assessments, validating payments, and issuing authentic certificates to learners who successfully complete the CEM, CEA, and GEMP programs. Fulfilling these responsibilities reliably is therefore a regulatory compliance matter, not merely an operational efficiency concern.

Despite these broader digital advancements, energy-focused companies such as EnergyWize Solutions Inc. continue to rely on traditional or semi-digital processes,

resulting in inefficiency and increased manual workload. All of this means that the lack of a single intelligent platform that can be used to manage training programmes, consultation services, verify payments and issue certificates is a major technological void that needs to be remedied to secure sustainable development, better services, and conformity to domestic energy certification requirements.

This chapter opens with a background, context, and rationale for the development of EnergyWize Solutions Inc.'s PowerWize: an AI-Enhanced Training and Consultation System equipped with Blockchain Payment Verification and Tokenized Certificates. The context of the project, problem statement, objectives, scope and limitation of the study and significance of the study is discussed below.

EnergyWize Solutions Inc. is an organisation that specialises in Solar Energy Solutions & Solar PV Consultations. It is an approved technical training and consultation provider sub-regulated by the Department of Energy and as such a Recognized Training Institution (RTI) to the diverse national energy certification programmes. This is one of the existing training programmes which is offered and run by EnergyWize Solutions Inc. The following are these: Certified Energy Manager (CEM) Training, Certified Energy Auditor (CEA) Training, and Government Energy Management Program (GEMP) Training and Workshop. The CEM Training Program aims to educate different energy professionals on detailed understanding about energy efficiency, Strategic Energy Management (SEM), Cost-Benefit Analysis (CBA), and compliance to Energy Efficiency and Conservation Act (RA 11285). The Energy Auditing Skills Training series, through the CEA Training Program, is based on ISO 50002 and DOE DC 2022-03-0006 standards, and includes Level 1, Level 2 and Level 3 energy auditing skills training for practitioners in the commercial and industrial sectors. The GEMP Training and Workshop have been designed for government

designated Energy Efficiency Professionals, such as Energy Efficiency Officers, Focal Persons and Coordinators in implementation of energy management systems, compliance with RA 11285, local energy efficiency planning and use of the GEMP On-line Submission System as stipulated in DOE DC 2023-05-0009.

These are the key processes available to support these training programs, namely: Training materials and modules, Quizzes and competency assessments, Solar PV system design consultant, Payment of training and payment enrollments, Awarding certificates to participants upon completion of CEM, CEA and GEMP training. For these types of certifications and the requirements of the DOE, the accuracy, security and timeliness is very crucial for each of these processes.

Currently, these processes are basically electronic in nature with EnergyWize Solutions Inc. using some simple digital tools, primarily electronic mail (email). Training material, quizzes and certificates are delivered to the programs of the CEM and the GEMP with email, but without any integrated or automated systems. While this system has allowed the company to function, they have had problems within their processes that involve efficiency, transparency and scalability, both in the amount of people enrolled in training and the number of people requesting consultations.

This email-based, manual workflow makes it difficult to track a participant's submitted quiz, monitor training completion across programs, and ensure certificates are issued to those who have completed the CEM, CEA, or GEMP requirements in a timely manner. Because each of these actions, submitting a quiz, confirming payment, issuing a certificate is performed manually, administrative staff time is spent on repetitive tasks, and there is a risk of information loss, delays, and errors that may have regulatory consequences given that these training programs are accredited by the DOE.

To make these concerns more concrete rather than relying solely on general descriptions such as "time-consuming" or "inefficient," Table 1.1 summarizes the current process, the identified problem, and the type of evidence that supports each concern, gathered through preliminary interviews and process observation with EnergyWize Solutions Inc. staff.

EXISTING PROCESS	CURRENT METHOD	IDENTIFIED PROBLEM	EVIDENCE TYPE
Training materials distribution	Email	Difficult to track completion	Staff interview
Quiz submission	Manual / email-based	Delayed monitoring of scores	Process observation
Payment verification	Manual reference checking	Verification delay, risk of error	Staff interview
Certificate issuance	Email (PDF)	Difficult third-party verification	Document review
Solar PV calculation	Manual computation by staff	Time-consuming; depends on staff availability	Staff interview
Records management	Separate spreadsheets/tools	Fragmented information across programs	Process observation

Table 1. Summary of Current Operational Concerns at EnergyWize Solutions Inc.

Furthermore, the current design lacks all the automated applications for a complex Solar PV consultation calculation like sizing the system, estimation the load, configuration of the panels and cost estimation. Additionally, those clients and program participants who request Solar PV consultation services also have to depend on manual calculations by program staff for consistency and potentially time-consuming. If the

company does not have an automated calculation tool, they are limited both in the services that they can offer their clients efficiently and professionally.

Moreover, the existing payments verification process fails to take advantage of any type of blockchain or cryptographic and security solutions, making it easy to doubt and scam payments for enrolment fees, or even lacking in immutable documents. A valid concern for the companies with respect to issuance of certificates is the lack of blockchain validation of the certificates issued for the CEM, CEA and GEMP programs as well as the lack of digital security of the certificates being issued, particularly when these are DOE accredited certificates for the profession. With the growing importance of digital credentials and assurance that's hard to tamper with for employers, government agencies and industry groups, not having a secure tokenized certificate system is a competitive liability and compliance risk for EnergyWize Solutions Inc.

All of these pains point to one major technical hurdle: provisioning of an integrated digital platform that will include AI to enhance the management of CEM, CEA, and GEMP training & evaluation programs, automate Solar PV consultation computations, allow for safe payments via blockchain for certificate confirmation, and deliver “tamperproof and tokenised” certificates. This impetus is the biggest driver in the development of the proposed PowerWize system.

The solution proposed in the proposed system called PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates seeks to address the above shortcomings and provide a fully Integrated Digital Solution to EnergyWize Solutions Inc. This will enable intelligent training delivery for the CEM, CEA and GEMP programs; automatic Solar PV consultation calculations and intelligent data-driven analytics for administration and

client. With its integration of blockchain technology, payment transactions and certificates produced will be more secure and verifiable, which would further support the trust of EnergyWize Solutions Inc. as a Department of Energy (DOE) accredited Recognized Training Institution (RTI).

As important is the context of the Energy sector's Digitalization (PESC) in which EnergyWize Solutions Inc. (EWS Inc.) found its operating demands for PowerWize's invention. With the surge in the numbers of CEM, CEA and GEMP certified professionals, manned and secured, due to RA 11285 and DOE policies, the digital management of training becomes more crucial than ever before with an intelligent and scalable system. PowerWize is intended to support EnergyWize Solutions Inc. in delivering more consistent, DOE-aligned training services, with reduced administrative overhead and improved data integrity and security.

The technology gap was identified by performing a comprehensive evaluation of the business processes at EnergyWize Solutions Inc, conducting interviews with relevant personnel, and reviewing existing evidence related to AI: based systems, Blockchain applications and Digital training platforms. The results of this gap analysis have confirmed the need and viability of building the proposed PowerWize system for capstone development of a real-world organizational system, implemented with the use of emerging technologies.

1.2 Research Objectives[EF1]

1.2.1 General Objective

The general objective of this study is to design, develop, and evaluate PowerWize — an AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates — to improve the operational

efficiency, data integrity, and service quality of EnergyWize Solutions Inc. in managing its DOE-accredited energy certification training programs, specifically the Certified Energy Manager (CEM), Certified Energy Auditor (CEA), and Government Energy Management Program (GEMP) Training and Workshop, as well as its Solar PV consultation services. The specific objectives are as follows:

1.2.2 Specific Objectives

1. To develop an AI-supported learning assistant for the CEM, CEA, and GEMP online training programs by:

    1.1 Analysing trainee performance and learning patterns to generate contextual explanations of complex course materials

    1.2 Providing intelligent quiz hints and personalized feedback based on individual knowledge gaps

    1.3 Guiding trainees through technical concept comprehension and program evaluation

    1.4 Assisting instructors in creating quiz questions aligned with learning objectives

2. To develop an AI-assisted Solar PV consultation computation engine that:

    2.1 Analyzes client-supplied data, including energy load requirements, site location, available roof or land area, and equipment specifications

    2.2 Generates Solar PV system sizing recommendations based on predefined engineering parameters and client-supplied data

    2.3 Reduces the need for manual computation by automating predefined Solar PV sizing and estimation procedures

3. To implement a blockchain-based payment verification module that:
   3.1 Records and validates payment transactions for CEM, CEA, and GEMP enrolment fees and Solar PV consultation fees
   3.2 Produces a blockchain-anchored, tamper-evident payment audit trail
4. To design and deploy a tokenized digital certificate issuance system that:
   4.1 Generates, issues, and securely stores blockchain-anchored certificates for participants who complete CEM, CEA, or GEMP training assessments
   4.2 Enables third-party verification of issued certificates
   4.3 Provides blockchain-based verification mechanisms intended to improve certificate integrity and traceability
5. To build an administrative dashboard that provides EnergyWize Solutions Inc. personnel with tools for:
   5.1 Managing user accounts and access levels
   5.2 Monitoring training completion and progress across all three programs
   5.3 Reviewing Solar PV consultation requests
   5.4 Generating operational reports for data-driven decision-making
6. To evaluate the system's functional performance, usability, and overall quality through functional testing and User Acceptance Testing (UAT), focusing on:

   6.1 Functional suitability
   6.2 Performance efficiency
   6.3 Interaction capability
   6.4 Reliability
   6.5 Security
   6.6 Maintainability

These evaluations shall be guided by selected quality characteristics of the ISO/IEC 25010:2023 software quality standard and the Technology Acceptance Model (TAM), and administered to administrators, trainers, CEM/CEA/GEMP training participants of EnergyWize Solutions Inc. Solar PV consultation clients are not part of this respondent group; they use the system's consultation module but are not surveyed as part of UAT.

Objective–Feature–Test–Evaluation Matrix

OBJECTIVE	SYSTEM FEATURE	FUNCTIONAL TEST	EVALUATION MEASURE
1. AI learning assistant	AI training module	Table 5 — quiz interaction, feedback delivery, completion scoring	UAT: Functional Correctness (Table 10), Perceived Usefulness (Table 13)
2. AI-assisted PV computation	PV computation engine	Table 6 — input validation, sizing/yield/cost output generation, benchmark comparison *(to be added — see Priority 4)*	UAT: Functional Correctness (Table 10), Perceived Usefulness (Table 13); benchmark accuracy/error rate
3. Blockchain payment verification	Blockchain module	Table 7 — transaction recording, alteration-attempt rejection	UAT: Reliability, Security (Table 10); Pass/Fail + security testing
4. Tokenized certificate issuance	Certificate/token module	Table 8 — eligibility, minting, third-party verification	UAT: Security (Table 10), Perceived Usefulness (Table 13); verification success rate
5. Administrative dashboard	Admin dashboard	Table 9 — reporting, transaction logs, user management	UAT: Functional Completeness/Appropriateness (Table 10)
6. Overall system quality	Entire system	Tables 5–9 — Black Box Testing (Pass/Fail)	ISO/IEC 25010:2023 (six characteristics, Tables 10–16), TAM

Table[EF2] 2. Objective–Feature–Test–Evaluation Matrix

1.3 Conceptual Framework

Based on the related literature and studies, the conceptual framework of this study was considered, which integrates relevant theories, models and variables related to the adoption of artificial intelligence technology in education and training systems, blockchain technology for payment proof and digital credentialing, and solar photovoltaic consultation systems. The framework is theoretically anchored and gives reason to the development and evaluation of PowerWize.

In the current study, the main organising framework used is Input-Process-Output (IPO). This model is widely adopted in systems development and engineering research, and offers a straightforward and logical way of thinking about how data and user inputs are processed in a sequence of computational and engineering transformations, to produce useful and usable outputs. The IPO model applies well to PowerWize, which is essentially a data transformation system, working with data inputs submitted by users, manipulated by its AI and blockchain capabilities, and generating outputs as certificates, verified transactions, and reports.

The conceptual framework is based on four important theories and tools. First, Artificial Intelligence in Training and Assessment is not merely a tool for delivering training content; rather, it functions as an intelligent learning assistant for the learners enrolled in CEM, CEA and GEMP programs. It is able to offer real time feedback, adaptive content suggestions, personalized cues, and instant performance-based feedback which allows trainees to more easily navigate content and enhance assessment results. AI e-assessment has proven beneficial for improving learner engagement and decreasing teacher workload in a digital learning context (Wang et al., 2024; Alotaibi, 2024). Second, AI-Assisted Solar PV Computation is based on the fields of expert

systems and decision support systems in engineering. Intelligent system sizing, intelligent sizing of panels, energy yield estimation, and cost estimation, without depending on manual engineering calculations, while considering the client-provided data, such as energy load, site conditions, available area, and equipment preferences, are assured by the AI system (Trindade & Cordeiro, 2021; Lemence & Tamayao, 2021). Third, with Blockchain Technology for Payment Verification, all financial transactions concerning the enrollment fee of training and consultation are kept secure with the characteristics of immutability, transparency and decentralization (Nakamoto, 2008; Autade, 2025). Fourth, based in the blockchain technology, Tokenized Digital Certificates is a solution for generating credentials that are verifiable, tamper-proof and immune to loss, theft, and falsification, thus solving the problems related to traditional paper-based certificates (Esguerra et al., 2024; Khati et al., 2023).

To ensure the framework is research-oriented and not only technology-oriented, the conceptual model is organized into two interconnected layers. The System Development Framework follows the Input–Process–Output structure described above, showing how training content, client data, and payment and user records are transformed by the AI, Solar PV, blockchain, and administrative modules into system outputs such as training feedback, PV reports, verified payments, and certificates. The Evaluation Framework then carries these system outputs through functional (black-box) testing, followed by evaluation against selected ISO/IEC 25010:2023 quality characteristics and the Technology Acceptance Model (TAM) through User Acceptance Testing (UAT), producing the study's acceptance and quality findings. Findings from this evaluation layer feed back into the development process, so that AI models, computation parameters, and system design can be refined based on evaluator

input. This two-layer structure makes explicit the research variables, evaluation criteria, and feedback mechanism that a single-layer IPO diagram does not show on its own.

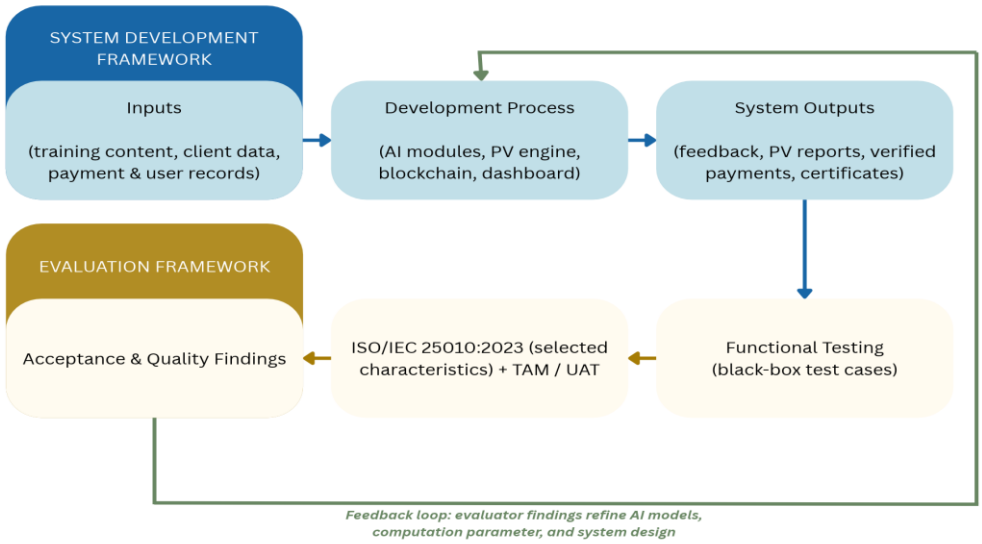


Figure 1. Two-layer conceptual framework of PowerWize

The Input layer includes training materials for CEM, CEA, and GEMP; parameters used for Solar PV consultation; client payment information; user information; and system configuration parameters. The Process layer covers the core technology operations: AI guidance and support functions for trainees across CEM, CEA, and GEMP content; the AI-assisted Solar PV computation engine that analyzes client data to suggest system designs; blockchain payment verification; certificate tokenization; and administrative data management. The Output layer includes personalized training feedback, Solar PV system design reports, blockchain payment records, tokenized training certificates, and operational analytics. The feedback loop allows for ongoing, iterative refinement of the system based on evaluation results.

1.4 Project Purpose

This section describes the significance of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates and articulates the potential value of the study and its findings to its identified target audiences. The development of PowerWize is driven by the need to address critical operational gaps within EnergyWize Solutions Inc. And to contribute to the broader advancement of technology-driven solutions in the renewable energy and digital training sectors.

What makes PowerWize relevant is the interplay of three facts: A sharp increase in the demand for professionally certified Energy practitioners in the Philippines, specifically for those possessing the CEM and CEA certifications and attending the GEMP Training program; Lack of adequate manual systems and semi-digital systems being used while managing the EnergyWize Solutions Inc. business; and the global trend towards secure, verifiable and intelligent digital systems. The management of training content, quizzes, consultation calculations, processing payments, issuing certificates for CEM, CEA, and GEMP participants via email-based workflows risks becoming inefficient, exposed to data breaches, administratively burdensome, and vulnerable to certificate fraud all of which could compromise the quality and credibility of the services offered by the company.

PowerWize, which combines artificial intelligence, automated computation, and blockchain technology into a streamlined platform, directly tackles these risk fronts. Additionally, with Solar PV consultation computation autosaved into the system, the need for manual engineering computation is reduced; the system's blockchain-based payment and certification modules are intended to improve the

integrity and verifiability of payments and certificates, including those issued under the CEM, CEA and GEMP programs. These capabilities are an essential improvement in the trustworthiness, professionalism, and scalability of EnergyWize Solutions Inc.'s service delivery.

In addition, the study adds to the body of knowledge of the application of AI and blockchain technology in the renewable energy and E-Management in Training field which has not received much research in academic and industry literatures in the Philippines. The results of this study provide a reference and model for other organizations interested in modernizing their activities with the intelligent platform of the digital era.

1.4.1 Intended Beneficiaries

EnergyWize Solutions Inc.; CEM/CEA/GEMP training participants and clients; researchers and academic institutions; and, prospectively, government and policy stakeholders.

1.4.2 Expected Benefits

The findings and outcomes of this study are expected to be beneficial to the following stakeholders:

- EnergyWize Solutions Inc. will gain a unified, automated platform that streamlines the management of their CEM, CEA and GEMP Training programs, Solar PV consulting process, payment verification and certificate issuing. It will make the process of administration less cumbersome, render fewer mistakes in operations, and improve its level of service for customers.

- Training Participants and Clients Individuals attending the CEM, CEA, and GEMP Training programs will have access to an easier to use, more structured and professional digital learning experience. The automated computation engine provides clients with more speedy and accurate system design recommendations and cost estimates when meeting for Solar PV consultations.

- The development of PowerWize may provide a reference point for the broader solar PV and energy auditing sectors in exploring how AI and blockchain can support professional training, consultation, and credentialing, potentially informing similar efforts by other practitioners in the industry.

- Researchers and Academic Institutions The research advances the body of knowledge on AI-enhanced learning systems, blockchain-based credentialing, and automation in engineering computations. Future researchers can use the frameworks and findings from this study as building blocks in order to innovate new ideas in the respective areas.

- It demonstrates how new and emerging technologies such as AI, blockchain, and automated computation can be incorporated into a real-world system that offers solutions to the problems and is a practical, real-life example to the future IT developer.

- Government and Policy Makers With the Philippines' growing momentum in renewable energy (RE) adoption and the implementation of energy efficiency (EE) regulations in Republic Act No. 11285, the need for energy managers and energy auditors who are trained and certified by a professional group is imperative. A tokenized certificate system, provided by PowerWize, could

establish a more robust and trustworthy credentialing ecosystem that can meet national GEMP, CEM, and CEA credentialing needs.

- It is important to note that this potential benefit to government and policy stakeholders is prospective: PowerWize has not been endorsed, adopted, or evaluated by the Department of Energy or any government agency as part of this study.

1.5 Scope and Limitation of the Study

1.5.1 Scope of the Study

This section defines the boundaries of the study by identifying what is covered under the scope of the PowerWize system and what falls outside its coverage as limitations. A clear delineation of scope and limitations is essential to set realistic expectations, guide the development process, and provide transparency to stakeholders regarding what the system is designed to accomplish.

The study is conducted in relation to a company, EnergyWize Solutions Inc., regarding the design, development and evaluation of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates. The system is designed to address the identified gaps in the current business in handling training programs, solar PV consultations, payments verification, and issuing certificates.

The study area is the following functional area and modules:

- AI-Driven Training and Assessment Module: responsible for training material upload and distribution, and access management for the CEM, CEA, and GEMP training programs. The AI serves as a digital learning aid, providing timely tips,

course suggestions, personalized scaffolding, and instant feedback for trainees during quizzes and module assessments.

- AI-Assisted Solar PV Consultation Computation Engine: supports solar PV system design — energy yield estimates and cost estimates — by analyzing client-supplied data, reducing the need for manual engineering computation and aiming to improve consistency and efficiency.
- Blockchain Based Payment Verification Module Record, verify and log all payment transactions for CEM and CEA enrollments, Solar PV consultation fee on a blockchain ledger.
- Third Party Verification Interface for Tokenized Digital Certificate System Provides automated generation and issuance of digital certificates backed by blockchain technology to the participants passing the CEM, CEA and GEMP training assessments.
- Administrative Dashboard Provides user account management, reporting on the progress in users' training, review of operations reports, generation of operational reports, and tracking available consultation requests.
- System Evaluation System functional testing and ISO/IEC 25010:2023 User Acceptance Testing (UAT) as a culminating experience for the BSIT program in the academic year 2025–2026.

1.5.2 Limitations of the Study

While PowerWize addresses the identified operational needs of EnergyWize Solutions Inc., the study acknowledges the following limitations that define the boundaries of what the system is not designed to cover:

- Platform Exclusivity Designed exclusively for EnergyWize Solutions Inc.; not a general-purpose training or consultation platform.
- Solar PV Computation Coverage Limited to standard residential and commercial grid-tied and off-grid configurations. Industrial-scale installations, hybrid systems, and advanced BESS design are excluded.
- Blockchain Payment Scope Limited to transactions within the PowerWize platform. Integration with third-party gateways (e.g., GCash, PayMaya) is not covered.
- Training Format Limitations Supports asynchronous, text-based, and multimedia content only. Real-time live video streaming, interactive virtual classrooms, and AR-based environments are not supported.
- Internet Dependency Requires a stable internet connection. Offline access is not supported.
- Certificate Coverage Limited to training completion certificates for CEM, CEA, and GEMP programs within EnergyWize Solutions Inc. Government-regulated professional licenses and academic degrees are outside scope.
- Post-Deployment Maintenance Long-term maintenance and scalability beyond the capstone period are outside the scope of the study.
- Generalizability Findings and conclusions are specific to EnergyWize Solutions Inc. And may not be directly generalizable to other organizations or sectors.
- AI Model Dependency and Reliability: the AI-supported learning assistant and quiz-feedback features depend on the quality and coverage of their training or reference data and may occasionally produce inaccurate, incomplete, or hallucinated responses; outputs should not be treated as authoritative without human review.

- Blockchain Network Dependency: the blockchain-based payment and certificate modules depend on the availability and performance of the underlying blockchain network, and may be affected by network congestion, transaction ("gas") costs, and wallet or private-key management by users. PowerWize currently operates on a Polygon testnet (e.g., Amoy/Mumbai) rather than mainnet; testnet transactions do not involve real currency or production-grade network conditions, and this should be explicitly disclosed as a limitation of the current evaluation.
- Cybersecurity Testing Scope: the study's security evaluation is limited to the security-related testing described in Chapter III; independent penetration testing and third-party security auditing were not conducted and are recommended for future work.
- Sample Size and Organizational Scope: User Acceptance Testing was conducted with a limited number of respondents from a single organization, EnergyWize Solutions Inc., which may limit how far the findings generalize to other organizations or larger user populations.
- Deployment Duration: the evaluation reflects a single point-in-time assessment; the study does not include longitudinal deployment data or a comparison against EnergyWize Solutions Inc.'s actual production operations over an extended period.
- Independent Engineering Validation: Solar PV computation outputs were benchmarked against predefined engineering formulas and sample cases as described in Chapter III; independent certification of PV sizing results by a licensed engineering body was not obtained.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Studies and Literature

This chapter presents a review of related literature and studies relevant to the development of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. The review is organized thematically, covering six major areas: (1) Artificial Intelligence in Training and Learning Systems; (2) AI-Assisted Solar PV System Design and Computation; (3) Blockchain Technology for Payment Verification and Security; (4) Blockchain-Based Digital Certificates and Credentialing; (5) Agile Software Development Methodology; and (6) Software Quality Evaluation using ISO/IEC 25010. Each theme is discussed through a synthesis of peer-reviewed journals, conference papers, and technical reports published primarily between 2021 and 2025. The chapter concludes with a Technology Gap Analysis comparing existing related systems against the proposed PowerWize system to justify the novelty and necessity of its development.

2.1.1 Selection of Related Literature

Literature was prioritized based on direct relevance to the study's six major themes AI in training and learning, AI-assisted Solar PV computation, blockchain payment verification, blockchain-based digital certificates, Agile software development, and ISO/IEC 25010:2023 software quality evaluation favoring recent (2021-2026), peer-reviewed, and credible academic, professional, or institutional sources.

2.1.2 Review and Analysis of Studies

Artificial Intelligence in Training and Learning Systems

The use of AI in education and training has become a major topic of research in recent years, with a particular focus on the potential for personalized learning, intelligent tutoring, and adaptive assessment. The researchers have looked at a variety of AI-supported mechanisms that seek to match the effectiveness of an individual human tutor by offering each learner timely and contextually appropriate guidance and feedback, based on their level of performance and learning style.

To incorporate AI into education, particularly in higher education, we must first understand the existing research on adaptive learning systems, as represented in this systematic review by Yuensook et al. (2026), performed according to PRISMA 2020 criteria and with a center-focus on current studies from 15 countries. Out of the studies they reviewed, 73% have been published in 2023 and 2024, pointing to the field's fast-growing nature. The majority of the key AI technologies found were machine learning (40%), natural language processing (33%) and hybrid (combining two or more technologies) (27%), all of which can enable real-time personalization and adaptive feedback mechanisms. Energy management training, analogous to programs provided under the CEM, CEA and GEMP, was most often implemented in STEM courses, language courses and hybrid learning environments. The review validates the necessity to integrate AI for the learning assistant that guides Energy Certification program trainees through the program content, as is being developed by PowerWize.

Alotaibi (2024) reviewed 60 papers on the topic of integrating AI with LMS in higher education published between 2014 and 2023, finding a steady annual increase of 16.53% in relevant publications, and concluded that integrating AI in higher

education LMS improves learner outcomes, content accessibility, and administrative efficiency. This gap in AI training platforms outside general academia affirms the need for a new AI learning assistant system, which is precisely what PowerWize introduces into EnergyWize Solutions Inc.'s own CEM, CEA, and GEMP Energy Certification Training programs. Alotaibi's findings are such that the most effective AI-LMS integration is through domain-aligned content personalization, which has been the power behind the design of PowerWize.

A medium-to-large effect size (Hedges' g = 0.70) indicates that AI-adaptive systems are scoring higher than the non-adaptive systems, with Wang et al. (2024) finding this result in a meta-analysis of 45 studies published from 2010 to 2022. The strong quantitative evidence backs the inclusion of AI-based adaptive learning in PowerWize, which suggests that the AI learning assistant should be expected to generate positive user impacts related to trainee performance over the non-adaptive training via email that EnergyWize Solutions Inc. is presently performing. This meta-analytic discovery is among the most solid empirical grounds for PowerWize's AI module.

To offer more empirical evidence regarding the effectiveness of automated personalized feedback in intelligent tutoring systems, Kochmar et al. (2021) provided substantial findings. The researchers demonstrated that large-scale dialogue-based tutoring systems can generate personalized pedagogical interventions, such as adaptive hints, Wikipedia-based explanations, and mathematical guidance, which significantly improved student performance outcomes. Their study further emphasized that iterative and performance-based feedback mechanisms enhance learner engagement and

success. Similar adaptive hinting and feedback processes are implemented in the PowerWize AI Assistant when energy trainees encounter difficulty answering quizzes.

More recently, Hariyanto et al. (2025) presented another systematic review on AI techniques for personalized learning covering 142 empirical studies from 2015 to 2025. An overview of three types of personalization (supervised, unsupervised, and reinforcement learning), and the two most consistently effective forms of adaptive feedback and knowledge tracing were identified as being related to learner performance. Their results inform the design of PowerWize's AI facilitator, which relies on an LLM API prompted with the trainee's current quiz context — rather than a dedicated knowledge-tracing algorithm — to personalize hints and quiz support throughout each of the training course modules (CEM, CEA, GEMP).

Recent literature on artificial intelligence in education categorizes AI applications into learner analytics and prediction, intelligent tutoring systems, adaptive assessment, and automated educational management (Crompton et al., 2023). These classifications inform the development of the PowerWize training module, which incorporates quiz analytics for learner profiling, ITS-style guidance through AI-assisted feedback, adaptive quiz mechanisms, and automated certificate issuance. Collectively, these features demonstrate the broad implementation of AI-powered educational support within the system.

Lin et al. (2023) showed how the work of AI in ITS could be applicable to the domain of sustainable education, highlighting knowledge tracing, behavior modeling, and adaptive content sequencing as the seminal AI mechanisms that have the greatest impact. For PowerWize, their work is important for the energy certification programs, CEM, CEA and GEMP are structured knowledge domains, in which accurate modeling

of learner mastery would be valuable for sequencing competence-building. PowerWize's current AI assistant does not implement a dedicated knowledge-tracing algorithm; it relies on LLM-based contextual responses instead, which is noted as a direction for future enhancement.

Zahab et al. (2024) noted the use of generative AI in adaptive learning, which refers to the use of large language model (LLM) tools to create personalized course outlines, quiz questions and customized study plans for each learner. The generative functions meet PowerWize's AI module, which uses LLM-class AI to produce contextually-relevant hints and explanations when trainees need more help during assessment in any of the 3 energy certification programmes.

Yu et al. (2025) carried out a thorough examination of intelligent tutoring systems based on AI, based on publications from 2010 to 2025, finding that gains of up to 20% of AI-tutored learners compared to traditional instruction were reported. The personalized feedback, adaptive difficulty adjustment, and real-time intervention capabilities were all cited as the three essential elements of AI tutoring that were central to any effective system, and which are all core to PowerWize's AI learning assistant. The discovery creates an objective standard for the performance of the AI module that will be used and tested throughout the study for PowerWize's AI module.

Okonkwo and Ade-Ibijola (2021) examined chatbots as intelligent tutoring tools in higher education, finding that AI chatbots deployed within LMS environments provided meaningful academic support by responding to student inquiries, generating adaptive quiz content, and recommending learning pathways. These capabilities are directly reflected in PowerWize's AI learning assistant module, which is designed to

respond to trainee queries, deliver quiz hints, and provide personalized content recommendations during CEM, CEA, and GEMP assessments.

Taken together, the literature on AI in training and learning systems provides robust theoretical and empirical justification for PowerWize's AI learning assistant module. The convergence of adaptive learning, intelligent tutoring, personalized feedback, and generative AI technologies validates the system's approach of deploying AI not merely as a content delivery mechanism but as an active, responsive learning partner for energy certification trainees.

Author/ Year	Technology	Application	Finding	Limitation	Relevance to PowerWize
Yuensook et al. (2026)	ML / NLP / hybrid (PRISMA review)	Higher-education adaptive learning	73% of reviewed studies published 2023–2024; ML, NLP, and hybrid approaches most common	Reviewed studies are mostly general higher-education/STEM contexts, not vocational energy-certification training	Supports the rationale for an AI learning assistant in CEM/CEA/GEMP training
Alotaibi (2024)	AI–LMS integration	Higher-education LMS	Steady 16.53%/year increase in publications; improved learner outcomes and admin efficiency	Focused on general academic LMS, not corporate/vocational certification platforms	Motivates extending AI–LMS integration to EnergyWize's non-academic training context
Wang et al. (2024)	AI-adaptive learning (meta-analysis)	Cross-domain adaptive learning	Medium-to-large effect size (Hedges' g = 0.70) favoring adaptive systems	Aggregates heterogeneous systems; does not isolate LLM-based assistants specifically	Provides quantitative backing for an adaptive AI assistant
Kochmar et al. (2021)	Dialogue-based intelligent tutoring	Personalized tutoring dialogue	Adaptive hints and explanations improved student performance	General/math tutoring domain, not energy-engineering certification content	Precedent for hint-based feedback design in PowerWize's AI assistant
Zahab et al. (2024)	Generative AI / LLM	Adaptive course and quiz	LLMs can generate personalized	Study does not address review of	Directly relevant to PowerWize'

Author/ Year	Technology	Application	Finding	Limitation	Relevance to PowerWize
		generation	outlines and quiz questions	generative outputs for factual accuracy	s API-based generative AI assistant; underscores the need for human review of AI-generated quiz content [EF3]

Table 3. Critical Comparison of Selected AI-in-Training Literature

AI-Assisted Solar PV System Design and Computation

The application of artificial intelligence and machine learning to solar photovoltaic system design has emerged as one of the most active research areas in renewable energy engineering. The complexity of PV system sizing involving interdependent variables such as site irradiance, load demand, panel efficiency, degradation rates, and financial parameters makes it a natural candidate for AI-assisted computation, where data-driven models can identify optimal configurations faster and more accurately than manual calculations performed by engineering staff.

Hu et al. (2024) have conducted a comprehensive review of the potential of AI in PV systems and found the most significant areas to be maximum power point tracking (MPPT), power forecasting and fault detection. The review outlined how AI integration can be used to optimize PV operational efficiency and solve key challenges in data processing and model management. From PowerWize's perspective, Hu et al. (2024) underscore the practicality of integrating AI into PV systems beyond just theory,

as this technology has been demonstrated to increase the effectiveness of system output, with documented improvements primarily in the case of different deployment environments.

Engel and Engel (2022) carried out a comprehensive survey of machine learning (ML) applications for solar plant design, forecasting, maintenance, and control and studied a set of benchmarks for solar power plant applications, which are based on traditional methods with and without ML. Through their analysis, they determined that ML algorithms, such as neural networks, support vector machines, and ensemble models, were better suited for dealing with the nonlinearity of solar plant systems compared to traditional algorithms. The paper's benchmarking framework proposes intelligent self-adaptive models for PV-system sizing, which is the methodology that the AI-assisted computation engine of PowerWize implements; both use ML-based sizing algorithms with the client provided data.

Jobayer et al (2023) carried out a systematic review of ML-aided PV parameter estimation studies, reviewing the publications from the year of 2020 to 2022, and identified the three most frequently used ML algorithms for PV system parameter estimation: neural networks (NN) (32.55%), random vector functional link (RVFL) (13.95%), and support vector machine (SVM) (9.30%). The review concludes that ML achieves a significant benefit in terms of both accuracy and computation over the analytics-based methods, offering direct empirical evidence for replacing EnergyWize Solutions Inc.'s existing manual consultations computation tools with an AI-based computation engine.

Said et al. (2026) introduced a new technique for PV system modeling by proposing a methodology of integrating machine learning with the design method for

the technical sizing and performance optimization of a PV system under different climate variables. The methodology was well accepted and showed very good agreement between the measured and predicted recovered solar energy on various locations with different latitude values, hence relatively high predictive accuracy and seasonality consistency. Similarly, this work offers a direct technical precedent for PowerWize's use of an integrated ML-design approach for creating accurate sizing and yield estimates based on site data provided by the client into PowerWize's AI-assisted computation engine.

Ndiaye et al. (2024) discussed AI-driven precision in solar forecasting, providing an overview of advancements in machine learning and deep learning technologies for forecasting solar irradiance and emphasizing how AI is increasingly becoming a tool for decision support in the energy planning and consulting sector. PowerWize's consultation engine powered by Artificial Intelligence does the same to forecast energy yields and financial return for energy solutions presented by their clients (Solar PV installations) which is intended to reduce reliance on manual engineering consultation, an area that currently places significant demand on EnergyWize Solutions Inc.'s staff time.

Taha et al. (2025) found that the linear regression, random forest, gradient boosting, CNN, LSTM models and CNN-lSTM hybrid models for short and long-term solar irradiance forecasting gave comparable predictive results. The study showed that the hybrid deep learning models are better than single-algorithm models on all time scales in particular with increasing forecast horizon. PowerWize's AI module takes advantage of this line of research to identify the suitable prediction models depending on the specific forecast needs of each Solar PV consultation request on the platform.

The decision support system to install a standalone solar energy system (SSES), built by the group Hafeznia et al. (2024), has combined the techno-economic, seismic and flood risk criteria into a single recommendation framework. This multi-criteria decision-support method is more of an analogue effort than with PowerWize's consultation engine, where the client can provide information regarding his/her energy requirement, site location, availability of space and preferences for energy equipment, and a personalised recommendation for a Solar PV system, incorporating more than one design criterion at a time, is synthesised.

Hanif et al (2024) proposed hybrid ANN and LightGBM framework to promote solar energy forecasting and evaluate the accuracy of these models for different climatic conditions. The hybrid model (combination of ANN and LightGBM) combines the advantages of ANN (pattern recognition) and LightGBM (lighting gradient boosting), and is a feasible model to use when calculating PV performance on the fly. Similar to the hybrid ML architecture, PowerWize's AI-assisted computation engine provides accurate and responsive solar system design recommendations through a Web-based consultation interface.

Lemence and Tamayao (2021) in the Philippines analysed energy use profiles with techno-economic potential for hybrid solar energy systems for rural health units via HOMER Pro simulations. For the solar-plus-hybrid configuration, the study produced a LCOE reduction of 37 to 42%, and a CO_2 reduction of around 59% compared to a system running solely on diesel. This locally calibrated research serves as an important benchmark tool in PowerWize's consultation-based calculations, which must consider local site conditions, grid availability and load profile of power

consumers in order to produce the system sizing and cost projection reports for the benefit of clients of EnergyWize Solutions Inc.

Mashaly et al (2025) showed how cost-efficient PV systems can be designed using simple but well-parameterized design tools, which are validated by simulation using PVsyst software packages, and how accurate the results of these designed systems are, through a novel study. The finding aligns with PowerWize's design philosophy of making an AI-assisted computation engine available for clients, many of whom are not known energy experts, to use, especially those using EnergyWize Solutions Inc. for the first time; simultaneously providing engineering accuracy great enough to give reliable recommendations on system design.

The literature on AI-assisted solar PV computation as a whole has demonstrated the ability of intelligent systems to automate, optimize and simplify PV installation design and consultation in various contexts. These findings directly validate PowerWize's AI-Assisted Solar PV Consultation Computation Engine which PowerWize has developed using the same validated ML and AI algorithms to support its clients to receive accurate and data-driven system recommendations without the need for manual expertise.

Author/ Year	Technology	Application	Finding	Limitation	Relevance to PowerWize
Hu et al. (2024)	AI/ML in PV systems (review)	MPPT, forecasting, fault detection	AI improves PV operational efficiency across deployment environments	Findings are largely utility/plant-scale, not validated for a consultation-engine use case	General rationale for applying AI to PV analysis
Engel & Engel (2022)	ML (neural networks, SVM, ensembles)	PV design, forecasting, control	ML outperforms traditional methods for nonlinear PV system behavior	Survey-level comparison; does not prescribe a single deployable algorithm	Motivates ML-based sizing, but does not by itself specify PowerWize's algorithm
Jobayer et al. (2023)	Neural networks, RVFL, SVM (systematic review)	PV parameter estimation	Neural networks most frequently used (32.55%) and most accurate	Literature review (2020–2022); no field validation with Philippine site data	Candidate algorithm family for PowerWize's PV engine
Said et al. (2026)	ML-integrated PV design	Sizing/performance optimization	High agreement between predicted and measured yield across latitudes	Validated on the authors' own dataset and sites, not EnergyWize's client base	Precedent for an ML-plus-engineering-formula hybrid design
Lemence & Tamayao (2021)	HOMER Pro simulation (non-ML)	Hybrid solar techno-economic analysis (Philippines)	37–42% LCOE reduction, ~59% CO2 reduction vs. diesel s	Simulation-based, not a trained ML model; limited to rural health-unit context	Locally relevant benchmark data, though methodologically different from PowerWize's ML approach

Table 4. Critical Comparison of Selected AI-Assisted Solar PV Literature [EF4]

Blockchain Technology for Payment Verification and Security

Blockchain has emerged as a significant innovation in the financial landscape, providing a secure and trusted method for conducting financial transactions in the

digital realm. The key characteristics of decentralization, immutability, transparency, and cryptographic security tackle the major shortcomings of traditional centralized payment-verification systems, such as record auditing disputes, record changes and forgery. Organizations such as EnergyWize Solutions Inc., whose struggles with disagreements over enrollment fees often always settled via the blockchain have been documented as operational risks, are particularly primed for the advantages that payment verification on blockchain offers.

An early concept paper introducing Bitcoin as a financial transaction system that provides trustless payment without an intermediary is provided in the seminal whitepaper by Nakamoto (2008) for the development of blockchain-based payment. The distributed ledger of cryptographically connected blocks guarantees the immutability and verification of transactions by all network elements. This basic architecture supports PowerWize's blockchain payment verification module which is based on the concept of decentralized and tamper-proof building transaction records, for verifying all payment transactions related to training enrolment in CEM, CEA and Solar PV consultation fee.

However, insights from transactional inefficiencies led to the development of Turing-compatible smart contracts, allowing users to store self-executing code on the blockchain that automatically validates and executes agreements like payments. Smart contracts were developed by Buterin (2014) to widen the scope of the blockchain past just recording transactions. Smart contracts are directly applicable to PowerWize's payment verification module, which capitalises on their logic to automatically verify financial membership payments, kick off the certificate issuance process when

payments are confirmed and accurately record all financial transactions without additional manual administrative involvement.

The study titled "A detailed survey of blockchain smart contract security threats and their mitigation strategies" by Detian et al. (2025) used a detailed survey of 166 research papers to analyze the threats to blockchain smart contracts and the measures that can be taken to mitigate them. The research revealed three smart contract security vulnerabilities that presented the greatest risks: reentrancy attacks, integer overflow, and access control violations and discussed the mitigation strategies such as static analysis, formal verification and dynamic testing. Along with these security concerns, PowerWize's development include them in the smart contracts it designs for payments which uses formal verification and role-based access control practices that prevent financial losses occurring in the consequences of high-profile smart contract exploits.

With decentralization and immutability of the blockchain database, Adanigbo et al. (2024) found that this integration boosts financial transaction security, transparency, and efficiency. The research substantiated the idea that the ability of blockchain to offer distributed consensus, cryptographic hashing, and smart contracts yields an evidentially better replacement for traditional settlement methods in the centralized financial system, plagued by fraud and single points of failure. The results of this study clearly support the selection of blockchain over traditional payment systems for the system used to verify people's participation enrollment fees. These results support PowerWize's decision to use blockchain technology, rather than a traditional centralized system, to verify participants' enrollment-fee payments.

In a related study, Gu and Dib (2025) showed how a combination of ensemble learning techniques, that combine random forest, XGBoost and recurrent neural

networks, can achieve high accuracy when detecting fraudulent transactions on the Ethereum blockchain. Their study is pertinent to PowerWize's payment module because they examined how easy it is to add an added security layer in a blockchain payment system that will aid in anomaly detection via Ai. This study provided a further layer of security that acts as an extra protection from the blockchain's immutability when looking for suspicious payment fee registration activities.

According to Autade (2025), blockchain technology solutions have been employed to create secure and transparent financial systems, with studies leading up to that time showing that companies implementing blockchain for financial transactions have seen major drops in the number of financial transactions that occur in the transaction dispute category and in administrative time. The documented benefits correspond directly to challenges of EnergyWize Solutions Inc.'s operation, which may include a lack of a verifiable payment record leading to a possible enrollment fee dispute and a history of transactions that can be hard to verify and maintain credibility within the company's regulatory standing as a DOE-accredited training provider.

In a systematic review of the landscape, Masud et al. (2024) reviewed 137 blockchain papers examining the integration of blockchain’s immutability feature with machine learning fraud detection, and found that hybrid blockchain-ML systems are consistently able to improve the security of digital financial transactions. Smart contract automation and decentralized identity verification have been determined as the top two defense strategies against payment fraud and both features are integrated into PowerWize's payment module architecture as AI powering transaction monitoring and the immutability of blockchain technology.

Tiwari et al. (2024) carried out a study of blockchain applications in the financial sector with 12 major blockchain platforms, finding that platforms like Ethereum and Polygon have the most advanced smart contract support and Layer-2 scaling, making them popular for financial transactions. The comparative analysis guides the choice of the platform for Polygon's modules, which includes its payment module and certificate module, with the gas-efficient Layer-2 architecture of Polygon being a fitting choice for a high-frequency payment verification process in EnergyWize Solutions Inc. (EWSI) transaction volume.

In the field of education credentialing, Esguerra et al. (2024) showed a proof-of-concept Blockchain system using Polygon in a real educational setting in the Philippines; this is a locally relevant precedent for the use of blockchain in the credentialing and payment space. Their efforts demonstrate the feasibility of implementing blockchain in Philippine entities with regulatory regimes and business structures comparable to that of EnergyWize Solutions Inc., a DOE-recognized training institution.

A complete survey about how to detect anomalies in blockchain networks, which categorizes anomaly detection methods and identifies the best practices for preventing integrity violations into blockchain networks was provided by Hassan et al (2022). The results of the survey regarding the vulnerabilities of the consensus mechanism and the shared patterns of transaction anomalies provide a valuable input to PowerWize's security design, meaning that checks for payers will be built to incorporate the most prevalent identified blockchain attack vectors, while maintaining a high enough blockchain transaction rate to support EnergyWize Solutions Inc.'s CEM, CEA, and GEMP enrollment system.

Taken together, this literature — from blockchain's foundational architecture to recent smart-contract security research and Philippine implementation studies — supports the use of blockchain technology to help ensure the integrity, transparency, and verifiability of payment records in PowerWize's operating environment

Author/Year	Technology	Application	Finding	Limitation	Relevance to PowerWize
Nakamoto (2008)	Blockchain / distributed ledger	Decentralized trustless payments	Immutable, verifiable transaction ledger	Original design targets currency issuance, not enterprise fee verification	Foundational architecture for PowerWize's payment ledger
Buterin (2014)	Smart contracts (Ethereum)	Self-executing agreements	Enables automated validation/execution of agreements	Smart-contract security risks are well documented (see Detian et al., below)	Basis for PowerWize's automated payment verification and certificate triggering
Detian et al. (2025)	Smart-contract security survey	Vulnerability analysis (166 papers)	Reentrancy, integer overflow, and access-control flaws are the top risks	Survey-level; mitigations not yet tested on PowerWize's own contracts	Defines the security checklist PowerWize's smart contracts should be tested against
Tiwari et al. (2024)	Platform comparison (12 blockchains	Financial blockchain platforms	Ethereum and Polygon offer the strongest smart-contract and Layer-2 support	Comparative/qualitative; does not benchmark PowerWize's actual transaction volume	Basis for PowerWize's choice of Polygon
Gu & Dib (2025)	Ensemble ML (random forest, XGBoost, RNN)	Fraud detection on Ethereum	High accuracy detecting fraudulent transactions	Tested on Ethereum mainnet data, not a training-fee payment context	Potential future enhancement; confirm whether this is implemented in PowerWize's current scope

Table 5. Critical Comparison of Selected Blockchain-Payment Literature *[EF5]*

Blockchain-Based Digital Certificates and Credentialing

Blockchain has empowered a centralized, tamper-proof alternative to paper-based certifications and centralized digital credential management systems, revolutionizing the entire process of issuing and verifying digital credentials. Users of verifiable professional credentials (VPCs) including those recognized by the DOE whose certification is a legal requirement have gained traction in recent years, driving the scholarly and industrial efforts to realize blockchain-based credentialing systems to ensure immediate third-party proof of a credential without needing credential-issuing institutions.

Khati et al. (2023) created an academic certificate sharing system based on blockchain that runs on the Ethereum network and integrates IPFS. The system mints academic certificates as ERC-721 non-fungible tokens (NFTs), where the access control is managed by blockchain addresses. Learners can share institutional credentials more directly with the employers and third party verifiers through the blockchain addresses, without the support of the intermediaries of institutions. This is the most similar design to PowerWize's tokenized certificate system, where completion certificates (CEM, CEA), issued by DOE or third party certificates, are also blockchain based and available to employers, government and other stakeholders to verify for the purposes of non-essential compliance.

Khati et al. (2025) conducted another user acceptance study that built on a PLS-SEM based-TAM framework with high levels of perceived usefulness and ease of use reported by both certificate issuers and recipients in the NFT-based certificate system. The high acceptance scores from Khati et al. (2025) support PowerWize's decision to use tokenized certificates as the credentialing format for EnergyWize Solutions Inc. 's

training completions, as this system was found to be useful by target users and was easy to use.

Esguerra et al. (2024) offered an immediate case study for the Philippines in creating a Smart Credentialing and Verification System (SCVS) for national certificates based on blockchain technology running on a proof-of-stake network, Polygon. The system met the ISO/IEC 25010 quality model, and gained a high level of functional suitability, security and usability scores. EnergyWize Solutions Inc. is a Philippine company that issues national energy certification (NEC) by the Department of Energy (DOE) which finds itself in a similar setting as the companies in Esguerra et al's 2024 article the context of doing so is akin to that of PowerWize's tokenized certificate module.

Given the wide range of educational applications of NFTs, Wu and Liu (2023) surveyed the current potential use cases in education and demonstrated the benefits of NFT-based educational credentials over a centralised PDF certificate, such as uniqueness, immutability, portability and self-custody. The review suggests that NFT credentials are well suited for situations where certificate portability and third-party verifiability are important, which is relevant to PowerWize's attempt to design certificates for completion that are based on NFTs and will be verifiable by external government bodies for both CEM and CEA certificates as well as GEMP certificates.

The feasibility of zero-knowledge proof blockchain verification of academic records was proved by Berrios Moya et al. in 2025, with a gas-fee reduction of about 94% when deployed on the Polygon zkEVM compared to the Ethereum mainnet deployment. The cost reduction is significant, allowing companies such as EnergyWize Solutions Inc. to make credentialing on the blockchain economically practical, as they

produce multiple batches of certificates each training cycle year-round through their programs, including certificates in energy efficiency for design (CEM), energy efficiency for assemblies (CEA), and green energy management packages (GEMP).

Nikolic et al. (2022) designed a blockchain application to generate digital certificates. They showed that their prototype application was implemented on the Ethereum blockchain to create a digital education certificate, which they presented as a crypto-signed artifact. The ability to prevent certificate forgery and also to remove manual verification processes was marked as top operational benefits of the certificate systems based on the blockchain benefits that are paramount to PowerWize as certificate holders' background directly influences their regulatory status.

As shown by Rahman et al. (2024), blockchain-based certificate verification systems proved to be efficient and effective solutions against credentialing fraud and non-digital certificate's inherent inefficiencies. The researchers noted the application of cryptographic hash functions and decentralized storage for creating an individual and verifiable certificate identity that can be verified by any party with access to the blockchain. Each certificate whether CEM, CEA or GEMP issued via PowerWize's system can be uniquely identified and not easily forged, highlighting the significant difference in the platform's technical implementation over the current PDF format-based certificate systems which are susceptible to forgery.

Among end-users Andrade and Amate (2026) successfully demonstrated the viability of issuing academic certificates in a decentralized way, with low latency (0.9 seconds) to confirm their issuance in the TRON network, and a System Usability Scale score of 76.67, above the 68-point industry average. The study's performance metrics offer a quantitative measurement for the performance goals that PowerWize sets for

issuing certificates, such as how fast transactions are confirmed and how certain end-users are in the certificate retrieval process.

IMS Global Learning Consortium (2022) is the entity that developed the Open Badges 3.0 specification which represents a world-first new collaboration between IMS Global Learning Consortium and W3C to make blockchain-based digital badges interoperable with global credential registries that are compliant with the W3C Verifiable Credentials Data Model. Building upon Open Badges 3.0 principles, PowerWize's certificate module will enable EnergyWize Solutions Inc.'s DOE accepted certificates to be blockchain-verifiable as well as interoperable with other globally accepted digital credential ecosystems to ensure EnergyWize Solutions Inc.'s DOE accepted solutions are globally portable.

Mazzocca et al. (2025) offered a thorough overview of decentralized identifiers and verifiable credentials, making the W3C DID standard the most comprehensive and authoritative framework for blockchain-based identity and credential management. PowerWize's tokenized certificate (ERC-721 NFT) architecture does not implement the W3C DID standard directly, but shares the same underlying goal of enabling cross-institutional certificate verification, and could draw on DID principles in a future iteration that pursues broader interoperability.

The examined literature shows a common consensus about blockchain-based digital certificates, as the paths for professional and academic certification globally are towards tokenized, blockchain-anchored credentials. Rooted in this literature, PowerWize's tokenized certificate system includes international standards for NFT and certificate credentials along with implementation experiences from the Philippines, all

underpinned by a secure and verifiable interoperable system that will meet the needs of EnergyWize Solutions Inc.'s DOE-approved training programs.

Author/Year	Technology	Application	Finding	Limitation	Relevance to PowerWize
Khati et al. (2023)	Ethereum + IPFS, ERC-721 NFTs	Academic certificate sharing	Certificates minted as NFTs enable direct learner-to-verifier sharing	Ethereum gas costs are high; single-institution pilot	Closest architectural analog to PowerWize's certificate module
Esguerra et al. (2024)	Polygon (proof-of-stake)	Philippine national credentialing (SCVS)	High ISO/IEC 25010 suitability, security, and usability scores	Single Philippine case study; not specific to the energy sector	Directly relevant precedent for a DOE-context Philippine deployment
Berrios Moya et al. (2025)	Polygon zkEVM, zero-knowledge proofs	Academic record verification	~94% gas-fee reduction vs. Ethereum mainnet	Implementation complexity of zk-proofs not detailed	Supports cost-effective certificate issuance at scale
Andrade & Amate (2026)	TRON network	Decentralized certificate issuance	0.9-second confirmation latency; SUS score of 76.67	Uses a different blockchain (TRON) than PowerWize's Polygon	Comparable performance/usability benchmark, not a direct precedent
IMS Global Learning Consortium (2022)	Open Badges 3.0 / W3C Verifiable Credentials	Interoperable digital credentials	Establishes a global interoperability standard	Adoption depends on issuer/verifier ecosystem support	PowerWize should confirm alignment with

Table 6. Critical Comparison of Selected Blockchain-Certificate Literature [EF6]

Agile Software Development Methodology

The software development methodology is a key factor that will determine the quality, flexibility and time line of a software system. Both in the professional software industry and in academia, Agile methodologies and in particular the Scrum framework

are now the preferred methods not just for their capacity to deal with changing requirements but also for their success in developing working software in small increments and for the fact that they keep stakeholders closely engaged throughout the project. Agile is especially vital for this kind of system, which combines artificial intelligence with blockchain and online consultation features. In an environment as diverse and multifaceted as PowerWize where AI, blockchain, and web-based consultation tools are all being put into practice flexibility and iterative quality control are particularly significant.

The fundamental document in Agile development is the "Manifesto for Agile Software Development" by Beck et al. (2001) which clarifies four values: individuals and interactions over processes and tools; working software over comprehensive documentation; customer collaboration over contract negotiation; responding to change over following a plan. These values serve as the foundation for the project's development process, characterized by incremental delivery, ongoing engagement of program EnergyWize Solutions Inc. stakeholders and adapting to changing system needs.

The definitive 2020 update to the Scrum Guide was published by Schwaber and Sutherland, which includes the adoption of the new principle: A single Product Goal and the reaffirmation that the shared Scrum Team is responsible for all facets of software delivery. The rules outlined in the 2020 Scrum Guide are used by PowerWize's capstone team as an operational mode when it follows the sprint-based development they use for the four major system building blocks, the AI training assistant, the Solar PV computation engine, the blockchain payment verification module, and the tokenized certificate system.

For the discussion of the critical success factors of Agile, Anifa et al. (2024) looked into the systematic literature review (SLR) of Agile methodology in different organizational settings, and found that sprint reviews, retrospectives, and stakeholder meeting and collaboration are the most frequent critical success factors emerged from the collection of articles. When the reviewers checked a wide range of the literature on software methodologies, they found that “in practice Agile methodologies have consistently outperformed traditional waterfall models in contexts with changing requirements, cross-functional teams and time-sensitive deliveries which are conditions that are applicable to PowerWize's capstone development.” The success factors identified by Anifa et al. (2024) are reflected in PowerWize's Agile-based development plan.

According to Mishra et al. (2023), a comparative study of structured and Agile software development revealed that the success rate of Agile projects is 88.2% and for waterfall 47%. The study found that Agile's good performance was due to a facet of the iterative delivery model the ability of the software teams to detect and correct defects early in the development process and to fine-tune based on changing stakeholder demands. The computer evidence is fundamental to PowerWize's methodology justifications stating that the current approach is not based on observers' preferences, but one that is empirically better for the current multi-technology and complex system being developed.

Dixit, et al. (2023) looked at Scrum as an agile software development process and compared metrics used for measuring sprint productivity, team velocity, and defect density for numerous instances of it. They have been able to conclude that teams which adopt Scrum with explicit sprint backlogs and daily standups had significantly higher

velocity and lower levels of defects than less structured variants of Agile. PowerWize's capstone team utilizes the sprint backlog and velocity tracking methods suggested by Dixit et al. (2023) to track progress in system development and to keep the development on track to meet the timeline for when each system module is delivered during the academic year.

Lee et al. (2023) stressed that it was possible for teams to deliver reusable components across various modules of the system using the application of practices from software product line engineering (SPLE-Scrum) with Scrum in a hybrid development approach for an online platform in the blockchain domain. It is definitely applicable to the development of PowerWize, where the blockchain layer is directly applicable to the payment verification module and the certificate tokenization module can be directly reused and developed already at an early stage of the sprint, benefiting development efficiency.

In the field of software project management, Rodrigues et al. (2024) investigated how agile methodologies became adopted in software project management education, revealing that all of the surveyed companies in the industry and educational institutions emphasized that agile software methodologies are now the most preferred methods over the waterfall method in software development projects. To carry out and educate students about PowerWize's use of Agile as a methodology and a means of developing the Agile skills of the capstone team, their case study made the software development community's product better quality and showed students and instructors the benefits of applying Scrum.

To deduce the critical success factors that can lead to the success of software process deployment in the context of Scrum, we systematically reviewed empirical

studies in different contexts as reported by Bayona-Oré et al. (2023). Every module is validated with EnergyWize Solutions Inc. during the sprint review each two weeks and the process improvement is done based on the retrospective, and these are already part of PowerWize's Agile development plan, which follows the identified success factors.

Existing literature around the use of the Agile software development methodology as a whole enables providing that the Scrum framework is the most suitable and well-explored solution to the development of complex systems that include multiple technologies such as PowerWize. The reviewed studies directly influence PowerWize's development plan through an iterative process based on sprints, and guide the implementation of best practices in the industry.

Software Quality Evaluation ISO/IEC 25010

Software Quality Evaluation, systematic demand from which comes evidence that a developed software system fulfills user requirements, functional requirements and non-functional requirements. The ISO/IEC 25010 standard has emerged worldwide as the standard reference for software evaluation, providing a robust and systematic description of software quality attributes that can be used to assess software systems with numerous attributes that affect end-user satisfaction and the efficiency of software within organizations.

The latest iteration of the SQuaRE product quality model, ISO/IEC 25010:2023, introduced safety as a new characteristic, and the rest of the product quality model were renamed: functional suitability, performance efficiency, compatibility, interaction capability (replaced with usability in the 2011 version), reliability, security, maintainability, flexibility (replaced with portability in the 2011 version), and safety. PowerWize's UAT protocol evaluates selected characteristics of the 2023 standard —

functional suitability, performance efficiency, interaction capability, reliability, security, and maintainability — consistent with the research objectives in Chapter I. Compatibility, flexibility, and safety are not directly measured in this study.

Prior to the revision in 2023, the predecessor ISO/IEC 25010:2011 (ISO, 2011) defined eight characteristics, and used it as the evaluation framework mostly applied in Philippine capstone research. Both versions are of significance in understanding studies conducted on related topics which were explored in this chapter since majority of the Philippine capstone precedents employed the 2011 version. The new characteristic of Safety is especially significant for PowerWize, which deals with delicate payment transaction information and is registered by DOE as a qualified specialist, situations that could result in repercussions from control data safeguarding breakdowns.

For the development platform, Mobile JKN, which is a digital health insurance management application, Lusiani and Princes (2024) tested the SEM-PLS model quantitatively using questionnaires distributed to the platform users in the range of Indonesia to get the ISO 25010 application quality scores. They investigated the impact of functional suitability, reliability, performance efficiency, usability and security on perceived application effectiveness. The energy training and consultation sector used the SEM-PLS methodology and Lusiani and Princes' (2024) questionnaire instrument as a blueprint for the development of PowerWize's questionnaire instrument for UAT.

Dela Rosa et al. (2022) evaluated a web-based database courses e-learning application based on ISO/IEC 25010 standard which yields an overall mean score of 4.24-4.41 that represents the rating 'Very Good' for each evaluator participant, namely student, faculty and expert evaluators. The study's occurrence using three groups of respondents (end-users, administrators and technical experts) along with an ISO 25010-

based Likert-scale questionnaire provides the most methodologically similar analog of PowerWize's UAT design because of the study's academic setting, Filipino users, and the web-based system type.

Estdale and Georgiadou (2018) gave guidance to the practical application of ISO/IEC 25010 evaluation for implementation in software quality assurance on how each quality characteristic could be translated into measurable parameters and acceptance criteria. The model they developed, which represents the ISO 25010 characteristics, directly influences the construction of PowerWize's UAT, with each quality characteristic having test cases that are appropriate and illustrative of the actual usage of CEM, CEA, and GEMP training participants.

The System Usability Scale (SUS) developed by Brooke (1996) is still the most widely used supplementary measure of system usability in evaluations based on ISO 25010. SUS is a 10-item questionnaire designed to measure system usability perceived by non-technical users and can be compared to published norms. PowerWize's UAT protocol is designed to include the SUS as a supplementary measure of interaction capability. SUS administration to respondents is scheduled as part of the upcoming survey rollout; results will be reported in Chapter IV once collected.

Lewis (2018) performed a review of the SUS after 20 years of empirical studies which found SUS to be valid, reliable and sensitive to usability difference, with SUS scores above 68 being classified as 'Above-average usability' and SUS scores above 80.3 being classified as 'Excellent'. These benchmarks will be used as the scale's acceptability criteria once PowerWize's UAT is administered, with a planned minimum SUS value of 70 for acceptability for deployment among CEM, CEA, and GEMP training participants.

A bibliometric and thematic analysis of 1,613 scientific publications mapped the evolution of ISO/IEC 25010 and found that functional suitability, usability, and reliability are the three characteristics that are most often assessed based on ISO 25010; and ISO 25010 is increasingly adopted in systems that involve the use of AI and blockchain. The acceptance of this trend is a direct acceptance of the fits of ISO/IEC 25010:2023 as an Evaluation framework for a technically complex system like PowerWize.

This ISO/IEC 25010:2023: Model for Evaluating DevOps Software Products was applied to a range of "DevOps software products" on security-sensitive and performance-critical systems, as demonstrated in the paper by Okonkwo (2025), in relation to the 2011 model to which it has also been extensively applied. The results of the study inform the design of PowerWize's UAT instrument which ensures that it covers the latest version of the standard and matches the Safety characteristic introduced in the 2023 version.

All of the literature provides a solid theoretical, methodological, and empirical basis for the quality evaluation approach implemented by PowerWize. The studies reviewed support the planned use of ISO/IEC 25010:2023 in combination with the System Usability Scale (SUS) to assess a web-based, AI-enhanced, blockchain-integrated training and consultation system for use by professionals issuing energy certification, once both instruments have been administered.

2.1.3 Thematic or Chronological Organization

The literature above is organized thematically across the six areas listed in 2.1.2, since a thematic grouping most clearly shows how each body of work maps onto PowerWize’s corresponding system module.

2.1.4 Relevance of the Present Project

Collectively, the reviewed literature establishes the technical and methodological basis for PowerWize's AI learning assistant, AI-assisted Solar PV engine, blockchain payment verification module, tokenized certificate system, Agile-based development process, and ISO/IEC 25010:2023-guided evaluation approach, as detailed in the comparison tables and synthesis below.

2.1.5 Citation and Referencing

APA 7th Edition is used for in-text citations and the reference list. A citation-reference audit identified remaining inconsistencies (Trindade & Cordeiro's in-text year, the Dixit et al. in-text year, the Ndiaye et al. source pairing, and a duplicate DevOps/ISO 25010 reference under two author-name formats) that require correction before final submission.

2.2 Related System and Technology Gap Analysis

2.2.1 Identification of Related Systems

This study aimed to design and develop PowerWize: an AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. The researchers conducted a technology gap analysis of ten existing related systems similar to the proposed platform to determine their strengths, limitations, and relevance to PowerWize.

2.2.1 System Feature Comparison

System	AI Training	PV Computation	Blockchain Payment	Tokenized Certificate	Admin Dashboard	Integrated (all four)
Moodle	✗	✗	✗	✗	✓	✗
Canvas	✗	✗	✗	✗ (PDF only)	✓	✗
TalentLMS	✓ (AI-generated content)	✗	✗	✗ (PDF only)	✓	✗
Coursera	✗	✗	✗	✗ (centrally hosted)	✓	✗
PVsyst	✗	✓	✗	✗	✗	✗
SAM (NREL)	✗	✓	✗	✗	✗	✗
HelioScope	✗	✓	✗	✗	✗	✗
Blockcerts	✗	✗	✗	✓	✗	✗
Credly	✗	✗	✗	✓ (centralized)	✗	✗
BitPay	✗	✗	✓	✗	✗	✗
PowerWize	✓	✓	✓	✓	✓	✓

Table 7. Technology Gap Comparison Matrix [EF7]

2.2.3 Analysis of Existing Systems

2.2.4 Technology Gap

As Table 2.5 shows, none of the ten reviewed systems combine AI-assisted training, AI-assisted PV computation, blockchain payment verification, and tokenized certificates in one platform; each specializes narrowly (see 2.3.2 Identified Gap for the full statement).

2.2.5 Proposed Response to the Gap

PowerWize responds by integrating all four capabilities within a single platform purpose-built for EnergyWize Solutions Inc. 's CEM, CEA, and GEMP programs (see 2.3.3 Contribution of the Proposed Project for the full discussion).

2.3 Synthesis

2.3.1 Integration of Findings

What is known. Prior research demonstrates that AI can support personalized training delivery and PV design decisions, that blockchain can help secure payment and credentialing records, and that Agile/Scrum together with ISO/IEC 25010 provide workable frameworks for developing and evaluating complex, multi-technology systems.

What is not yet known. Limited evidence addresses how AI-assisted training, AI-assisted Solar PV consultation, blockchain-based payment verification, and blockchain-based credentialing perform when integrated into a single platform — particularly one built for a Philippine DOE-accredited energy training and consultation provider.

2.3.2 Identified Gap

What existing systems lack. As Table 2.5 shows, none of the ten reviewed systems combine all four capabilities; each specializes narrowly in either training delivery, PV computation, payment processing, or credentialing.

Although prior studies have separately demonstrated the feasibility of AI-assisted training, machine-learning-based PV computation, blockchain-based payment records, and blockchain-based credentialing, limited evidence currently addresses their integration into a single platform purpose-built for a Philippine DOE-accredited energy training and consultation provider. This gap is the basis for the development and evaluation of PowerWize in the present study.

2.3.3 Contribution of the Proposed Project

What PowerWize contributes. PowerWize integrates an API-based generative AI learning assistant, a machine-learning-based Solar PV consultation engine, Polygon-based blockchain payment verification, and ERC-721 tokenized certificates within a single platform built specifically for EnergyWize Solutions Inc.'s CEM, CEA, and GEMP programs.

How the study evaluates it. The integrated system is evaluated through black-box functional testing and User Acceptance Testing guided by selected ISO/IEC 25010:2023 characteristics and the Technology Acceptance Model, as detailed in Chapter III.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Materials and Methods

This chapter presents the methods and procedures that the researchers employed in designing, developing, testing, and evaluating the proposed PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. It includes the research design, population of the study, sampling design, data collection instruments, project design, system testing and evaluation procedure, validation of questionnaire, and statistical treatment.

3.1.1 Research and Fact-Finding Activities

The primary data collection instrument used in this study was a structured questionnaire designed to gather relevant information from all respondent groups during the User Acceptance Testing (UAT) phase of PowerWize's evaluation. To ensure that the data collection process was comprehensive, the researchers also employed several complementary fact-finding techniques prior to and during system development.

Before the development and implementation of the system, a preliminary interview was conducted with selected administrators and staff of EnergyWize Solutions Inc. to gather insights about their current operational processes, level of awareness on the limitations of existing email-based workflows, and expectations regarding a unified digital training and consultation system. The interview helped the researchers identify common challenges in training material management, payment

verification, and certificate issuance across the CEM, CEA, and GEMP programs, and determine which features would be most needed and beneficial in the proposed PowerWize system. The information collected from this activity served as the foundation for designing the system's functionality and ensuring alignment with the actual needs of EnergyWize Solutions Inc.

The researchers conducted online research to review existing training management systems, Solar PV design tools, blockchain payment platforms, and digital credentialing systems. Several existing systems were identified including Moodle, Canvas, PVsyst, SAM, Blockcerts, and Credly that provide partial functionality relevant to PowerWize's intended use case. However, a significant technology gap was identified: no existing system simultaneously integrates AI-assisted energy certification training, AI-driven Solar PV consultation computation, blockchain payment verification, and blockchain-based tokenized certificate issuance within a single platform tailored for DOE-accredited training providers. This gap analysis validated the necessity and novelty of developing PowerWize as an integrated solution for EnergyWize Solutions Inc.

3.1.2 System Development Activities

The researchers organized development into iterative Agile activities: requirements gathering, interface and database design, module implementation, functional testing, stakeholder review, refinement, and preparation for deployment. The training, Solar PV consultation, payment verification, certificate, and administrative functions were developed and reviewed in increments. Section 3.9 discusses the development model and its phases in greater detail.

3.1.3 Resources and Technologies

The development of PowerWize utilizes a combination of web-based technologies, artificial intelligence, blockchain integration, and database management tools to support its training, consultation, payment verification, and digital certification functions. The system is developed using Python 3.12.10 and the Flask framework for backend processing, while Jinja2 templates, HTML, CSS, and JavaScript are used to create the web-based user interfaces. PostgreSQL serves as the primary relational database management system, with psycopg2 facilitating database connectivity and data operations.

For artificial intelligence functionalities, the system integrates the Groq API for cloud-based AI processing and LM Studio for optional local AI inference. FastEmbed, NumPy, and retrieval-augmented generation (RAG) techniques support contextual information retrieval and AI-assisted learning. Additionally, Tesseract OCR and supporting document-processing libraries facilitate text extraction and file handling.

Blockchain functionalities are implemented using Web3.py, Solidity, and OpenZeppelin smart contracts deployed on the Polygon Amoy testnet. These technologies support blockchain payment verification and the issuance of non-transferable ERC-721 tokenized certificates. PayMongo is integrated for processing card and QR Ph payments, while Gmail SMTP supports email notifications. The primary development environment consists of Windows 11 Pro and Ubuntu, with Git for version control and Docker Compose for database services. These technologies provide the technical foundation for integrating the major functionalities of PowerWize into a centralized web-based platform.

3.2 Research Design

3.2.1 Rationale for the Selected Research Design

This study's primary research design is Developmental Research Design, which focuses on the systematic design, creation, and evaluation of technological solutions. Within this primary design, a descriptive-evaluative approach was used to assess the usability and effectiveness of the developed system, through surveys and guided system demonstrations with selected administrators, trainers, and participants of EnergyWize Solutions Inc. Separately, Agile served as the software development methodology used to build the system iteratively — this is a development methodology, not the study's research design, and the two are described separately in this chapter to avoid conflating them.

According to Palmerola (2024), developmental research is commonly applied in technology-driven projects to systematically build and evaluate systems through iterative cycles, ensuring that the final product is both functional and reliable. In line with this, the researchers applied developmental research to create the software components of PowerWize including the AI learning assistant module, AI-assisted Solar PV computation engine, blockchain payment verification module, tokenized certificate issuance system, and administrative dashboard refining each component based on continuous testing and feedback throughout the Agile development process.

Bejar, Arciso, and Buladaco (2023) emphasized that descriptive analysis is effective in evaluating user perceptions such as usefulness, ease of use, and intention to adopt new technologies, making it suitable for determining how administrators, trainers, and training participants engage with the proposed PowerWize system. Similarly, Kosie and Lew-Williams (2022) highlighted that descriptive research

focuses on systematically documenting conditions, behaviors, and user experiences, which aligns with the goal of this study to measure how PowerWize improves the management of CEM, CEA, and GEMP training programs and Solar PV consultation services at EnergyWize Solutions Inc. By integrating both developmental and descriptive approaches, the study ensures not only the creation of an innovative digital solution but also its empirical validation in terms of usability, acceptance, and contribution to operational efficiency.

3.3 Population of the Study

The population of the study consists of selected stakeholders from EnergyWize Solutions Inc. who are the intended users and evaluators of the proposed PowerWize system. These respondents represent the different user groups that will directly interact with the system's major features and functions during actual operations. The respondents include the super user, administrators, secretary, trainers, and trainees involved in the organization's training and consultation services. Specifically, the study covers administrators and staff responsible for managing training operations, payment verification, and certificate issuance for the DOE-accredited Certified Energy Manager (CEM), Certified Energy Auditor (CEA), and Government Energy Management Program (GEMP) training and workshops. It also includes trainees and participants enrolled in these programs. Solar PV consultation clients are not part of this respondent population; they are served by the system's consultation module but are not surveyed as part of UAT.

Since the total population consists of only 48 respondents who are all accessible to the researchers, the study will employ total or complete enumeration as the sampling technique. This ensures that all intended users of the proposed system are properly represented in the evaluation and assessment of PowerWize.

3.4 System Tester

In addition to the primary respondents, a group of ten (10) IT professionals was selected through purposive sampling to serve as system testers and conduct technical evaluations of PowerWize's functionality and performance. These IT professionals were deliberately chosen based on their expertise in information technology, software development, and systems evaluation to ensure objective assessment of the system's technical components, including the blockchain payment module, AI computation outputs, and certificate tokenization mechanisms. Eligibility for selection required at least two years of professional experience in software development, quality assurance, information systems, or a related IT field.

The use of purposive sampling is supported by Tongco (2007), who explained that purposive sampling is effective in selecting informants with specialized knowledge relevant to the objectives of the study. Foley (2023) further emphasized that purposive sampling enables researchers to obtain high-quality and relevant information from participants with direct expertise and experience.

3.4.1 Profile of System Testers[8]

NO.	TESTER NAME	EDUCATIONAL BACKGROUND/ CREDENTIALS	CURRENT POSITION/COMPANY
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Table 8. System Testers

Table 2 is intended to document the qualifications and roles of the ten purposely selected IT professionals. Their names, educational backgrounds, professional positions, and relevant experience must be entered only after they are verified from actual tester records; no tester credentials are assumed in this study.

3.5 Respondents for User Evaluation

The intended user-evaluation population consists of 48 EnergyWize stakeholders: one super user, one administrator, one secretary, 20 trainers, and 25 trainees. These groups represent the principal organizational and training users of

PowerWize. Solar PV consultation clients may interact with the consultation feature but are not included as a separate UAT respondent group in the stated population.

3.5.1 Distribution of Respondents

RESPONDENTS	POPULATION	SAMPLE SIZE	PERCENTAGE
Super User	1	1	2.08%
Admin	1	1	2.08%
Secretary	1	1	2.08%
Trainers	20	20	41.67%
Trainees	25	25	52.08%
TOTAL	**48**	**48**	**100%**

Table 9. Respondents for the User Evaluation

Table 1 presents the distribution of respondents who participated in the user evaluation of PowerWize. The study involved a total of 48 respondents composed of 1 super user, 1 administrator, 1 secretary, 20 trainers, and 25 trainees. Among the respondents, trainees comprised the largest group with 52.08% of the total population, followed by trainers with 41.67%, while the super user, administrator, and secretary each represented 2.08% of the respondents. Since the total population was relatively small and accessible, the researchers utilized total or complete enumeration to ensure that all intended users of the system were included in the evaluation process.

3.6 Sampling Design

This study employed total enumeration for the primary respondents since the total population consisted of only 48 accessible participants, including administrators, trainers, and trainees of EnergyWize Solutions Inc. Through total enumeration, all

intended users of the system were included in the evaluation process to ensure comprehensive and reliable feedback.

The use of total enumeration is appropriate when the target population is relatively small and manageable. This approach allows researchers to gather complete data from all relevant respondents, minimizing sampling bias and ensuring that the perspectives of all intended system users are represented.

3.7 Data Collection Instrument

The researchers conducted direct observation of EnergyWize Solutions Inc.'s existing operational processes to gather first-hand data on the current state of the company's training program management, payment processing, and certificate issuance workflows. Specifically, the researchers observed how training materials are distributed via email, how quiz submissions and payment confirmations are manually processed by administrative staff, and how certificates for CEM, CEA, and GEMP program completions are generated and sent to participants. This observational data provided practical, context-specific insights that guided the functional design of PowerWize's modules and ensured that the system addresses the actual pain points experienced by EnergyWize Solutions Inc. in its daily operations.

A structured survey questionnaire was the primary instrument used during the User Acceptance Testing phase of the study. The questionnaire was designed based on the combination of User Acceptance Testing (UAT) criteria and ISO/IEC 25010:2023 software quality characteristics, gathering respondent feedback across six quality dimensions: functional suitability, performance efficiency, interaction capability, reliability, security, and maintainability. A five-point Likert scale was applied, with response options ranging from Strongly Disagree (1) to Strongly Agree (5). The

questionnaire was administered to all 48 respondents following a guided system demonstration and supervised hands-on interaction with PowerWize's core modules.

3.7.1 Instrument Development

The researchers developed a structured UAT questionnaire using selected software-quality characteristics from ISO/IEC 25010:2023 and technology-acceptance constructs, including perceived usefulness, perceived ease of use, attitude toward using, and behavioral intention to use. The instrument also provides space for open-ended comments. The ten-item System Usability Scale (SUS) is identified as a supplementary measure, but its administration and results must be confirmed before reporting a SUS score.

3.7.2 Instrument Validation

According to Mohamad Adam Bujang et al. (2022), many widely used validated questionnaires must undergo a rigorous validation process that includes content validation and psychometric testing before being applied to specific respondent populations. Since different respondent groups may interpret questionnaire items differently based on their role and background, careful validation procedures are necessary to ensure the accuracy, clarity, and consistency of the evaluation instrument.

One form of validity is face validity, which refers to the extent to which a measurement tool appears appropriate and relevant to what it is intended to measure. It is assessed through expert judgment and structured review to ensure that the questionnaire items are clear, understandable, and contextually appropriate for the study's respondents. According to Patel and Desai (2023), face validity is particularly important in the early stages of validating evaluation instruments used in technology

adoption studies, as it enhances both the perceived relevance and usability of survey tools among non-technical respondents.

The PowerWize UAT questionnaire was submitted for review to three validators: (1) a faculty member from the College of Computer Studies of Laguna State Polytechnic University with expertise in information systems evaluation; (2) a registered energy professional with experience in DOE-accredited CEM, CEA, and GEMP training program delivery; and (3) an IT practitioner with professional experience in blockchain application development and web-based system evaluation. The validators assessed each questionnaire item for clarity, relevance to the ISO/IEC 25010:2023 quality characteristics, alignment with the TAM evaluation framework, and appropriateness for the target respondent groups. Based on the validators' feedback and recommendations, necessary revisions were made to selected questionnaire items to improve clarity and contextual fit before the instrument was finalized for distribution. This validation was limited to face and content validity through expert review; a content validity index, pilot testing, and post-collection reliability statistics such as Cronbach's alpha were not conducted and are noted as a limitation of the current validation process.

3.7.3 Administration of the Instrument

The intended administration procedure begins with an explanation of the study and the respondents’ voluntary participation, followed by a guided demonstration and hands-on use of role-appropriate PowerWize functions. Respondents then complete the structured questionnaire independently. Responses are checked for completeness, encoded by respondent group, and summarized without disclosing personal identities. Any actual response counts lower than 48 must be reported accurately rather than treated as a complete census.

3.8 Project Design

PowerWize was designed as an integrated platform for managing CEM, CEA, and GEMP training activities, supporting Solar PV consultation calculations, recording payment-verification information, issuing tokenized certificates, and providing administrative oversight. Project design connects the operational problems identified during fact-finding to functional requirements, interface workflows, data records, and module interactions.

3.8.1 System Requirements

Functional requirements include user authentication and authorization; training enrollment, learning content and quiz handling; AI-generated post-assessment feedback; Solar PV consultation input and computation; payment verification; certificate issuance and verification; and administrative monitoring. Nonfunctional requirements include understandable interfaces, appropriate response times, data integrity, access control, recoverability, and traceable error handling. The final requirements list must match the implemented scope and approved specifications.

3.8.2 System Design

The design translates these requirements into user-role workflows, interface layouts, database entities, API interactions, and validation rules. Training users follow an enrollment-to-assessment workflow; consultation users submit site and load information for calculation; authorized personnel review payments, certificates, and reports. Interface prototypes and data-flow or process diagrams should be referenced by their actual figure numbers when available.

3.8.3 System Architecture

The proposed system uses a layered structure: users access role-specific interfaces; the application layer processes authentication, training, assessment, consultation, payment, and certificate requests; and the data and blockchain layers retain application records and verification references. Sensitive account and transaction information must be protected by role-based access, input validation, and appropriate handling of credentials. The final architecture diagram should reflect the frameworks and services actually implemented.

3.9 Software Development Model

According to Highsmith (2022), the Agile Model is a software development methodology that focuses on iterative progress, collaboration, and adaptability. It is designed to deliver functional components in small increments, allowing continuous improvements based on feedback from stakeholders. Each development cycle or sprint typically includes planning, designing, coding, testing, and reviewing, making it flexible and highly responsive to changing requirements. This approach is ideal for the development of PowerWize, which integrates multiple complex and interdependent technologies artificial intelligence, blockchain, and automated engineering computation that benefit greatly from the iterative quality assurance and continuous stakeholder alignment provided by the Agile framework.

The Agile development process of PowerWize is structured into seven iterative phases: Requirement and Analysis, Design, Implementation, Testing, Development, Maintenance, and Launch. As illustrated in Figure 2, these phases follow a cyclical and iterative sequence, allowing the research team to revisit and improve earlier work based on findings and feedback from subsequent phases. This cyclical structure ensures that

the final system is both technically robust and operationally aligned with the needs of EnergyWize Solutions Inc

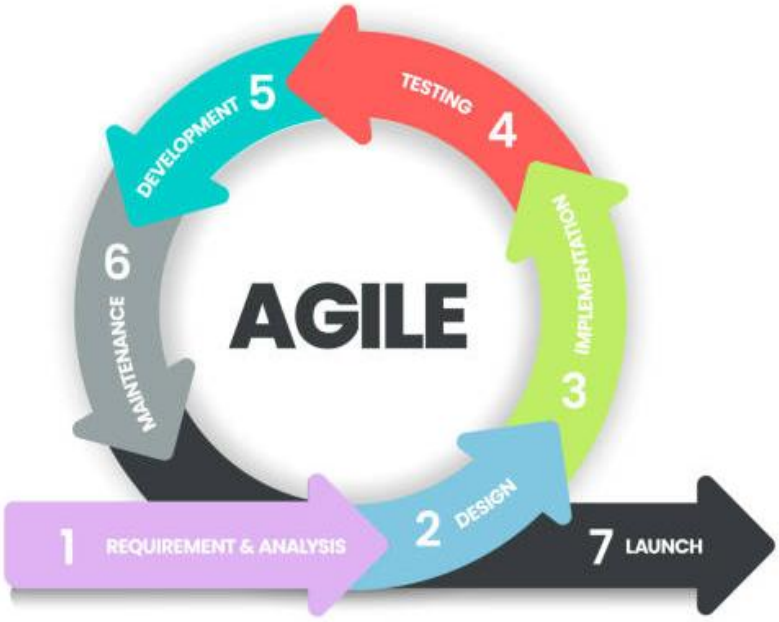


Figure 2. Agile Model

The following table summarizes the seven Agile development phases applied in the construction of PowerWize:

PHASE	KEY ACTIVITIES
Phase 1 Requirement and Analysis	Stakeholder interviews with EnergyWize Solutions Inc. management and staff; definition of system scope, user roles (admin, trainer, CEM/CEA/GEMP trainee, PV client), and functional requirements; technology stack selection (AI frameworks, blockchain platform, web development tools); initial sprint planning and project timeline establishment; product backlog creation and prioritization.
Phase 2 Design	System architecture design; database schema and entity relationship modeling; UI/UX wireframing for all modules (AI training module, Solar PV consultation engine, blockchain payment module, certificate issuance system, admin dashboard); blockchain smart contract schema design; AI model integration specification; design review with EnergyWize Solutions Inc. stakeholders.
Phase 3 Implementation	Sprint-based iterative coding of all system modules; integration of AI learning assistant for CEM, CEA, and GEMP trainees; development of AI-assisted Solar PV computation engine; implementation of blockchain payment verification smart contracts; development of tokenized certificate issuance system; construction of administrative dashboard. Continuous integration and biweekly sprint reviews with stakeholders.
Phase 4 Testing	Unit testing of individual modules; integration testing across all system components; performance testing of the blockchain transaction layer; security testing of payment and certificate modules; validation of Solar PV computation outputs against engineering benchmarks; regression testing after each sprint correction cycle. Black Box Testing applied by 10 IT professional testers.
Phase 5 Development (Refinement)	Iterative refinement of all implemented modules based on testing findings; resolution of identified defects and performance issues; re-development and re-testing of affected modules before reintegration; sprint backlog updates reflecting completed items and refinement tasks; continuous alignment with EnergyWize Solutions Inc. operational requirements.
Phase 6 Maintenance	Post-deployment monitoring; bug fixes and performance optimization based on UAT findings; updates to AI models and blockchain smart contracts as required; documentation updates; preparation of final system documentation and deployment package for EnergyWize Solutions Inc.; finalization of user manuals and technical guides for all modules.
Phase 7 Launch	Formal deployment of the finalized PowerWize system; final User Acceptance Testing (UAT) conducted with all 48 respondents (EnergyWize admin, trainers, CEM/CEA/GEMP participants, and Solar PV consultation clients); collection of structured feedback through UAT questionnaire and System Usability Scale (SUS); final revisions incorporated; official system handover to EnergyWize Solutions Inc. for full operational use

Table 10. Summary of Agile Development Phases for PowerWize[EF9]

Table 3 presents the seven Agile development phases of PowerWize and their corresponding key activities. The iterative and cyclical nature of the Agile framework

ensures that each phase from initial requirement and analysis through implementation, testing, refinement, maintenance, and final launch contributes to the progressive quality, functionality, and stakeholder alignment of the final system delivered to EnergyWize Solutions Inc. The inclusion of the Launch phase formalizes the system's transition from development and testing into full operational deployment, ensuring that all 48 UAT respondents have validated the system's readiness before it is officially handed over for use.

3.10 System Architecture

3.10.1 Hardware and Software Requirements

The hardware and software requirements of PowerWize were identified based on its web-based architecture, artificial intelligence services, blockchain-related features, and development environment. The resources presented in Table 3A reflect the actual tools and development workstation used by the researchers. These specifications describe the development and testing setup rather than independently established minimum requirements for end users or production deployment.

Category	Requirement	Specification/Description	Purpose in the System
Hardware	Motherboard and Processor	MSI MS-7A38 (B450 series); AMD Ryzen 5 3600, 6 cores and 12 threads.	Primary workstation for backend development, application integration, and testing.
Hardware	System Memory	16 GB DDR4-3200 RAM (2 × 8 GB Kingston modules).	Supports concurrent development processes, application execution, and local database use.

Hardware	Graphics Processor	AMD Radeon RX 9060 XT with 16 GB VRAM.	Supports the optional locally hosted AI model through LM Studio and ROCm; not required for cloud-based AI access.
Hardware	Data Storage	Samsung 870 EVO 500 GB SATA SSD; Ramsta 256 GB NVMe SSD; Seagate 1 TB HDD.	Stores project files, development tools, database data, and the shared dual-boot project folder.
Hardware	Operating Environment	Windows 11 Pro, build 26200, and Ubuntu in a dual-boot configuration.	Windows served as the primary development OS; Ubuntu supported Docker Compose and Linux-based testing.
Software	Backend Language and Framework	Python 3.12.10; Flask and blueprints; Flask-WTF; Werkzeug; Gunicorn (1 worker, 4 threads, gthread).	Implements application routes, request validation, CSRF protection, and WSGI application serving.
Software	Database and Connection	PostgreSQL 16 (Windows) and PostgreSQL 18 (Linux Docker container); psycopg2-binary; direct SQL without ORM.	Stores and retrieves application records; database schema initialized through init_db() in src/db/core.py.
Software	Cloud-Based AI	Groq API: llama-3.3-70b-versatile and llama-3.1-8b-instant for text; Llama 4 Scout for vision; gpt-oss-120b for benchmarking.	Provides cloud AI inference for intelligent features and comparative response-time testing.
Software	Optional Local AI	LM Studio at 127.0.0.1:1234; qwen/qwen3-vl-4b; ROCm	Enables optional local AI inference on the dedicated graphics processor.

		llama.cpp runtime, Vulkan fallback.	
Software	Embeddings and Retrieval	fastembed with ONNX CPU inference; BAAI/bge-small-en-v1.5 (384 dimensions); NumPy cosine similarity on PostgreSQL REAL[] fields.	Supports retrieval-augmented responses without a separate vector database or ANN index.
Software	OCR and Document Processing	Tesseract and pytesseract; pypdfium2; pdf2image with Poppler; PyMuPDF; Pillow; openpyxl; qrcode[pil].	Processes text in uploaded documents, renders PDFs, handles spreadsheet data, and generates QR codes.
Software	Blockchain and Certificates	web3.py; Polygon Amoy testnet (chain ID 80002); Solidity ^0.8.20; OpenZeppelin soulbound ERC-721; Base64 on-chain metadata.	Supports blockchain interactions and non-transferable, verifiable digital certificates during testnet development.
Software	Payments and Messaging	PayMongo (cards, QR Ph, and webhooks); Gmail SMTP; ngrok for webhook testing and demonstrations.	Handles payment-service integrations, outgoing email, and temporary public access to locally hosted webhook endpoints.
Software	Web Interface	Jinja2 templates; plain CSS; vanilla JavaScript; Chart.js, SweetAlert2, Flatpickr, Font Awesome 6.5.1, PDF.js 3.11.174, html5-qrcode 2.3.8, and paymongo.js.	Displays role-based interfaces, charts, interactive forms, PDF content, QR scanning, and payment controls without a frontend build step.
Software	Configuration and Tools	python-dotenv; Windows PowerShell and batch	Supports configuration, environment management,

		launchers; Docker Compose; Git; graphify.	database containers, version control, and development utilities.
Software	Verification Tools	Python py_compile, application import checks, and custom checking scripts.	Supports developer-level checks; no dedicated automated test framework was identified in the supplied environment.

Table 11. Hardware and Software Requirements of PowerWize[EF10]

3.10.2 System Components and Modules

The major components are (1) account authentication and role-based administration; (2) training modules, assessments, and AI-supported feedback; (3) Solar PV consultation data entry and computation; (4) blockchain payment verification; (5) tokenized certificate generation and verification; and (6) administrative reporting. Users are given access only to the functions appropriate to their roles. Each component is tested independently and through its connections with related modules.

3.10.3 Database Structure

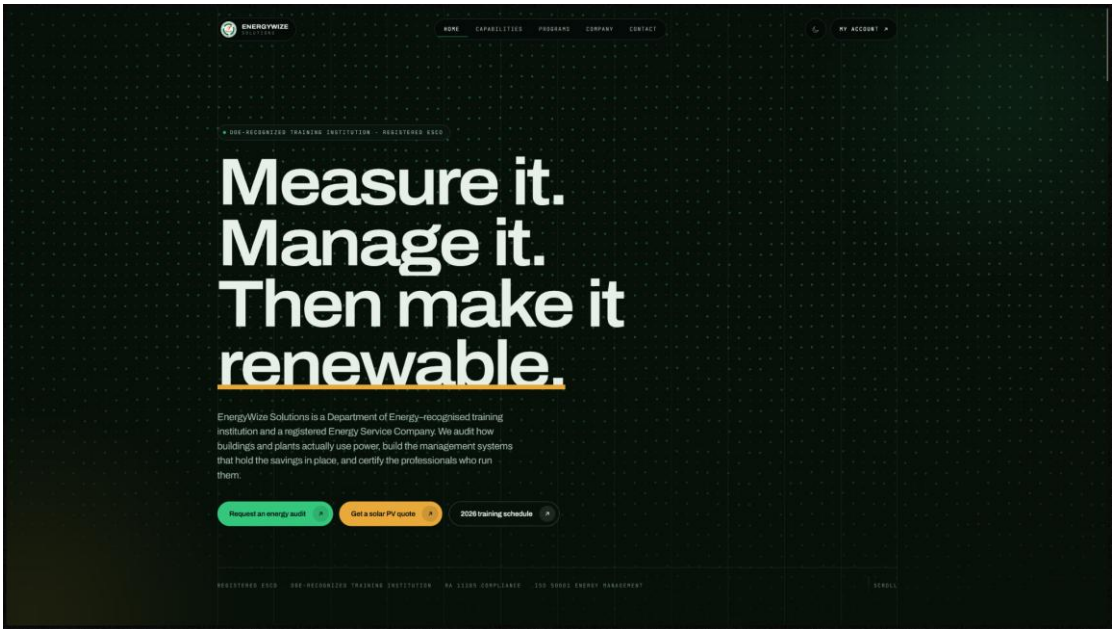
The PowerWize system utilizes PostgreSQL as its relational database management system to organize, store, and manage essential information across its integrated modules. The database structure supports user accounts and roles, training programs, learning materials, quizzes, trainee progress, Solar PV consultation requests and computation results, payment transactions, and tokenized digital certificates. Each table is organized to maintain data consistency, minimize redundancy, and establish relationships between related records through primary and foreign keys. Database operations are handled using Python and the psycopg2 driver through direct SQL queries. This structured approach enables efficient data management, retrieval, and

coordination among the system's training, consultation, payment verification, certification, and administrative functions.



Figure X. Entity-Relationship Diagram (ERD)

3.10.4 User Interface Design



The interface design emphasizes consistent navigation, clear form labels, readable feedback messages, and role-appropriate dashboards. Training participants should be able to locate lessons, assessments, and certificate status; trainers should be able to manage learning content and review progress; and administrative users should be able to monitor transactions and requests. Error messages should explain missing or invalid entries without exposing internal system details.

3.10.5 System Communication and Interaction

The presentation layer sends authorized requests to the application backend through defined API endpoints. The backend validates submitted information, processes training and consultation workflows, reads or writes relevant database records, and interacts with blockchain services for payment and certificate verification. A successful blockchain transaction is not treated as proof of a valid business payment without matching the expected amount, recipient, transaction status, and corresponding system record. Network or service failures must return understandable status messages.

3.11 System Testing and Evaluation Procedure

The procedure follows four steps: (1) prepare test accounts, sample records, expected outputs, and a controlled test environment; (2) execute black-box scenarios with qualified IT testers; (3) record defects, apply corrections, and retest failed cases; and (4) conduct guided UAT with intended users using the validated questionnaire. Test inputs include both valid and invalid credentials, quiz responses, Solar PV inputs, transaction references, and certificate eligibility conditions. The study should retain test logs and questionnaire summaries as evidence.

This section outlines the researchers' action plan from system testing to user evaluation. According to AcqNotes (2024), testing and evaluation are essential processes used to assess the performance, reliability, and overall functionality of a system. These processes involve systematically examining and validating the developed system to ensure that it meets the specified requirements and performs effectively in real-world operational conditions.

The developed PowerWize system was subjected to thorough testing and evaluation to assess the quality of all its features and to ensure that every module functions as specified in the system requirements. The testing procedure was designed to verify: (a) the accuracy and responsiveness of the AI learning assistant in delivering personalized guidance and quiz feedback to CEM, CEA, and GEMP training participants; (b) the accuracy of the AI-assisted Solar PV computation engine in generating system sizing, energy yield estimates, and cost projections; (c) the integrity and immutability of blockchain-recorded payment transactions; and (d) the correct generation and secure issuance of tokenized digital certificates upon training completion.

To assess user satisfaction and gather structured feedback, the system was presented and demonstrated to all 48 respondents — EnergyWize Solutions Inc. administrators, trainers, and CEM/CEA/GEMP training participants — in a controlled evaluation environment. Respondents interacted directly with the system through guided usage scenarios, enabling them to evaluate PowerWize based on real experience with its core modules.

The system testing and evaluation procedure followed two major stages:

- System Testing conducted by 10 qualified IT professionals who evaluated the technical functionality and performance of each PowerWize module using Black Box Testing methodology.
- System User Evaluation conducted by all 48 respondents who assessed the system's usability, effectiveness, and overall acceptability using a structured UAT questionnaire based on ISO/IEC 25010:2023 quality characteristics.

3.12 System Testing

3.12.1 Test Cases

According to Rice (2022), system testing is one of the most critical levels of software testing as it ensures that the complete system meets the required specifications at an acceptable level of quality. It verifies whether all integrated components function correctly as a whole. Furthermore, Soylu (2022) emphasized that although many studies focus on improving testing approaches, only a few provide a comprehensive overview of testing practices and trends applicable to complex, multi-module systems such as PowerWize.

In this study, system testing involved evaluating the overall functionality and performance of the developed PowerWize system in an actual testing environment. The researchers conducted a series of structured tests to ensure that all software components, the AI training assistant module, AI-assisted Solar PV computation engine, blockchain payment verification module, tokenized certificate issuance system, and administrative dashboard operate efficiently, accurately, and securely.

To determine the functionality and performance of the system, the researchers carried out the following steps:

- Design and development of the testing tool
- Application of the testing method

Testing Tool

Testing tools, according to Collins (2023), are software programs or utilities used to verify and validate a system or product. These tools provide features such as defect identification, test execution, result analysis, and validation to ensure that the system meets its required specifications. They are essential in evaluating the system's performance, quality, and ability to handle errors effectively. To ensure comprehensive testing of PowerWize, the researchers utilized structured test case sheets and systematic evaluation checklists applied by the ten IT professional system testers.

Based on industry practices, Al-Azzam and Shatnawi (2022) highlighted the growing importance of structured and systematic testing approaches in improving testing accuracy, efficiency, and coverage especially in systems that involve real-time data, multi-module integration, and security-sensitive components such as blockchain transactions and AI-generated outputs. These approaches help reduce human error and improve the reliability and consistency of testing results.

For this reason, the researchers focused on evaluating PowerWize based on its major functionalities rather than its internal code structure. The study applied the Black Box Testing Model, where the system is tested by providing inputs and observing outputs without examining how the internal processes work. This method allowed the researchers to assess whether the system behaves as expected from the perspective of its intended users administrators, trainers, training participants, and Solar PV consultation clients.

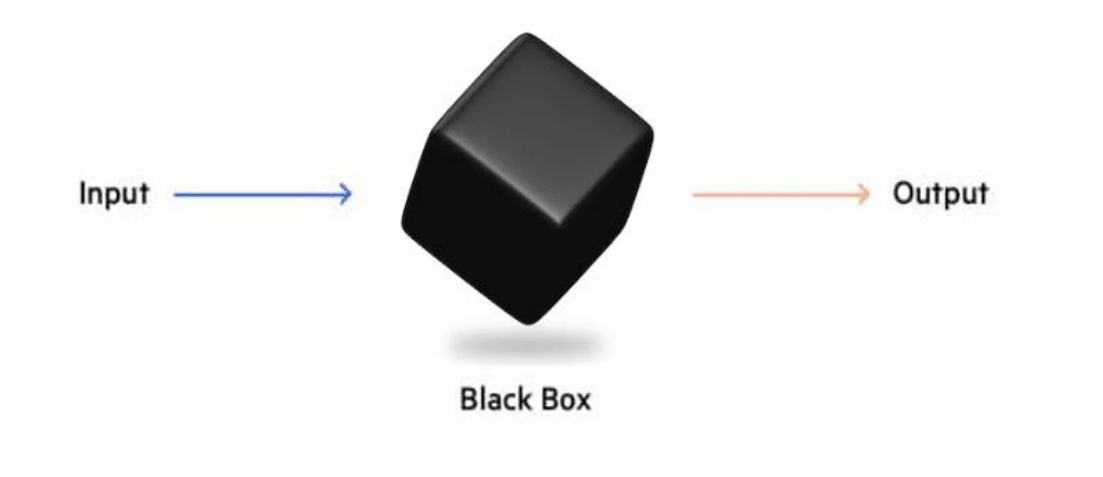


Figure 3. Black Box Testing Model

The figure above shows the Black Box Testing approach, where inputs (training content access, quiz submissions, Solar PV consultation data, payment details, and certificate requests) are entered into the PowerWize system and the resulting outputs (AI feedback, computation reports, blockchain payment records, and tokenized certificates) are observed and evaluated against the system's predefined expected results. This demonstrates that the system meets its functional requirements from the user's operational perspective.

In the context of this study, the following user roles interact with the PowerWize system:

Administrators and Staff are responsible for managing training program content, monitoring participant progress across CEM, CEA, and GEMP programs, reviewing consultation requests, processing payment confirmations, and issuing blockchain-anchored tokenized certificates. They have full access to the administrative dashboard and all system management functions.

Training Participants (CEM, CEA, and GEMP trainees) are the primary users of the AI-driven training module. They access training content, complete AI-assisted quiz assessments, receive personalized feedback and hints from the AI learning assistant, make enrollment fee payments verified through the blockchain, and receive tokenized digital certificates upon successful program completion.

Solar PV Consultation Clients submit their site data and energy load requirements through the consultation interface and receive AI-generated system sizing, panel configuration, energy yield estimates, and cost projection reports. They also make consultation fees through the blockchain payment verification module.

Each user role has specific access controls and permissions to ensure system security, data integrity, and proper handling of sensitive payment and certificate data. The structured test cases developed by the researchers include step-by-step procedures, defined inputs, and expected results for each core system function.

STEP NO.	TEST PROCEDURE	EXPECTED RESULT
1	Open the PowerWize platform and navigate to the login page	Login interface is displayed correctly
2	Enter valid administrator credentials	System authenticates and grants admin dashboard access
3	Enter invalid credentials	System displays appropriate error message and denies access
4	Logout from the platform	User session is terminated securely

Table 12. Testing for User Login and Authentication

STEP NO.	TEST PROCEDURE	EXPECTED RESULT
1	Admin uploads CEM training materials and quiz questions	Materials are saved and published to the CEM trainee dashboard
2	Trainee accesses the training module and begins a quiz	Quiz loads correctly with AI learning assistant active
3	Trainee submits an incorrect answer	The AI assistant gives corrective feedback and study recommendations after the trainee finishes the quiz.
4	Trainee completes all quiz questions and submits	System records completion and calculates score accurately

Table 13. Testing for AI-Driven Training Module and AI Learning Assistant

STEP NO.	TEST PROCEDURE	EXPECTED RESULT
1	Client submits Solar PV consultation data (load, site, area)	System accepts input data without errors
2	AI computation engine processes the submitted data	System generates system sizing, panel configuration, energy yield estimate
3	Client views generated Solar PV system design report	Report is accurate, complete, and displays cost projection
4	Admin retrieves consultation report from dashboard	Report is accessible and correctly stored in the system

Table 14. Testing for AI-Assisted Solar PV Computation Engine

STEP NO.	TEST PROCEDURE	EXPECTED RESULT
1	Participant submits enrollment fee payment	System initiates blockchain payment verification process
2	Blockchain smart contract validates the payment	Transaction is recorded on the blockchain ledger with a unique hash
3	Admin verifies the payment record	Immutable payment record is viewable and tamper-proof on the ledger
4	Unauthorized party attempts to alter a payment record	System rejects the alteration; blockchain integrity is maintained

Table 15. Testing for Blockchain Payment Verification Module

STEP NO.	TEST PROCEDURE	EXPECTED RESULT
1	Trainee completes CEM/CEA/GEMP training and passes assessment	System flags participant as eligible for certificate issuance
2	Admin triggers certificate issuance for the eligible trainee	Blockchain-anchored NFT certificate is minted and issued
3	Trainee accesses their tokenized digital certificate	Certificate is viewable, downloadable, and linked to blockchain record
4	Third-party verifier attempts to verify the certificate	System returns verified certificate details from blockchain; forgery-proof

Table 16. Testing for Tokenized Digital Certificate Issuance

STEP NO.	TEST PROCEDURE	EXPECTED RESULT
1	Admin logs in and opens the administrative dashboard	Dashboard loads with correct overview of all training programs and consultations
2	Admin generates a training progress report for CEM participants	Accurate and complete report is generated and downloadable
3	Admin views real-time payment transaction logs	All blockchain payment records are displayed correctly
4	Admin manages user accounts (create, update, deactivate)	User management functions execute correctly without data loss

Table 17. Testing for Administrative Dashboard

Testing Method

A testing method is defined as a precise and structured process intended to produce reliable, repeatable, and operationally viable test results. According to AZoM (2021), a testing method must be clearly defined and consistently applied to accurately measure and evaluate system characteristics or performance indicators. These principles apply equally to software system evaluation, where consistency and clarity are essential in validating multi-module system performance across diverse user roles and use cases.

To ensure clear and unbiased evaluation results for the PowerWize system, the researchers adopted a dichotomous testing approach. According to QA Expertise (2023), dichotomous testing uses a pass-or-fail scale to evaluate system performance based on predefined criteria, reducing ambiguity in interpreting results and improving consistency in determining whether each system function meets its expected outcome. By simplifying evaluation into binary results Passed or Failed the researchers were able

to systematically identify which features of PowerWize function correctly and which require further development or correction.

This approach was applied across all test cases covering the five core PowerWize modules: the AI training assistant, AI-assisted Solar PV computation engine, blockchain payment verification module, tokenized certificate issuance system, and administrative dashboard. For every test performed, the result was classified as either Passed meaning the system produced the expected output or Failed meaning the system did not meet the required condition and required corrective action.

Entry and Exit Criteria, Defect Severity, and Special Validation Requirements

The following subsections, drawn from the researchers' approved System Testing Plan, define measurable entry/exit criteria, defect severity classification, and validation procedures for the AI assistant, Solar PV engine, blockchain module, and certificate system — directly addressing the adviser's concern that the original test cases confirmed only that a result was produced, not that the result was correct or that failure modes were handled.

Category	Entry Criteria	Exit Criteria
Environment	Test build installed; test accounts available; test data prepared.	The environment remains stable, or all changes are documented.
Functional testing	All critical modules accessible for testing.	All critical test cases executed; no unresolved Critical defects.
Solar PV	Benchmark cases and expected results approved.	All benchmark cases tested; any error is within predefined tolerance or documented as a limitation.
Blockchain	Test blockchain/network mechanism available.	Payment/certificate records tested for recording and verification; tamper cases documented.
Security/access	Roles and permissions configured.	No unresolved Critical/High authorization or sensitive-data exposure defects.
Regression	Corrected defects available for retest.	Corrected critical defects pass retest.
UAT readiness	System testing is substantially complete.	Major workflows are stable and usable for the 48-user UAT.

Table 18. Entry and Exit Criteria for System Testing

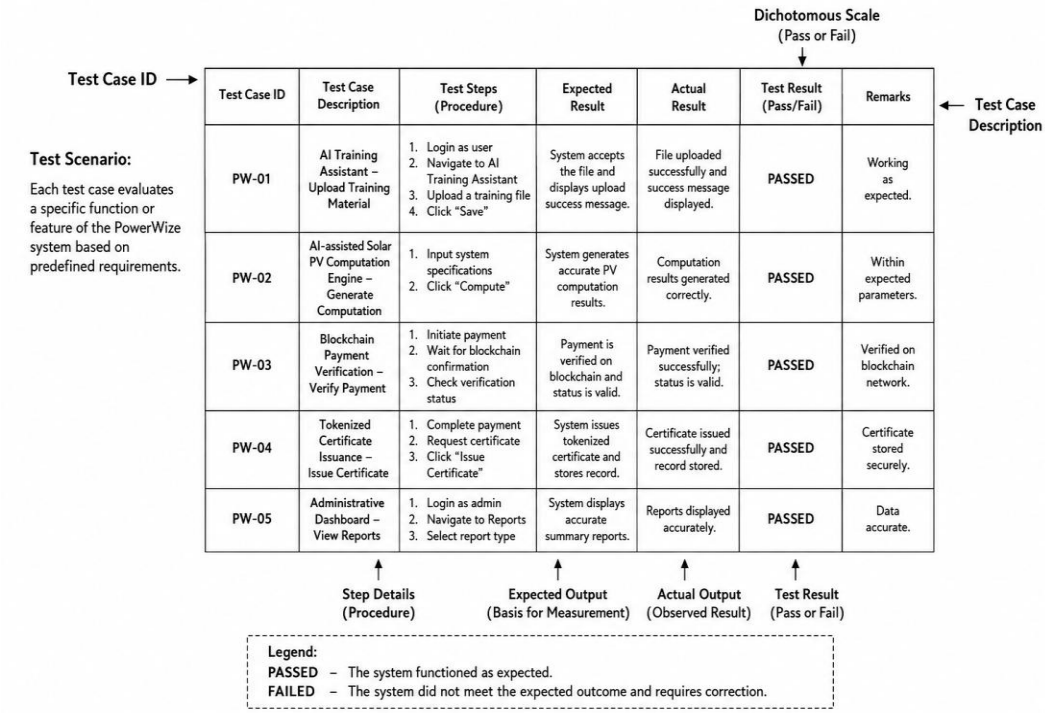


Figure 4. Dichotomous Testing Format for PowerWize System Evaluation

System User Evaluation

Software evaluation is a systematic and comprehensive process used to determine how effectively a system meets its intended purpose within a specific operational environment. According to Azeroual et al. (2022), evaluating software systems involves assessing key factors such as data quality, perceived usefulness, and reliability especially in institutional implementations where the system handles sensitive data and regulatory processes. These factors are essential in determining whether a system is suitable for real-world deployment and meets the expectations of its intended users.

In this study, system user evaluation was conducted to gather the opinions, perceptions, and responses of PowerWize's intended users — EnergyWize Solutions Inc. administrators, trainers, and CEM/CEA/GEMP training participants — regarding the developed system. The collected feedback served as the basis for determining

PowerWize's usability, effectiveness, and overall acceptance among its target user population, and for identifying any areas requiring final improvement before full deployment.

The system user evaluation process consisted of the following stages:

- Evaluation Tool development of the UAT questionnaire based on ISO/IEC 25010:2023 quality characteristics
- Evaluation Method administration of the questionnaire through guided system demonstrations and supervised hands-on interaction
- Validation of Questionnaire review and validation by three qualified validators
- Distribution of Questionnaire administration to all 48 target respondents

3.12.2 Testing Instrument

The testing instrument consists of structured black-box test-case sheets recording the module, test objective, input or action, expected result, actual result, pass/fail status, defect severity when applicable, and tester remarks. The ten IT professionals apply these sheets to the documented functional scenarios. The completed sheets, not the planned scenarios alone, form the evidence for Chapter IV.

3.12.3 Validation of Testing Instrument

Before use, the test cases are checked against the documented functional requirements, expected outputs, role permissions, and entry/exit criteria. The researchers review ambiguous instructions and revise cases where the expected outcome is not measurable. Testers record discrepancies and repeat affected cases after

corrections. Evidence of formal expert validation should be stated only when signed validation records are available.

3.12.4 Interpretation of Test Results

A test case is marked Pass when the observed output meets its predefined expected result and Fail when it does not. Failed cases are logged with the relevant defect severity and corrective action. The pass rate may be calculated as (number of passed cases / number of executed cases) × 100. Unexecuted cases are reported separately. Actual counts, retest outcomes, and unresolved defects belong in Chapter IV and must not be inferred from blank test-case tables.

3.13 System User Evaluation

3.13.1 User Evaluation Instrument

In system evaluation, various methods are used to measure user satisfaction and system performance, incorporating both qualitative and quantitative approaches such as surveys, guided interviews, and direct observation. According to Trochim (2025), effective evaluation involves stakeholder engagement, iterative feedback collection, and sensitivity to user needs ensuring that systems are assessed in real-world operational contexts rather than purely technical environments.

To support the evaluation of PowerWize, the researchers utilized User Acceptance Testing (UAT) to develop appropriate evaluation questions for each user group. According to Tong et al. (2022), applying user acceptance testing procedures is important to help minimize software failures and avoid unnecessary operational losses. The UAT framework was therefore applied to collect structured user feedback and

determine the level of acceptance of PowerWize among EnergyWize Solutions Inc.'s administrators, trainers, and training participants.

In addition, the researchers applied the ISO/IEC 25010:2023 standard as the primary quality evaluation framework. According to the International Organization for Standardization (2023), this standard defines software product quality through nine characteristics: functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety providing a comprehensive and internationally recognized basis for systematic software quality assessment. By combining UAT criteria with ISO/IEC 25010:2023 quality metrics, the researchers developed an evaluation instrument capable of producing accurate, reliable, and contextually valid assessments of PowerWize's quality across all user groups.

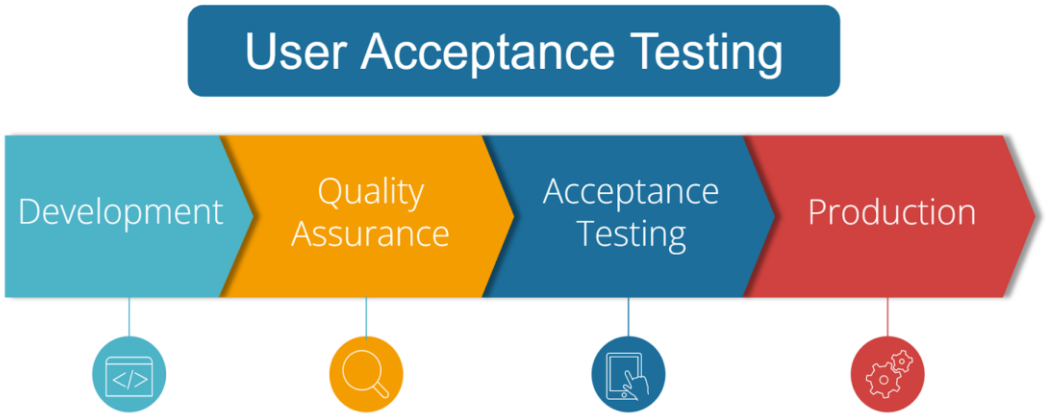


Figure 5. User Acceptance Testing (UAT) Process adapted from Lanars (2021)

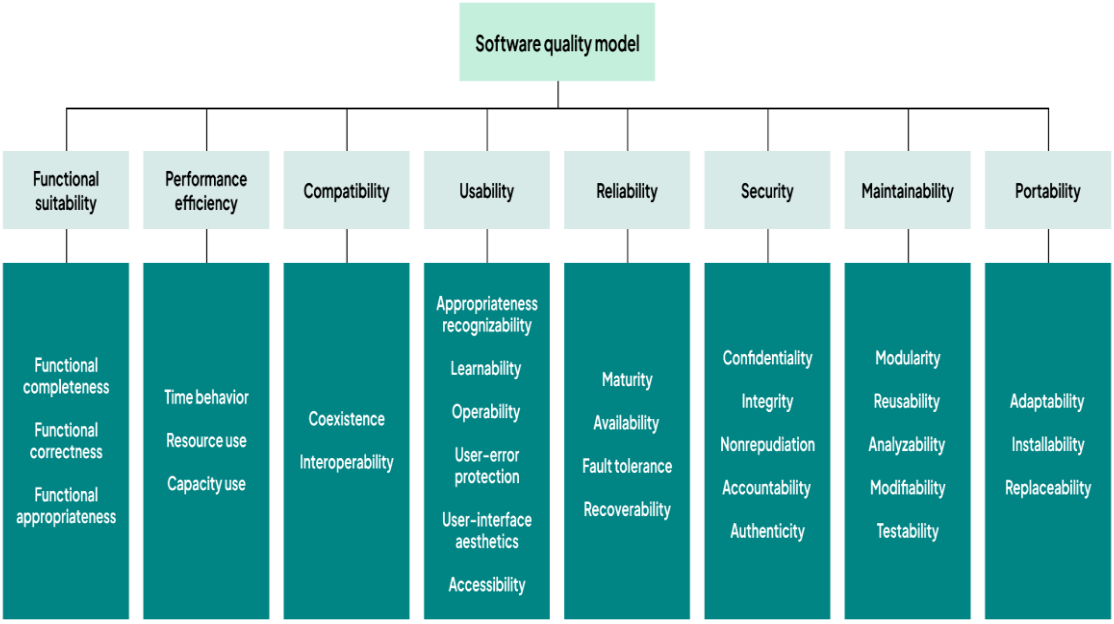


Figure 6. ISO/IEC 25010 Software Quality Metrics

The following tables present the UAT questionnaire developed for each evaluation category, mapping specific evaluation items to their corresponding UAT criteria and ISO/IEC 25010:2023 quality characteristics:

QUALITY FACTORS	SOURCE
1. Functional Completeness – PowerWize provides all the necessary functions needed to support CEM, CEA, and GEMP training program management.	ISO/IEC 25010:2023
2. Functional Correctness – The AI learning assistant provides accurate quiz feedback and the Solar PV computation engine produces correct system sizing outputs.	ISO/IEC 25010:2023
3. Functional Appropriateness – The system's modules are well-suited to the operational context of EnergyWize Solutions Inc. as a DOE-accredited training provider.	ISO/IEC 25010:2023
4. Reliability – The system consistently performs its functions without failures during CEM, CEA, and GEMP training sessions and blockchain transaction processing.	ISO/IEC 25010:2023
5. Security – The blockchain payment records and tokenized certificates are protected against unauthorized access, alteration, or forgery.	ISO/IEC 25010:2023

Table 19. Quality Factors – Functional Suitability and Reliability[EF11]

PERFORMANCE EFFICIENCY	SOURCE
1. The system responds quickly when accessing training materials, submitting quizzes, or processing blockchain payments.	ISO/IEC 25010:2023
2. The system can handle multiple simultaneous users — trainees, admins, and PV clients — without performance degradation.	ISO/IEC 25010:2023
3. The AI-assisted Solar PV computation engine generates results within acceptable response times.	ISO/IEC 25010:2023

Table 20. Performance Efficiency[EF12]

PERCEIVED EASE OF USE	SOURCE
1. The system is easy to learn and operate for training participants with varying levels of digital literacy.	Davis, 1989; ISO/IEC 25010:2023
2. The AI learning assistant's guidance and hints are easy to understand and act upon during quiz assessments.	Davis, 1989
3. The interface for Solar PV consultation data entry is clearly organized and user-friendly.	ISO/IEC 25010:2023
4. The certificate retrieval and verification process is straightforward and accessible to third-party verifiers.	ISO/IEC 25010:2023

Table 21. Perceived Ease of Use – Interaction Capability[EF13]

PERCEIVED USEFULNESS	SOURCE
1. The system's AI training assistant effectively helps trainees understand complex CEM, CEA, and GEMP content.	Davis, 1989
2. Using the blockchain payment module makes enrollment fee verification more trustworthy and transparent.	Davis, 1989
3. The AI-assisted Solar PV computation engine provides recommendations that are useful for clients' decision-making.	Davis, 1989
4. The tokenized digital certificates issued by PowerWize are more credible and verifiable than traditional paper-based certificates.	Davis, 1989

Table 22. Perceived Usefulness[EF14]

USER EXPERIENCE	SOURCE
1. I am satisfied with my overall experience using the PowerWize system.	Al-Qeisi et al., 2022
2. The system provides a smooth and responsive interaction across all modules.	Al-Qeisi et al., 2022
3. I would be willing to use PowerWize again for future training programs or Solar PV consultations.	Al-Qeisi et al., 2022

Table 23. User Experience[EF15]

ATTITUDE TOWARD USING	SOURCE
1. I have a positive overall impression of the PowerWize system.	Davis, 1989
2. The system is helpful and reliable for managing CEM, CEA, and GEMP training programs.	Venkatesh & Davis, 2000
3. PowerWize enhances the professional credibility of EnergyWize Solutions Inc. as a DOE-accredited training institution.	Venkatesh & Davis, 2000

Table 24. Attitude Toward Using[EF16]

BEHAVIORAL INTENTION TO USE	SOURCE
1. I plan to continue using PowerWize for future CEM, CEA, and GEMP training cycles.	Venkatesh & Davis, 2000
2. I would recommend PowerWize to other energy training providers or Solar PV consultation companies.	Venkatesh & Davis, 2000
3. I am willing to fully adopt PowerWize as the primary platform for managing energy certification training and Solar PV consultations.	Venkatesh & Davis, 2000

Table 25. Behavioral Intention to Use[EF17]

System Usability Scale (SUS)

The System Usability Scale (SUS) is a supplementary ten-item usability questionnaire. It should be reported only if respondents actually completed all ten items. It is distinct from the researcher-made five-point UAT instrument; its 0–100 score is not interpreted using the UAT weighted-mean categories

1. I think that I would like to use this system frequently.

2. I found the system unnecessarily complex.

3. I thought the system was easy to use.

4. I think that I would need the support of a technical person to be able to use this system.

5. I found the various functions in this system were well integrated.

6. I thought there was too much inconsistency in this system.

7. I would imagine that most people would learn to use this system very quickly.

8. I found the system very cumbersome to use.

9. I felt very confident using the system.

10. I needed to learn a lot of things before I could get going with this system.

If administered, SUS responses are scored by subtracting 1 from each odd-numbered item response and subtracting each even-numbered response from 5. The sum of the ten adjusted values is multiplied by 2.5, producing a score from 0 to 100. The number of valid SUS respondents and their actual scores must be documented before conclusions are drawn.

3.13.2 Instrument Source and Validation

The researcher-made UAT items were based on selected ISO/IEC 25010:2023 software-quality characteristics and Technology Acceptance Model concepts. The manuscript identifies three intended reviewers: an Information Systems faculty member, an energy-sector professional familiar with CEM/CEA/GEMP training, and an IT/blockchain practitioner. Their review should assess clarity, relevance, and

alignment with system functions. Signed validation forms and revision records should be attached if available; this description does not by itself establish psychometric reliability.

3.13.3 User Evaluation Procedure

After the researchers introduce the purpose and instructions, each respondent is guided through functions relevant to the assigned user role and given an opportunity to operate the system. The questionnaire is then distributed, collected, checked for missing responses, and encoded. Responses are summarized by evaluation criterion and, where appropriate, user group. Open-ended comments are reviewed to identify recurring usability concerns and recommended improvements. The final manuscript must distinguish the planned 48 respondents from the number who actually completed the evaluation.

3.13.4 Evaluation Criteria

The UAT evaluation covers selected ISO/IEC 25010:2023 characteristics—functional suitability, performance efficiency, interaction capability, reliability, security, and maintainability—as applicable to the questionnaire, together with perceived usefulness, ease of use, user experience, attitude, and intention to use. The five-point scale and its verbal interpretations are given in Table 17. A weighted mean of at least 3.41 is the stated descriptive acceptance criterion; this threshold is not evidence that every feature passed technical testing.

3.14 Business Plan for the Integration of the Technology

The proposed organizational integration begins with a pilot implementation for authorized EnergyWize personnel and a limited set of training participants. During this stage, the team should configure user roles, prepare approved course materials, verify consultation formulas against reference cases, and test payment and certificate workflows using nonproduction transactions. After pilot review, EnergyWize may plan user orientation, data migration, backup procedures, technical support responsibilities, and periodic security and software maintenance.

Operational adoption also requires agreement on hosting, service subscriptions, blockchain transaction costs, privacy and retention rules, and responsibility for approving training certificates. Moving from a Polygon test network to a production network requires a separate security, cost, and compliance review. This chapter does not claim that a commercial launch, cost-benefit study, or financial forecast has already been completed

3.15 Pretest–Posttest Design

A pretest–posttest design is not part of the documented study. The project follows a developmental design with descriptive user evaluation after system interaction, and the manuscript does not identify a baseline instrument administered before PowerWize use. Therefore, no pretest–posttest comparison or improvement score is reported. If the researchers later implement such a design, they must document matched participants, comparable instruments, administration dates, and an appropriate analysis before adding results.

3.16 Statistical Treatment

3.16.1 Descriptive Statistics

Descriptive statistics are used to summarize the respondent distribution and the completed UAT ratings. Frequency and percentage describe counts; weighted mean summarizes five-point responses; and standard deviation describes variation. Calculations must be based on valid collected responses, with missing responses identified where applicable.

The data collected from survey questionnaires were analyzed using descriptive statistics. The following statistical tools were applied to present, analyze, and interpret the evaluation results for PowerWize:

Frequency and percentage distribution are used to describe respondent categories and the distribution of questionnaire responses. The denominator for each computation is the number of valid responses for the corresponding item or group, not automatically the intended population of 48.

Formula:

$$\% = (f / N) \times 100$$

Where: % = percentage; f = frequency of responses; N = total number of respondents

Weighted Mean

The weighted mean was used to determine the average responses of all participants in the survey questionnaire. This statistical tool measures the overall level of acceptance, usability, and effectiveness of PowerWize across each evaluation category, providing a single representative score per quality dimension that enables direct comparison across respondent groups and across UAT criteria.

Formula:

$$\bar{x}w = \Sigma(f \times x) / N$$

Where: $\bar{x}w$ = weighted mean; f = frequency of responses per scale value; x = Likert scale weight; N = total number of respondents

Likert Scale

The five-point response scale is interpreted according to Table 17. To avoid inconsistency, all narrative references in this chapter use Table 17 rather than Table 18. Weighted-mean interpretations are descriptive labels and do not replace functional test results.

SCALE	VERBAL INTERPRETATION	RANGE OF MEAN	DESCRIPTION
5	Strongly Agree	4.21 – 5.00	Very Effective / Excellent
4	Agree	3.41 – 4.20	Effective / Very Satisfactory
3	Neutral	2.61 – 3.40	Moderately Effective / Satisfactory
2	Disagree	1.81 – 2.60	Less Effective / Needs Improvement
1	Strongly Disagree	1.00 – 1.80	Not Effective / Unacceptable

Table 26. Likert Scale for UAT Questionnaire Interpretation[EF18]

Table 17 presents the UAT weighted-mean interpretation: 4.21–5.00, Very Effective/Excellent; 3.41–4.20, Effective/Very Satisfactory; 2.61–3.40, Moderately Effective/Satisfactory; 1.81–2.60, Less Effective/Needs Improvement; and 1.00–1.80, Not Effective/Unacceptable. The study uses 3.41 as its descriptive acceptability threshold.

Standard Deviation

Standard deviation describes the spread of valid numerical responses around the mean for an item or criterion. Lower values indicate more similar ratings, whereas higher values indicate greater variation. The researchers must specify whether sample or population standard deviation is used and apply that choice consistently.

Formula:

$$SD = \sqrt{[\ \Sigma(x - \bar{x})^2 / (N - 1)\ ]}$$

Where: SD = Standard Deviation; x = individual response; $\bar{x}$ = mean of responses; N = number of respondents

3.16.2 Inferential Statistics

Inferential tests are not specified for the present descriptive evaluation. The study aims to develop and assess the system with its identified users, rather than test a population-level causal hypothesis. Consequently, p-values and claims of statistical significance should not be reported unless a justified hypothesis, sampling design, and suitable dataset are added.

3.16.3 Correlation and Regression Analysis

Correlation and regression are not required by the stated objectives and are not documented as performed. The UAT criteria are summarized descriptively; no relationship or predictive effect among acceptance constructs is claimed.

3.16.4 Analysis of Variance

Analysis of variance (ANOVA) is not part of the documented analysis. Although respondents belong to different user groups, the manuscript does not specify an ANOVA hypothesis or sufficient group-level data. Comparisons among user groups, if presented, should be labelled descriptive unless an appropriate statistical test is formally planned and conducted.

3.16.5 Qualitative Data Analysis

Written comments from the open-ended UAT question are grouped by recurring ideas, such as navigation, clarity of instructions, performance, and suggested features. The researchers read the responses, assign short codes, combine related codes into themes, and summarize themes without identifying respondents. Themes are used to explain and contextualize numerical ratings; quotations should be reproduced only with appropriate permission.

CHAPTER IV

Results and Discussion

This chapter presents the results and discussion of the development, implementation, testing, and evaluation of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. The chapter highlights the system's major components, including the AI-enhanced training module, AI-assisted Solar PV consultation computation, blockchain payment verification, tokenized digital certificates, and administrative dashboard. The results are systematically presented in accordance with the objectives of the study, covering the system architecture, hardware and software requirements, database structure, user interfaces, system testing results, and user evaluation. Tables, architectural diagrams, and system screenshots are provided to support the presentation, analysis, and interpretation of the findings.

4.1 System Architecture and Development Results

This section details the concrete outcomes derived from designing, engineering, and deploying LSPU ScholarHub to satisfy Objective 1. Transitioning from theoretical frameworks to practical implementation, this part illustrates how the traditional, manual workflow of the Scholarship and Financial Assistance Office (SFAO) at LSPU-SCC was converted into an integrated web platform. Guided by requirements gathered from administrative staff and student scholars, development was carried out using an Agile iterative approach. The resulting software features a web dashboard for office administrators and a scholar portal built on a modern stack, connected to a relational database. The following sub-sections delineate the system's architecture, hardware and

software specifications, relational database schema, and the tailored user interfaces engineered for each user role.

4.1.1 System Architecture

The System Architecture presents the overall structural design and integration of the PowerWize platform across its major layers: presentation, application, and data and external service integration. It illustrates the interaction between the user-facing web interfaces, the Flask-based application core, PostgreSQL database services, artificial intelligence components, blockchain-based certificate and payment verification functions, and third-party communication and payment services. This architectural framework demonstrates how PowerWize supports its primary functions such as AI-assisted training, Solar PV consultation computation, blockchain payment verification, tokenized certificate issuance, and administrative monitoring while maintaining organized data flow, role-based access control, and system security.

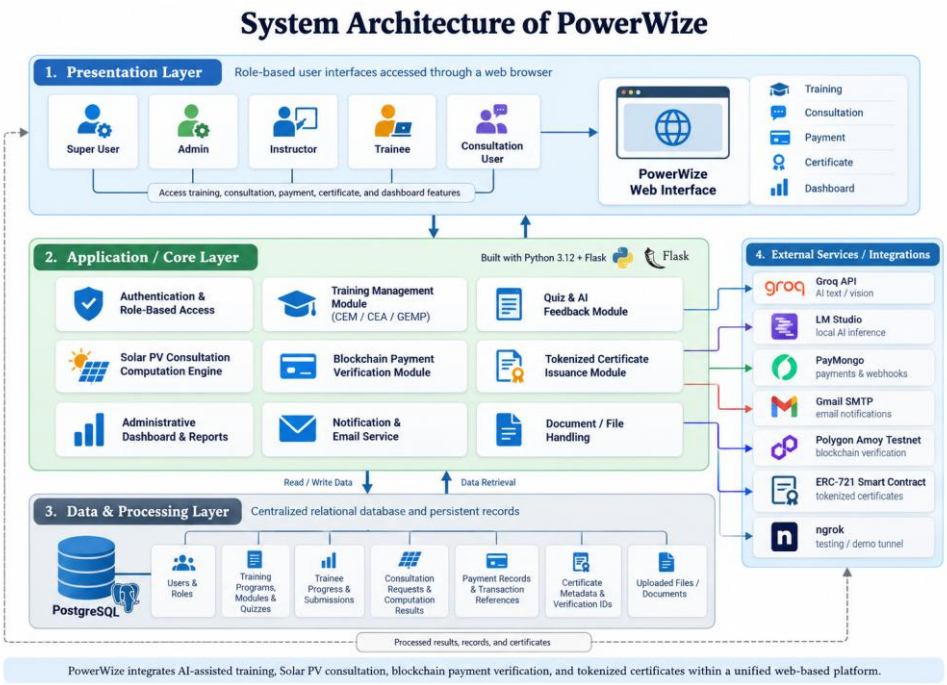


Figure X. System Architecture of PowerWize

The technical implementation of PowerWize adopts a layered system architecture that organizes the platform into interconnected components responsible for user interaction, application processing, and persistent data management. As shown in Figure 7, the architecture illustrates how trainees, instructors, administrators, and other authorized users interact with the system through browser-based interfaces, while the backend coordinates business logic, artificial intelligence processing, database transactions, and external service communication. This layered structure establishes a clear separation of responsibilities among system components and supports efficient and maintainable system operation.

The **Presentation Layer** serves as the user-facing component of the system and provides role-specific access to PowerWize through a web browser. This layer includes the interfaces for trainees, instructors, administrators, and the super user. Through these interfaces, users can access training materials, answer quizzes, review feedback, submit Solar PV consultation data, verify payments, manage certificates, and monitor operational records. The use of role-based interfaces ensures that each type of user is presented only with functions relevant to their responsibilities while maintaining a consistent and accessible interaction with the system.

The **Application Layer** performs the main processing and business functions of PowerWize. It is implemented using Python 3.12.10 and the Flask framework with blueprints, Flask-WTF, and Werkzeug. This layer includes the Authentication and User Management Module, AI-Driven Training and Quiz Module, Solar PV Consultation Computation Module, Blockchain Payment Verification Module, Tokenized Certificate Module, Notification and Email Services, and the Administrative Dashboard. The AI-assisted functions communicate with configured providers such as Groq API and LM

Studio for language and vision processing, while the retrieval component supports contextual responses through embeddings and similarity matching. This layer acts as the central coordinating engine of the system by receiving requests from users, processing inputs, and returning the appropriate outputs.

The **Data and Document Processing Layer** provides the persistent storage and supporting information-processing services required by the platform. PostgreSQL serves as the relational database management system for storing user accounts, roles, training programs, modules, quizzes, trainee submissions, consultation records, payment references, and certificate metadata. Document and image processing tools such as Tesseract OCR, pypdfium2, pdf2image, PyMuPDF, Pillow, and openpyxl support file handling and data extraction where needed. This layer ensures that data used across the different modules remain centralized, consistent, and retrievable for monitoring, reporting, and verification purposes.

PowerWize also integrates with several **External Services and Infrastructure Components** that extend the capabilities of the platform. PayMongo is used for payment-related processing and webhook support, Gmail SMTP is used for email notifications, and ngrok is used during testing and demonstration for public webhook access. For blockchain operations, the system communicates with the Polygon Amoy testnet through `web3.py` and the deployed Solidity-based certificate smart contract. These external components allow PowerWize to support tamper-evident transaction recording, tokenized certificate issuance, and external service interoperability without overloading the internal application core.

As shown in Figure 7, the architecture establishes a structured flow of information from the Presentation Layer to the Application Layer, then to the Data

Layer and integrated external services, with processed outputs returned to the appropriate interfaces. This design enables PowerWize to function as an integrated training and consultation platform rather than as separate isolated modules. The architecture supports the system's objectives of streamlining DOE-accredited training operations, automating Solar PV consultation computation, strengthening payment verification, issuing tokenized certificates, and improving overall administrative efficiency, while also providing a structured foundation for maintainability, scalability, and future enhancement.

4.1.2 Hardware and Software Requirements

This section presents the hardware and software resources used to develop and test PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. The platform combines browser-based interfaces, a Python and Flask backend, PostgreSQL database services, and AI, payment, and blockchain integrations. Tables 6 and 7 report the documented development resources and implemented technologies rather than unverified minimum production requirements.

The development workstation used Windows 11 Pro and Ubuntu in a dual-boot configuration, an AMD Ryzen 5 3600 processor, and 16 GB of RAM. Its AMD Radeon RX 9060 XT graphics card supported optional local AI inference through LM Studio. The GPU is not required for Groq cloud inference. Internet connectivity supports the cloud AI service, PayMongo payments, email notifications, and Polygon Amoy testnet communication.

Hardware Requirement	Actual Specification	Availability	Purpose/Use
Development workstation	MSI MS-7A38 motherboard (B450 series)	Used	Primary desktop environment for application development and system testing
Processor	AMD Ryzen 5 3600; 6 cores and 12 threads	Used	Application execution, local processing, and development activities
Memory (RAM)	16 GB Kingston DDR4-3200 (2 × 8 GB)	Used	Concurrent development tools, application processes, and databases
Graphics card	AMD Radeon RX 9060 XT; 16 GB VRAM	Used (optional AI)	Local LLM inference through LM Studio and ROCm; not needed for Groq inference
Network connectivity	Internet connection for cloud services and blockchain testnet	Used	Groq API, PayMongo webhooks, SMTP notifications, and Polygon Amoy access
Client access device	Compatible browser-equipped desktop, laptop, or mobile device; minimum specifications not established	Web access supported	Role-based access to the PowerWize interface; production minimums remain unverified
Storage	Samsung 870 EVO 500 GB SATA SSD; Ramsta 256 GB NVMe SSD; Seagate 1 TB HDD	Used	Operating systems, software installation, application files, and shared dual-boot project storage

Table 27. Hardware and Software Requirements

Table 27 summarizes the actual development hardware used for PowerWize. The workstation provided sufficient resources for the activities performed during the capstone project, including optional local AI inference. Since external cloud-based inference can be selected, the dedicated graphics card is not a general requirement for users of the web interface. The study did not establish minimum processor, memory, or server specifications for a permanent production deployment.

Software Requirement	**Implemented Specification**	**Status**	System Use
Operating systems and tools	Windows 11 Pro (build 26200); Ubuntu dual-boot; Git; Docker Compose	Used	Development, version control, and local container operations
Backend framework	Python 3.12.10; Flask with blueprints; Flask-WTF; Werkzeug	Implemented	Routing, server-side processing, CSRF protection, and request handling
Application server	Gunicorn; configured with one worker and four threads in Procfile	Configured	WSGI application serving configuration; production host not specified
Relational database	PostgreSQL 16 (Windows); PostgreSQL 18 (Linux Docker)	Implemented	Persistent storage of system records and transactions
DB access and configuration	psycopg2-binary; direct SQL; init_db() schema setup; python-dotenv	Implemented	Database operations without ORM and environment configuration
Web interface	Jinja2; HTML; CSS; vanilla JavaScript	Implemented	Role-based browser interfaces with no frontend build framework
Frontend libraries	Chart.js; SweetAlert2; Flatpickr; Font Awesome; PDF.js; html5-qrcode; paymongo.js	Used	Charts, dialogs, date inputs, icons, PDF viewing, QR scanning, and payment UI
Cloud-based AI	Groq API; Llama 3.3 70B, Llama 3.1 8B, Llama 4 Scout vision	Integrated	Cloud text and vision model inference
Local AI	LM Studio; Qwen3-VL-4B via ROCm llama.cpp runtime (Vulkan fallback)	Optional; used	Local inference alternative to external APIs
Embeddings and retrieval	fastembed; ONNX CPU; BAAI/bge-small-en-v1.5; NumPy cosine similarity	Implemented	384-dimensional embeddings and retrieval over PostgreSQL REAL[] values
OCR and documents	Tesseract/pytesseract; pypdfium2; pdf2image/Poppler; PyMuPDF; Pillow; openpyxl; qrcode[pil]	Used	Document extraction, PDF processing, spreadsheets, and QR generation
Blockchain	web3.py; Polygon Amoy testnet (chain ID 80002); Solidity ^0.8.20; OpenZeppelin ERC-721	Integrated	Blockchain interaction and non-transferable certificate tokenization; on-chain metadata
Payments and messaging	PayMongo (cards, QRPH, webhooks); Gmail SMTP	Integrated	Payment-related processing and email notifications
Testing / access tools	ngrok; py_compile; import checks; custom validation scripts	Used for testing	Public webhook/demo tunnel and implementation verification

Table 28. Hardware and Software Requirements

Table 28 [EF19]identifies the software technologies used in PowerWize. Flask processes application requests, PostgreSQL stores relational data through direct SQL using psycopg2, and Jinja2, CSS, and JavaScript provide the web interface. Groq and optional LM Studio support AI inference, while PayMongo, Gmail SMTP, and Polygon Amoy provide external integrations. The records do not establish production server hardware or a fixed hosting provider; a Supabase example URL alone does not confirm deployment. Moreover, requirements.txt does not pin all package versions.

4.1.3 Database Structure

The developed PowerWize platform uses PostgreSQL as its relational database management system to organize and maintain the application records required by EnergyWize Solutions Inc. During development and testing, PostgreSQL 16 was used in the Windows environment and PostgreSQL 18 was configured in the Linux Docker environment. The Flask application communicates directly with the database using the psycopg2 driver and SQL queries rather than an object-relational mapping (ORM) framework. The database initialization routine defined in the application establishes the required schema within the configured environment.

The database structure supports several interconnected areas of operation. User and access information is maintained for authentication and role-based system use. Training-related information includes CEM, CEA, and GEMP programs, learning modules, quiz records, trainee progress, and assessment submissions. Consultation data covers the information entered for Solar PV requests together with the corresponding computation results. Additional record groups support payment transactions and

references, certificate metadata and verification identifiers, and the administrative information needed for monitoring and reporting.

The system also uses PostgreSQL for information retrieval associated with its AI-assisted functions. Its retrieval component stores numerical embedding values in PostgreSQL REAL[] columns and performs similarity calculations using NumPy, instead of relying on a separate vector database. The storage of operational data and retrieval-related information within the configured database environment supports the coordination of the system's training and consultation functions.



Figure 8. Database Structure of PowerWize

Figure 8 presents the implemented PostgreSQL database structure of PowerWize, consisting of 28 interconnected tables organized into seven functional groups: Intake, Identity, Billing, Courseware, Assessments, Submissions, and Solar/RAG. The Intake group manages training registrations, scheduled batches, and

account activation tokens, while the Identity group maintains user information, access permissions, password recovery records, and audit logs. The Billing group stores payment transactions, discount codes, corporate payment information, and transaction events.

The Courseware, Assessments, and Submissions groups support the management of learning modules, instructional materials, trainee progress, quizzes, practical examinations, and assessment responses. The Courseware group also maintains certificate records containing identifiers and blockchain-related transaction references. Meanwhile, the Solar/RAG group manages Solar PV consultation requests, computation results, and knowledge-based information retrieval used by the system's AI-supported features.

The database establishes relationships among relevant records through primary and foreign keys, allowing the different modules to access and manage related information. Through this relational structure, PowerWize supports centralized data organization, consistent record management, and coordination among its training, consultation, payment, certification, and administrative functions.

4.1.4 User-Interface and System Modules

The PowerWize user interface was developed as a browser-based environment for EnergyWize Solutions Inc. personnel, CEM, CEA, and GEMP training participants, and Solar PV consultation clients. The interface uses Jinja2 templates, CSS, and JavaScript, with Flask handling user requests and application processing. Rather than relying on a separate frontend framework, the system presents role-appropriate pages and functions through its integrated web application. This design connects training, consultation, payment, certificate, and administrative workflows within one platform.

The system applies role-based access to distinguish the activities of the Super User, Administrator, Instructor, and Trainee, while providing consultation-related functions for clients. The interface descriptions below focus on the developed modules and their intended operations. The corresponding screenshots should be taken from the implemented PowerWize application so the figures accurately represent the actual pages and controls.

4.1.4.1 Administrative and Instructor Web Interfaces

The Administrative Dashboard provides authorized personnel with a centralized view of operations relevant to the PowerWize system. Administrative functions cover user account management, monitoring of CEM, CEA, and GEMP training activities, review of Solar PV consultation requests, payment-related records, digital certificate information, and operational reporting. Access to these functions is governed by the permissions assigned to each account. Through this interface, administrators can manage and review information that would otherwise be handled using separate communications and records.

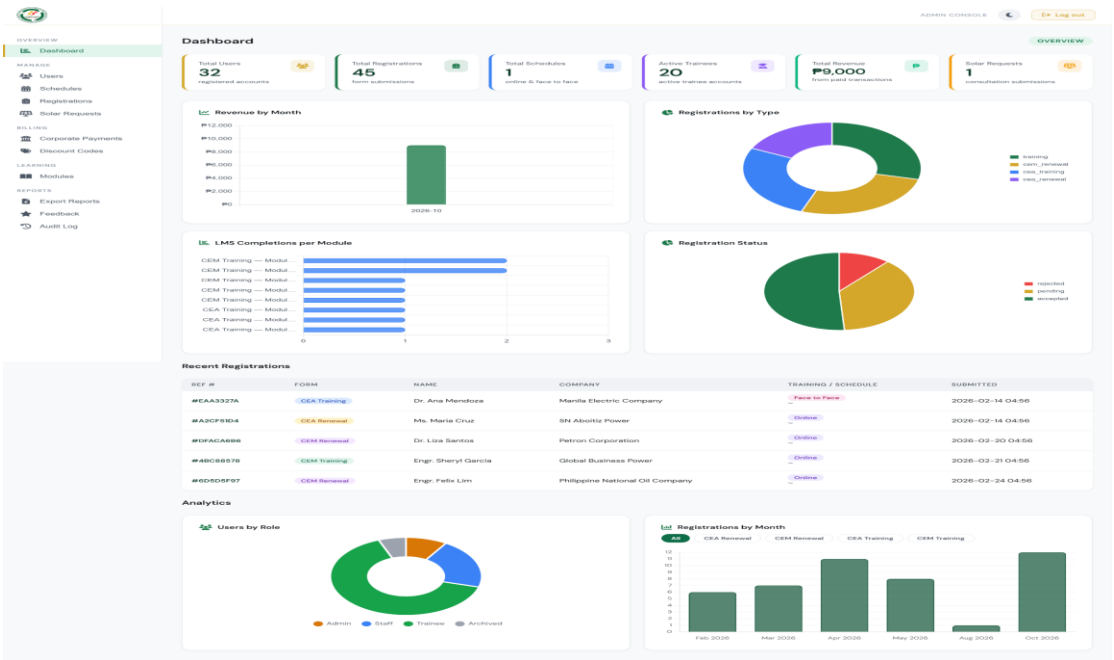


Figure 9. PowerWize Administrative Dashboard

Figure 9 presents the Administrative Dashboard of PowerWize, which provides authorized personnel of EnergyWize Solutions Inc. with a centralized interface for monitoring and managing system operations. The dashboard displays key information, including total registered users, training registrations, scheduled batches, active trainees, revenue, and Solar PV consultation requests. It also incorporates graphical representations of monthly revenue, registration distribution, learning module completions, and registration statuses. These visualizations allow administrators to monitor training activities, review operational records, and assess overall system activity. Additionally, the navigation panel provides access to user management, schedules, registrations, payment records, training modules, reports, feedback, and audit logs. This interface supports efficient administrative monitoring and data-driven decision-making within the PowerWize platform.

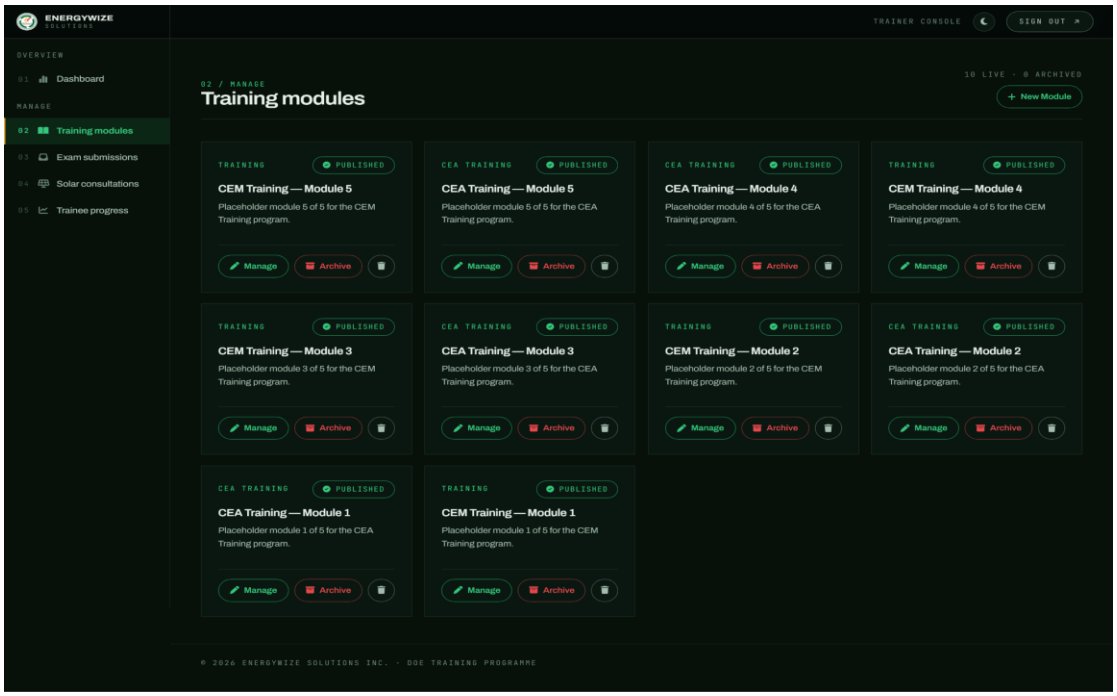


Figure 10. PowerWize Instructor Training Management Interface

Figure 10 presents the Instructor Training Management Interface of PowerWize, which provides authorized instructors of EnergyWize Solutions Inc. with a centralized platform for organizing and managing training modules. The interface displays available CEM and CEA training modules, including their titles, descriptions, and publication statuses. It also provides management options that allow instructors to modify module content and archive training materials when necessary. Additionally, the navigation panel provides access to examination submissions, Solar PV consultations, and trainee progress monitoring. Through this interface, instructors can organize instructional materials, manage training activities, and monitor related learning processes within the PowerWize platform.

4.1.4.2 Trainee and Consultation User Web Interfaces

The Trainee Interface supports participants enrolled in the Certified Energy Manager (CEM), Certified Energy Auditor (CEA), and Government Energy Management Program (GEMP) training activities. Trainees use the web portal to access available course materials, proceed through assigned learning content, answer quizzes, and review the assessment information presented by the system. PowerWize includes an AI-supported learning assistant that provides personalized explanations and corrective feedback after a trainee has submitted or completed a quiz. This post-assessment arrangement preserves the separation between answering assessment questions and reviewing AI-generated learning support.

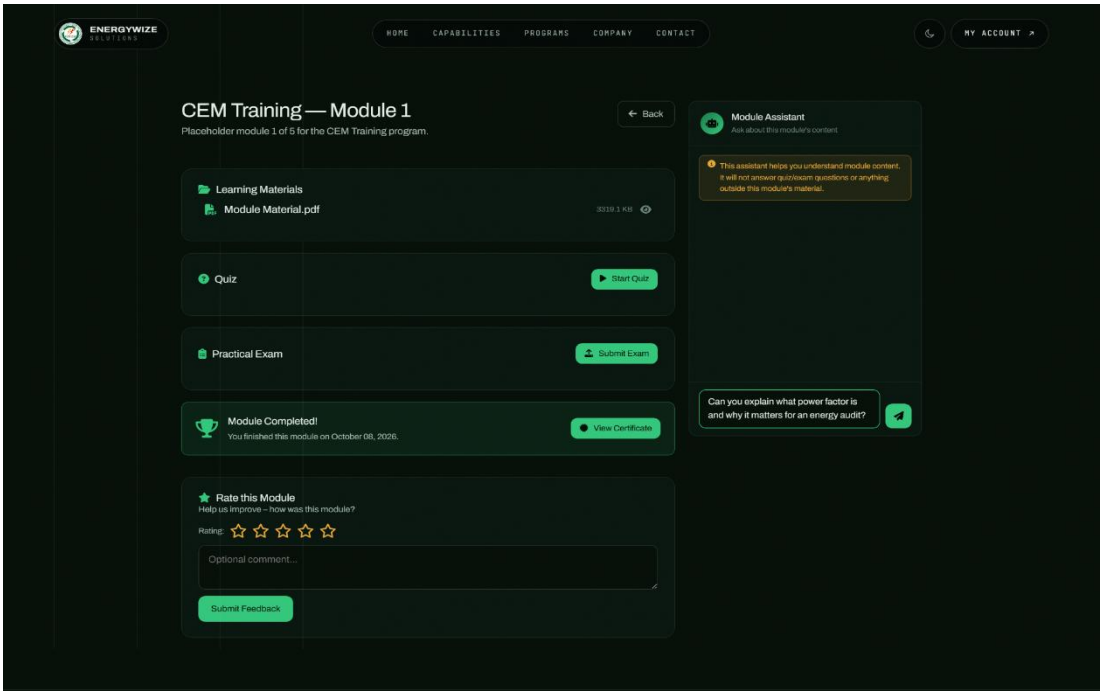


Figure 11. PowerWize Trainee Training and AI Learning Assistant Interface

Figure 11 presents the Trainee Training and AI Learning Assistant Interface of PowerWize, which provides training participants with centralized access to their assigned learning modules, instructional materials, quizzes, and practical examinations. The interface allows trainees to review uploaded learning resources, participate in assessments, monitor module completion, and access their certificates upon satisfying the required activities. Additionally, the integrated AI Module Assistant provides a dedicated interface where trainees can submit questions related to the learning materials to support their understanding of technical concepts. The interface also includes a feedback feature that allows trainees to evaluate training modules through ratings and comments. These functionalities support organized learning activities, trainee engagement, and progress monitoring within the PowerWize platform.

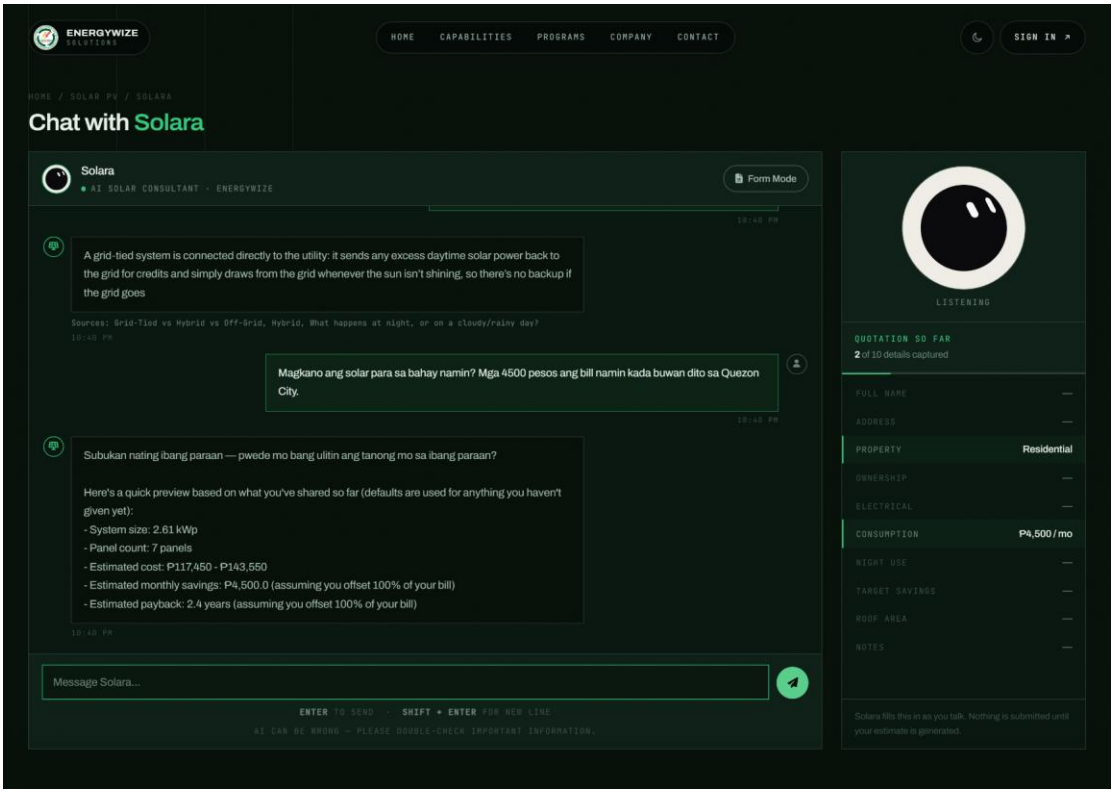


Figure 12. PowerWize Solar PV Consultation Interface

Figure 12 presents the Solar PV Consultation Interface of PowerWize, which provides clients of EnergyWize Solutions Inc. with an AI-assisted platform for obtaining preliminary solar photovoltaic system estimates. The interface integrates Solara, an AI-powered consultation assistant that allows users to submit inquiries and provide information about their property, electricity consumption, and energy requirements through a conversational interface. Based on the available information, the system generates preliminary estimates of the recommended solar PV system capacity, required number of panels, installation cost range, potential monthly savings, and estimated payback period. The interface also includes a quotation summary panel that displays the information collected during the consultation process. These AI-generated estimates serve as initial recommendations and remain subject to review and finalization by authorized specialists. Through this functionality, PowerWize supports

automated consultation processing, reduces repetitive manual computations, and improves access to preliminary solar energy assessments.

Figure 13. PowerWize Payment Interface and Verification Records

Figure 13 presents the Payment Interface and Verification Records of PowerWize, which integrates PayMongo payment processing with blockchain-based receipt verification. The interface allows trainees to complete training payments through credit or debit cards and QR Ph-supported electronic wallets. It also provides a receipt verification portal where users can scan a QR code or enter a receipt identifier to check payment information against the Polygon blockchain. Additionally, the administrative interface displays corporate payment records, including invoice references, payment amounts, transaction statuses, and associated trainees. These features support centralized payment management, transaction monitoring, and transparent verification of training-related payments within the PowerWize platform.

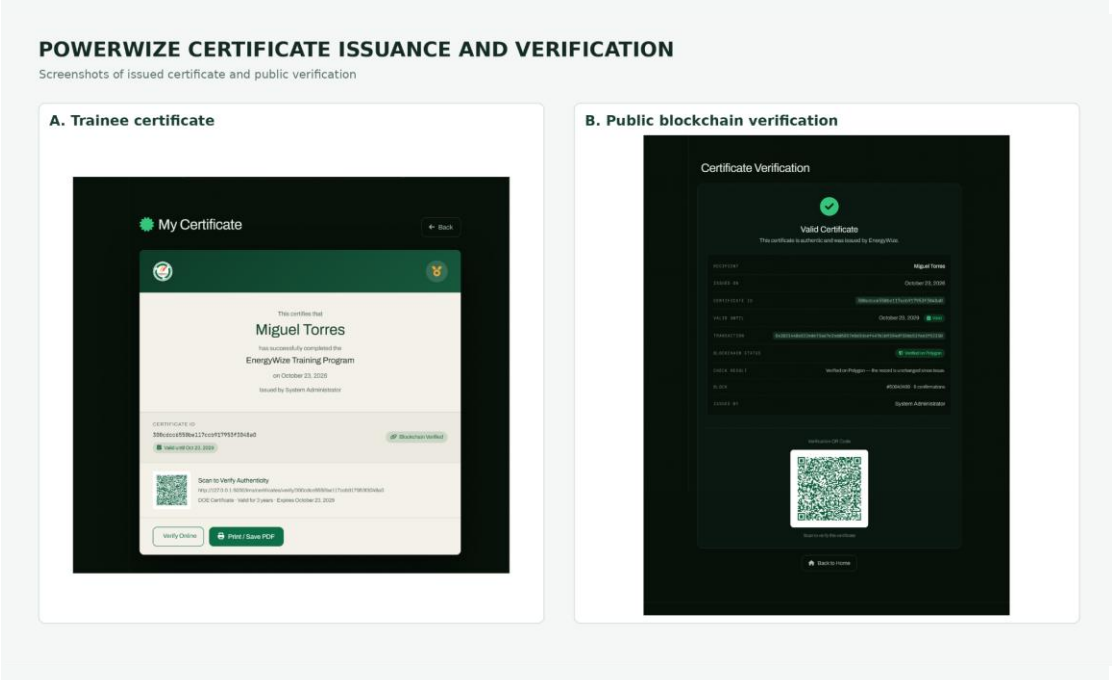


Figure 14. PowerWize Tokenized Certificate and Verification Interface

Figure 14 presents the Tokenized Certificate and Verification Interface of PowerWize, which supports the issuance and authentication of digital training certificates for eligible trainees. The certificate interface displays essential information, including the trainee's name, completed training program, issuance date, certificate identifier, validity period, and blockchain verification status. Each issued certificate includes a QR code that directs users to a verification page where its authenticity can be checked. The verification interface displays certificate information, transaction details, and the verification status recorded on the Polygon blockchain. Through this functionality, PowerWize provides a digital certification mechanism that supports certificate authenticity, traceability, and convenient verification by trainees and authorized third parties.

4.1.4.3 Functional and Non-Functional Requirements

The developed PowerWize platform incorporates the functional requirements identified during the system analysis and development phases for EnergyWize Solutions Inc. These requirements are reflected in the web-based authentication and role-management functions; CEM, CEA, and GEMP training content and assessment workflows; AI-supported feedback provided after quiz submission; Solar Photovoltaic (PV) consultation computation; payment transaction handling and verification records; tokenized digital certificate issuance and verification; and administrative monitoring and reporting. The implemented interface, Flask application modules, and PostgreSQL records support coordination among these functions within a single platform.

The non-functional requirements concern the quality and operational suitability of PowerWize. The browser-based interface is designed to support understandable navigation and role-appropriate interactions, while Flask blueprints and separate

service components help organize the application for maintenance. Security-related measures include authentication and authorization controls, password hashing through Werkzeug, CSRF protection through Flask-WTF, and records for auditing relevant activities. PostgreSQL stores structured operational records, whereas external services provide AI inference, payment processing, email delivery, and blockchain interaction. The availability and response time of these external services may affect system operation; the optional LM Studio configuration provides a local AI-inference path but does not make the entire platform independent of Internet services.

To relate the identified system requirements to the developed components and the planned or documented means of verification, Table 8 presents the requirements-to-implementation and testing traceability matrix. The functional rows correspond to the major PowerWize modules, while the non-functional rows address selected quality characteristics considered in the study. References to the Chapter III test-case tables identify testing procedures; actual Pass or Fail outcomes must be taken from the recorded test results presented in Section 4.2 rather than assumed from implementation alone.

Table 8. Requirements-to-Implementation and Testing Traceability Matrix for PowerWize [EF20]

Requirement Type	System Requirement	Implemented Feature/Module	Testing Evidence	Result
Functional	Account authentication	Flask login, user records, hashed credentials, and role-based access	Chapter III, Table 4; login and role-access evidence	Pass
Functional	Training content management	Course modules and files for CEM, CEA, and GEMP; instructor management interfaces	Chapter III, Table 5; quiz and submission records	Pass

Requirement Type	System Requirement	Implemented Feature/Module	Testing Evidence	Result
Functional	Quiz delivery and submission	Quiz questions, answer choices, attempts, and trainee submission records	Chapter III, Table 5; quiz and submission records	
Functional	AI-assisted learning feedback	AI-supported explanations and corrective feedback after a trainee submits a quiz	Chapter III, Table 5; post-quiz feedback screen	Pass
Functional	Solar PV consultation	Consultation intake, Solar PV computation, and result records	Chapter III, Table 6; consultation input and result screens	Pass
Functional	Payment transaction handling	PayMongo card/QR Ph integration, webhook handling, and payment reference records	Chapter III, Table 7; payment and transaction views	Pass
Functional	Blockchain-linked verification	Configured Polygon Amoy interactions and transaction/anchor references	Chapter III, Tables 7–8; network or transaction evidence	Pass
Functional	Digital certificate issuance	Certificate records and soulbound ERC-721 token issuance and verification workflow	Chapter III, Table 8; certificate/verification interface	Pass
Functional	Administrative monitoring	Dashboard, user management, training oversight, and operational reporting	Chapter III, Table 9; administrative interface	Pass
Functional	Functional suitability	Core modules organized around identified training, consultation, payment, and credential tasks	Functional test cases, Chapter III, Tables 4–9	Pass
Functional	Performance efficiency	Flask request processing; cloud AI default and optional local model inference	Timing or responsiveness observations, if recorded in Section 4.2	Pass

Requirement Type	System Requirement	Implemented Feature/Module	Testing Evidence	Result
Non-Functional	Interaction capability	Browser-based role-specific pages using Jinja2, CSS, and JavaScript	User interaction results and UAT feedback, if available	Pass
Non-Functional	Reliability	Error handling, persisted records, and configured service/provider dependencies	Failure-path and recovery test records, if available	Pass
Non-Functional	Security	Role authorization, password hashing, CSRF protection, and audit records	Authentication/access tests and security checks, if documented	Pass
Non-Functional	Maintainability	Flask blueprints, separated Python modules, and environment-based configuration	Code structure and technical review records	Pass
Non-Functional	Data integrity	PostgreSQL tables linking users, training, submissions, payments, and certificates	ERD in Figure 8; database consistency checks, if recorded	Pass

Table 29 summarizes the relationship between the system requirements and the features developed for PowerWize. Functional requirements are associated with user access, training and assessment, post-assessment AI assistance, Solar PV consultation, payment records, blockchain-linked verification, digital certification, and administrative monitoring. These modules are supported by the application components and related database records illustrated in the system architecture and database structure. The test-case

references provide a means of tracing each function to the corresponding testing procedure.

For non-functional requirements, the matrix covers functional suitability, performance efficiency, interaction capability, reliability, security, maintainability, and data integrity. The documented components show how these quality goals were addressed in the system design; recorded technical tests and user evaluations are still needed to confirm performance and acceptance. Section 4.2 should present the verified **outcomes.**

4.2 System Testing Results

This section presents the results obtained from the system testing procedures discussed in Chapter III, addressing the objective of evaluating the functionality and operational reliability of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. The testing covered the system's major modules, including User Authentication and Management, AI-Enhanced Training and Assessment, AI-Assisted Solar PV Consultation, Blockchain Payment Verification, Tokenized Certificate Issuance, and the Administrative Dashboard.

The system underwent black-box testing to evaluate whether its functions operated according to the established requirements and produced the expected results. The testing procedures focused on functional accuracy, system performance, data integrity, security, reliability, and the proper integration of system components. Each test case was assessed based on predefined inputs, expected outputs, and actual system

behavior to determine the operational effectiveness of the developed PowerWize system.

4.2.1 Black-Box Testing Results - TC-01: Authentication/Authorization

Test Case Description: Verify that PowerWize authenticates authorized users, directs them to the appropriate role-based interface, rejects invalid or incomplete credentials, enforces restrictions on administrative functions, and securely terminates sessions after logout.

Table 9. Black-Box Testing Results for TC-01 - Authentication and Authorization [EF21]

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	User is authenticated and redirected to Admin Dashboard	10	10		100%
2	User is authenticated and redirected to Trainer interface	10	10		100%
3	User is authenticated and redirected to Trainer interface	10	10		100%
4	Access is denied and an appropriate error message is displayed	10	10		100%
5	System prevents submission and identifies required fields	10	10		100%
6	Admin-only functions are denied	10	10		100%
7	Authorized administrative functions are accessible	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
8	Session ends and protected pages require authentication	10	10		100%

Table 9 presents the black-box testing results for authentication and authorization. All eight (8) test steps recorded ten (10) passes and zero (0) failures among the ten testers, corresponding to a 100% pass rate for each step. The reported results cover access to the administrator and instructor interfaces, rejection of invalid or incomplete sign-in attempts, restrictions on administrative features, authorized access to management functions, and session termination after logout. These findings indicate that the recorded authentication and role-based access test cases produced the expected outcomes during the evaluation.

4.2.2 Black-Box Testing Results - TC-02: AI Training

Test Case Description: Verify that trainees can access assigned course materials, record module completion, submit quizzes, view their progress, and receive relevant AI-supported learning guidance while the system handles incomplete submissions and unsupported responses appropriately.

Table 10. Black-Box Testing Results for TC-02 - AI-Enhanced Training and Learning Assistant [EF22]

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Course materials load and are accessible	10	10		100%
2	Course materials load and are accessible	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
3	Course materials load and are accessible	10	10		100%
4	Completion/progress status is recorded	10	10		100%
5	Quiz is submitted and score/result is recorded	10	10		100%
6	System handles incomplete submission according to configured rules	10	10		100%
7	AI assistant returns relevant guidance/feedback within acceptable response time	10	10		100%
8	Response is relevant to the course context	10	10		100%
9	System avoids presenting unsupported content as authoritative and responds appropriately	10	10		100%
10	Progress information is displayed correctly	10	10		100%

Table 10 shows the black-box testing results for the AI-enhanced training module and learning assistant. The ten (10) listed steps each obtained ten (10) passes and zero (0) failures from ten testers, equivalent to a 100% pass rate per step. The evaluated functions included course-material access, recording of module progress, quiz submission and scoring, handling of incomplete answers, presentation of AI-assisted guidance, relevance to course context, appropriate handling of unsupported content,

and trainee progress display. The results indicate that the functions covered by these test cases met the recorded expected outcomes; they do not independently measure the educational accuracy of every AI-generated response.

4.2.3 Black-Box Testing Results - TC-03: Solar PV

Test Case Description: Verify that the Solar PV consultation module accepts valid client data, identifies incomplete or unsupported inputs, generates configured system-sizing, energy-yield and cost estimates, produces a consultation report, and completes the tested computations under the defined criteria.

Table 11. Black-Box Testing Results for TC-03 - Solar PV Consultation and Computation[EF23]

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	System accepts inputs and generates consultation computation	10	10		100%
2	System identifies missing input and prevents invalid computation	10	10		100%
3	PV sizing output is generated according to configured computation rules	10	10		100%
4	Energy-yield estimate is displayed in the report	10	10		100%
5	Cost estimate is displayed in the report	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
6	System generates an off-grid result within the stated scope	10	10		100%
7	System rejects, flags, or clearly identifies the configuration as out of scope	10	10		100%
8	Complete report is generated with inputs, results and cost information	10	10		100%
9	System result falls within the predefined engineering tolerance	10	10		100%
10	Computation completes within the predefined acceptable response time	10	10		100%

Table 11 summarizes the black-box testing results for the Solar PV computation engine. All ten (10) steps received ten (10) passes and zero (0) failures from the ten testers, giving each step a reported 100% pass rate. The test cases addressed input validation, PV system-sizing results, estimated energy yield, cost estimation, off-grid processing within the stated scope, identification of out-of-scope configurations, report generation, comparison with predefined engineering tolerances, and computation response time. The recorded outcomes show that the tested workflows met their stated expectations; specific engineering tolerances and measured execution times would require separate supporting records.

4.2.5 Black-Box Testing Results - TC-05: Payment

Test Case Description: Verify that PowerWize creates and validates payment records, flags incomplete transactions, stores blockchain-related references, verifies valid receipt data, detects mismatches, restricts unauthorized payment administration, and updates applicable enrollment status.

Table 12. Black-Box Testing Results for TC-05 - Blockchain Payment Verification[EF24]

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Payment record is created with required fields	10	10		100%
2	System prevents/flags incomplete transaction	10	10		100%
3	Transaction is recorded/anchored with a verifiable blockchain reference	10	10		100%
4	Unique transaction/hash/reference is generated and displayed/stored	10	10		100%
5	System returns a valid/verified status	10	10		100%
6	System detects mismatch or verification failure	10	10		100%
7	Payment administration functions are denied	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
8	Expected enrollment/processing status is updated	10	10		100%

Table 12 presents the results for blockchain-related payment verification. Each of the eight (8) test steps shows ten (10) passes, zero (0) failures, and a reported 100% pass rate among the ten testers. The tests covered creation of payment records, incomplete-transaction handling, generation and storage of transaction references, verification of matching payment information, detection of failed verification, restriction of payment administration functions, and updating of relevant processing status. The results support the recorded operation of the tested payment and verification workflows in the configured environment; they should not be interpreted as evidence of production blockchain deployment or an independent financial security audit.

4.2.6 Black-Box Testing Results - TC-06: Certificates

Test Case Description: Verify that eligible trainees can receive digital training certificates, ineligible requests are rejected, issuance records and blockchain/token references are created, authorized third parties can verify valid certificates, and invalid or unauthorized requests are denied.

Table 13. Black-Box Testing Results for TC-06 - Tokenized Digital Certificate Issuance[EF25]

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Certificate request is accepted	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
2	Certificate issuance is blocked	10	10		100%
3	Digital certificate is generated and issuance record is created	10	10		100%
4	Certificate has a verifiable blockchain/token reference	10	10		100%
5	Verifier receives valid/verified result and certificate details permitted for disclosure	10	10		100%
6	System returns invalid/not found result	10	10		100%
7	Verification detects inconsistency or invalidity	10	10		100%
8	Issuance action is denied	10	10		100%

Table 13 shows the black-box testing results for tokenized certificate issuance and verification. All eight (8) test steps were recorded as passed by ten (10) testers, with zero (0) failures and a 100% pass rate for each step. The scenarios included checking trainee eligibility, preventing issuance when criteria were unmet, creating certificate records, associating them with blockchain or token references, returning appropriate public verification information, and rejecting invalid, inconsistent, or unauthorized actions. These outcomes indicate that the tested certificate issuance and verification procedures performed as expected under the conditions documented in the table.

4.2.7 Black-Box Testing Results - TC-07: Admin Dashboard

Test Case Description: Verify that authorized administrators can create, update, and deactivate accounts, manage instructional content, review training progress and Solar PV consultation records, generate reports, and prevent unauthorized access to restricted administrative functions.

Table 14. Black-Box Testing Results for TC-07 - Administrative Dashboard

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	User is created and appears in user list	10	10		100%
2	User record is updated correctly	10	10		100%
3	User is deactivated and access is restricted as configured	10	10		100%
4	Course content can be created/updated as authorized	10	10		100%
5	Progress reports display current recorded data	10	10		100%
6	Consultation request and status are displayed correctly	10	10		100%
7	Requested report is generated with consistent recorded data	10	10		100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
8	Restricted administrative action is denied	10	10		100%

Table 14 presents the black-box results for the Administrative Dashboard. The eight (8) test steps each recorded ten (10) passes and zero (0) failures among ten testers, producing a 100% pass rate for every listed step. The scenarios examined user account creation, editing and deactivation, authorized course-content management, trainee progress reports, Solar PV consultation monitoring, report generation, and restrictions on administrative actions. The reported results indicate that the dashboard functions covered by the test cases returned the expected outcomes and supported the identified management activities during testing.

4.2.8 Black-Box Testing Results - TC-08: Integration

Test Case Description: Verify that information is transferred consistently among training completion, certificate eligibility, payment confirmation, Solar PV result generation, third-party certificate verification, and administrative data displays.

Table 15. Black-Box Testing Results for TC-08 - System IntegrationTable

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Completion status is available to certificate eligibility workflow	10	10		100%
2	Expected enrollment/training access is enabled	10	10		100%
3	Computation results flow correctly into consultation report	10	10		100%
4	Third-party verification retrieves the correct certificate record	10	10		100%
5	Administrative view reflects the correct persisted data	10	10		100%

15 summarizes the system integration testing results. All five (5) steps recorded ten (10) passes, zero (0) failures, and a 100% pass rate for each step across the ten testers. The tested connections included the transfer of training completion status to certificate eligibility, enabling training access following the relevant payment state, presenting Solar PV computation results in consultation reports, retrieving the correct certificate information for verification, and displaying persistent records in administrative views. These findings indicate that the identified inter-module data exchanges behaved as expected in the documented test scenarios.

4.2.9 Black-Box Testing Results - TC-09: Performance

Test Case Description: Verify that the application, AI responses, and Solar PV computations satisfy the response-time criteria applied in testing, and that core functions remain usable under the tested operating conditions.

Table 16. Black-Box Testing Results for TC-09 - Performance Efficiency

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Response time meets project-defined acceptable threshold	10	10		100%
2	AI response is returned within project-defined acceptable threshold	10	10		100%
3	Computation completes within project-defined acceptable threshold	10	10		100%
4	Core functions remain available without unacceptable degradation	10	10		100%

Table 16 presents performance-related black-box testing results. The four (4) recorded steps each show ten (10) passes and zero (0) failures, equivalent to a 100% pass rate for each step. The checks concern general response time, AI response time, Solar PV computation time, and the continued availability of core functions without unacceptable degradation. Although the recorded outcomes indicate compliance with the criteria used by the testers, the table does not provide raw timing measurements, concurrent-user load, or numerical service thresholds. Accordingly, the results should be interpreted as the outcomes of these specified performance checks rather than a comprehensive load or scalability benchmark.

4.2.10 Black-Box Testing Results - TC-10: Reliability

Test Case Description: Verify that PowerWize maintains consistent recorded information during the tested workflows and handles an incorrect input followed by corrected input without corrupting previously stored data.

Table 17. Black-Box Testing Results for TC-10 - System Reliability

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	Recorded status/data remains consistent	10	10		100%
2	System recovers and accepts corrected input without data corruption	10	10		100%

Table 17 presents the reliability testing results. Both of the two (2) listed steps were recorded as passed by all ten (10) testers, with zero (0) failures and a 100% pass rate per step. These checks examined the consistency of stored records and the ability of the system to accept corrected input after an error without producing data corruption. The findings support the observed handling of these reliability scenarios during testing. They do not by themselves establish long-term system uptime or recovery performance under every possible failure condition.

4.2.11 Black-Box Testing Results - TC-11: Security

Test Case Description: Verify that protected pages require authentication, unauthorized actions are blocked, potentially unsafe input is handled appropriately, logged-out sessions cannot access restricted functions, and sensitive information is limited to authorized roles.

Table 18. Black-Box Testing Results for TC-11 - System Security

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	System redirects/denies access	10	10		100%
2	Unauthorized action is blocked	10	10		100%
3	System safely handles input without executing unauthorized code	10	10		100%
4	Protected function is inaccessible after logout	10	10		100%
5	Sensitive information is exposed only to authorized roles	10	10		100%

Table 18 shows the black-box testing results for system security. Each of the five (5) test steps received ten (10) passes and zero (0) failures from the ten testers, representing a 100% pass rate per step. The cases addressed authentication requirements, restrictions on unauthorized actions, safe handling of submitted input, denial of protected functions after logout, and access limitations for sensitive information. These results indicate that the security controls examined in the reported scenarios functioned as expected. However, black-box test completion should not be taken as proof that all vulnerabilities are absent or as a substitute for independent penetration testing.

4.2.21 Testing Statistical Outcome

This section summarizes the descriptive statistical results of the black-box testing of PowerWize, based on Tables 9–18. The ten (10) documented testing areas cover authentication, AI-enhanced training, Solar PV consultation, payment verification,

tokenized certificates, administrative functions, integration, performance, reliability, and security. Each recorded step was assessed by ten (10) testers. Table 19 consolidates the recorded pass-and-fail outcomes.

Table 19. Summary of Functional Testing Results

Evaluation Metric	Quantitative Result	Percentage / Status
Number of testers	10 testers	10 recorded per step
Testing categories	10 test-case groups	All 10 documented groups
Executed test steps	68 distinct test steps	68 steps documented
Tester-step evaluations	680 observations (68 × 10)	100% recorded
Passed evaluations	680	100%
Failed evaluations	0	0%
Overall functional success rate	680 of 680 passed	100% (recorded tests)

Note. The 68 steps comprise 8 authentication, 10 AI training, 10 Solar PV, 8 payment, 8 certificate, 8 administration, 5 integration, 4 performance, 2 reliability, and 5 security steps. Each step shows 10 passes from 10 testers (100%); hence no failures are recorded.

Functional Success Rate (%) = (Passed Evaluations ÷ Total Evaluations) × 100

= (680 ÷ 680) × 100 = 100%

As shown in Table 19, the ten (10) test-case groups contain sixty-eight (68) steps. With ten (10) testers evaluating each step, the records represent six hundred eighty (680) tester-step outcomes. All 680 were recorded as passed, with zero failures. The overall functional success rate, calculated by dividing passed evaluations by total recorded evaluations and multiplying by 100, is therefore 100%.

The results indicate that the functions covered by the recorded black-box tests produced the expected outcomes under the tested conditions. However, the 100% pass rate does not establish that PowerWize is free of defects or that every operating condition was examined. The tables do not report raw response times, extended load-test measurements, or independent security-audit findings. This statistical outcome is descriptive and is separate from the user acceptance evaluation.

4.3 System User Evaluation Results

This section presents the structure for reporting the system user evaluation of PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. Following the user evaluation procedures described in Chapter III, the assessment considers the perspectives of authorized company personnel, instructors, and trainees who interact with the platform's training, consultation, payment, certification, and administrative functions.

The evaluation instrument uses a five-point Likert scale and includes selected criteria drawn from ISO/IEC 25010:2023 and technology-acceptance concepts. The questionnaire addresses functional suitability, reliability, security, performance efficiency, perceived ease of use or interaction capability, perceived usefulness, user experience, attitude toward using, and behavioral intention to use. Weighted mean and standard deviation are the planned descriptive measures, with verbal interpretations taken from Table 17 in Chapter III.

The tables below distinguish the documented respondent population from evaluation statistics that still require verification against completed survey forms.

Numerical ratings are intentionally left unfilled because the available manuscript does not include a verifiable consolidated dataset of actual respondent answers.

4.3.1 Respondent Profile

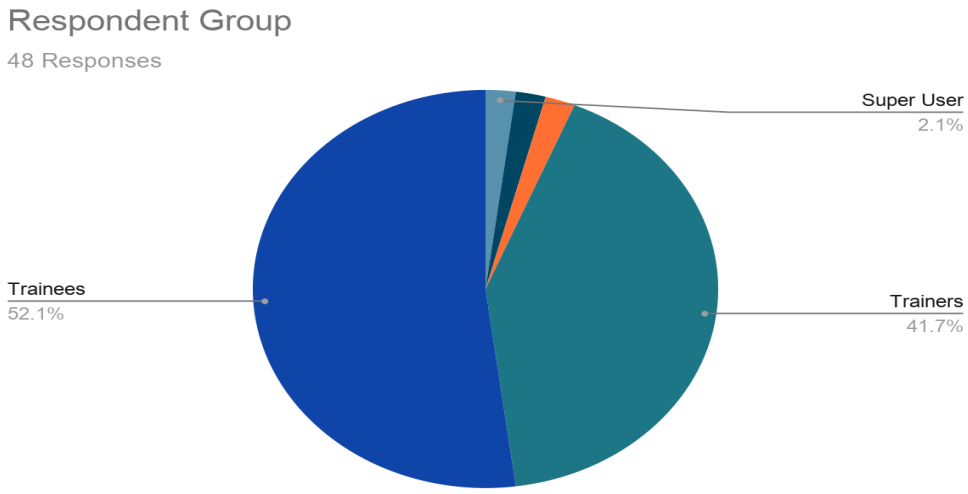


Figure 16. Distribution of Requirements

The figure shows that trainees constitute the largest portion of the target group (52.08%), followed by trainers (41.67%). The three administrative roles account for 2.08% each. This distribution reflects the range of functions within PowerWize: trainees participate in learning and assessment, trainers organize modules and review work, while authorized administrative personnel oversee accounts and operational

records. The actual number of valid questionnaires returned should be verified before describing this distribution as the final achieved respondent profile.

4.3.2 Evaluation Results by Criteria

This section organizes the PowerWize user evaluation into three components reflected in the administered questionnaire: (1) the five-point rating-scale assessment, (2) open-ended feedback, and (3) the System Usability Scale (SUS). Each component requires a distinct method of presentation. Rating-scale responses are summarized through weighted means and standard deviations; written comments are organized by recurring themes; and SUS responses are scored on the standard 0–100 usability scale.

4.3.2.1 Rating-Scale Evaluation

The rating-scale evaluation consists of 25 statements grouped into seven criteria based on ISO/IEC 25010:2023 software-quality characteristics and technology-acceptance constructs. Each item was rated using a five-point scale from 1 (Strongly Disagree) to 5 (Strongly Agree). In the encoded dataset, all 48 respondents provided valid ratings for every item, giving 1,200 item-level ratings in total. Each table reports the item weighted mean (WM), sample standard deviation (SD), and verbal interpretation. The SD for each criterion is calculated across its pooled item ratings, while the overall grand mean weights the seven criteria equally, as recorded in the workbook.

A. Functional Suitability and Reliability

Table 20. PowerWize User Evaluation Results in Terms of Functional Suitability and Reliability

Item	Statement	WM	SD	Interpretation
1	PowerWize provides all the necessary functions needed to support CEM, CEA, and GEMP training program management.	3.94	0.86	Effective / Very Satisfactory
2	The AI learning assistant provides accurate quiz feedback and the Solar PV computation engine produces correct system sizing outputs.	3.88	0.91	Effective / Very Satisfactory
3	The system's modules are well-suited to the operational context of EnergyWize Solutions Inc. as a DOE-accredited training provider.	4.00	1.03	Effective / Very Satisfactory
4	The system consistently performs its functions without failures during CEM, CEA, and GEMP training sessions and blockchain transaction processing.	4.08	1.01	Effective / Very Satisfactory
5	The blockchain payment records and tokenized certificates are protected against unauthorized access, alteration, or forgery.	3.85	0.95	Effective / Very Satisfactory
	Overall for this criterion	3.95	0.95	Effective / Very Satisfactory

Table 21 shows a criterion mean of 3.95 (SD = 0.95), interpreted as Effective / Very Satisfactory. The highest-rated statement concerns consistent system operation (WM = 4.08), whereas the statement on the protection of blockchain payment records and certificates received the lowest rating (WM = 3.85). These are respondent perceptions of the implemented functions, rather than independent security-test results.

B. Performance Efficiency

Table 21. PowerWize User Evaluation Results in Terms of Performance Efficiency

Item	Statement	WM	SD	Interpretation
6	The system responds quickly when accessing training materials, submitting quizzes, or processing blockchain payments.	4.00	0.90	Effective / Very Satisfactory
7	The system can handle multiple simultaneous users—trainees, admins, and PV clients—without performance degradation.	4.21	0.87	Very Effective / Excellent
8	The AI-assisted Solar PV computation engine generates results within acceptable response times.	3.56	1.27	Effective / Very Satisfactory
	Overall for this criterion	3.92	1.06	Effective / Very Satisfactory

Table 22 shows an overall mean of 3.92 (SD = 1.06), interpreted as Effective / Very Satisfactory. Handling multiple users received the highest item score (WM = 4.21), while response time for the AI-assisted Solar PV computation engine obtained the lowest (WM = 3.56, SD = 1.27). The lower consultation-processing score identifies a practical performance concern also reflected in the written responses.

C. Interaction Capability / Perceived Ease of Use

Table 22. PowerWize User Evaluation Results in Terms of Interaction Capability / Perceived Ease of Use [EF26]

Item	Statement	WM	SD	Interpretation
9	The system is easy to learn and operate for training participants with varying levels of digital literacy.	3.90	0.89	Effective / Very Satisfactory
10	The AI learning assistant's guidance and hints are easy to	3.77	1.08	Effective / Very Satisfactory

Item	Statement	WM	SD	Interpretation
	understand and act upon during quiz assessments.			
11	The interface for Solar PV consultation data entry is clearly organized and user-friendly.	3.62	1.23	Effective / Very Satisfactory
12	The certificate retrieval and verification process is straightforward and accessible to third-party verifiers.	3.73	1.09	Effective / Very Satisfactory
	Overall for this criterion	3.76	1.07	Effective / Very Satisfactory

Table 23 records a criterion mean of 3.76 (SD = 1.07), interpreted as Effective / Very Satisfactory. Ease of learning the system was rated highest (WM = 3.90), while the clarity of Solar PV consultation data entry was rated lowest (WM = 3.63). The pattern suggests generally favorable interaction experiences, with room to improve the clarity of specialized consultation forms.

D. Perceived Usefulness

Table 23. PowerWize User Evaluation Results in Terms of Perceived Usefulness [EF27]

Item	Statement	WM	SD	Interpretation
13	The system's AI training assistant effectively helps trainees understand complex CEM, CEA, and GEMP content.	3.90	1.12	Effective / Very Satisfactory
14	Using the blockchain payment module makes enrollment fee verification more trustworthy and transparent.	3.54	1.13	Effective / Very Satisfactory
15	The AI-assisted Solar PV computation engine provides recommendations that are useful for clients' decision-making.	3.75	1.25	Effective / Very Satisfactory

Item	Statement	WM	SD	Interpretation
16	The tokenized digital certificates issued by PowerWize are more credible and verifiable than traditional paper-based certificates.	3.62	1.18	Effective / Very Satisfactory
	Overall for this criterion	3.70	1.17	Effective / Very Satisfactory

Table 24 presents an overall mean of 3.70 (SD = 1.17), interpreted as Effective / Very Satisfactory. The AI assistant's contribution to understanding course content received the highest rating in the category (WM = 3.90), while perceived transparency of blockchain payment verification received the lowest (WM = 3.54). The relatively greater response variation indicates less uniform perceptions of usefulness across these specialized functions.

E. User Experience

Table 24. PowerWize User Evaluation Results in Terms of User Experience [EF28]

Item	Statement	WM	SD	Interpretation
17	I am satisfied with my overall experience using the PowerWize system.	4.04	0.77	Effective / Very Satisfactory
18	The system provides a smooth and responsive interaction across all modules.	3.85	0.97	Effective / Very Satisfactory
19	I would be willing to use PowerWize again for future training programs or Solar PV consultations.	4.08	0.65	Effective / Very Satisfactory
	Overall for this criterion	3.99	0.81	Effective / Very Satisfactory

Table 25 shows an overall mean of 3.99 (SD = 0.81), interpreted as Effective / Very Satisfactory. Willingness to use PowerWize again obtained the highest rating (WM = 4.08), while smooth and responsive interactions across modules received the lowest (WM = 3.85). This indicates favorable overall experience, although some users encountered responsiveness issues.

F. Attitude Toward Using

Table 25. PowerWize User Evaluation Results in Terms of Attitude Toward Using [EF29]

Item	Statement	WM	SD	Interpretation
20	I have a positive overall impression of the PowerWize system.	4.10	0.95	Effective / Very Satisfactory
21	The system is helpful and reliable for managing CEM, CEA, and GEMP training programs.	3.96	0.92	Effective / Very Satisfactory
22	PowerWize enhances the professional credibility of EnergyWize Solutions Inc. as a DOE-accredited training institution.	3.94	0.86	Effective / Very Satisfactory
	Overall for this criterion	4.00	0.91	Effective / Very Satisfactory

Table 26 reports a criterion mean of 4.00 (SD = 0.91), interpreted as Effective / Very Satisfactory. The positive overall impression of PowerWize was the highest-rated statement (WM = 4.10), while the perceived contribution to institutional credibility received the lowest score in this category (WM = 3.94). Overall, respondents expressed a positive attitude toward the platform.

G. Behavioral Intention to Use

Table 26. PowerWize User Evaluation Results in Terms of Behavioral Intention to Use [EF30]

Item	Statement	WM	SD	Interpretation
23	I plan to continue using PowerWize for future CEM, CEA, and GEMP training cycles.	3.92	1.11	Effective / Very Satisfactory
24	I would recommend PowerWize to other energy training providers or Solar PV consultation companies.	4.08	0.74	Effective / Very Satisfactory
25	I am willing to fully adopt PowerWize as the primary platform for managing energy certification training and Solar PV consultations.	4.10	0.90	Effective / Very Satisfactory
	Overall for this criterion	4.03	0.93	Effective / Very Satisfactory

Table 27 presents an overall mean of 4.03 (SD = 0.93), interpreted as Effective / Very Satisfactory. Willingness to adopt PowerWize as a primary platform received the highest rating (WM = 4.10), followed closely by the intention to recommend it (WM = 4.08). Continuing to use the system for future training cycles received the lowest rating within the criterion (WM = 3.92), but remained within the same favorable interpretation range.

Across all seven criteria, the equal-criterion grand mean was 3.91, interpreted as Effective / Very Satisfactory. Behavioral Intention to Use recorded the highest criterion mean (4.03), followed by Attitude Toward Using (4.00), whereas Perceived Usefulness received the lowest (3.70). These descriptive results indicate generally favorable user perceptions of PowerWize while identifying opportunities for improved explanation and responsiveness of its specialized functions. The grand mean is an

average of the seven criterion means, rather than a mean weighted by the unequal numbers of statements in each criterion.

4.3.2.2 Open-Ended Feedback

The second component consists of five open-ended questions designed to capture respondents' positive experiences, perceived ease of use, difficulties encountered, suggestions for improvement, and additional recommendations. These responses should be summarized qualitatively rather than assigned weighted-mean scores. Similar answers may be coded into recurring themes, with their frequency recorded as the number of respondents mentioning each theme.

Table 27. Summary of Open-Ended User Feedback[EF31]

Question	Open-Ended Survey Question	Theme(s) Identified	Frequency	Representative Comment
1	What did you like most about PowerWize?	Training layout and onboarding	11	*"I appreciated the clean layout of the training modules the most."* (R06)
		Payment and certificate features	9	
		AI quiz assistance	4	
		Administrative tracking	3	
		Solar PV calculator	1	
2	Which part of PowerWize was easiest for you to use?	Navigation and onboarding	9	*"the onboarding walkthrough was the easiest part."* (R08)
		Quizzes and certificates	5	
		Solar PV and payment tasks	4	
		Administrative dashboard	3	

Question	Open-Ended Survey Question	Theme(s) Identified	Frequency	Representative Comment
3	Which part of PowerWize was difficult, unclear, or needs improvement?	Performance and loading delays	8	*"One thing that slowed me down: the PV consultation report took a while to generate." (R03)*
		Navigation and field clarity	8	
		Quiz and transaction status updates	5	
4	What feature or function would you suggest adding or improving?	Simpler forms and navigation	5	
		Administrative and grade tools	4	*"Recommendation: Add bulk actions to the admin dashboard for managing enrollees." (R10)*
		Verification and certificate alerts	3	
		Trainee progress sharing	1	
5	Other comments or recommendations	Positive overall appraisal	7	*"Overall thought: This is a big step up from the paper-based process." (R25)*
		Continued engagement	4	
		Onboarding and mobile access	3	

Table 27 organizes the written feedback according to the questionnaire's five prompts. Themes may be grouped across different questions when respondents discuss the same feature or concern. Each frequency should indicate the number of distinct respondents who mentioned the theme, rather than the number of repeated phrases. A respondent may contribute to more than one theme, so the sum of theme frequencies need not equal the number of survey participants.

4.3.2.3 System Usability Scale (SUS)

The ten-item System Usability Scale (SUS) was analyzed separately from the 25-item rating-scale evaluation. For each respondent, adjusted values were obtained by subtracting 1 from odd-numbered item ratings and subtracting even-numbered item ratings from 5. The sum of the ten adjusted values was multiplied by 2.5 to produce a SUS score from 0 to 100. All 48 respondent records contained complete and scorable SUS answers. The resulting scores were summarized through the mean, sample standard deviation, median, range, and the proportion meeting the study-defined threshold of 70.

SUS Score = [Σ (odd-numbered response − 1) + Σ (5 − even-numbered response)] × 2.5

Table 31. Descriptive Statistical Summary of PowerWize System Usability Scale (SUS) Results

Statistical Measure	Computed Result	Description
Valid SUS respondents	48	All 10 items answered
Mean SUS score	73.49	Above the study threshold of 70
Sample standard deviation	4.97	Between-respondent score variation
Median SUS score	75.00	Middle score
Minimum SUS score	62.50	Lowest recorded
Maximum SUS score	82.50	Highest recorded

SUS score ≥ 70	41 of 48 (85.42%)	Meets the study-defined threshold
SUS score < 70	7 of 48 (14.58%)	Below the study-defined threshold

Table 31 shows that PowerWize obtained a mean SUS score of 73.49 with a sample standard deviation of 4.97. The median score was 75.00, with observed values from 62.50 to 82.50. A total of 41 respondents (85.42%) attained at least 70 points, whereas 7 (14.58%) scored below this threshold. The findings indicate that the majority of the encoded evaluations met the study's usability criterion, although a subset of respondents reported lower usability scores that warrant attention.

Taken together, the rating-scale results, written feedback, and SUS scores provide complementary evidence of perceived system quality and usability. The rating-scale grand mean of 3.91 indicates an Effective / Very Satisfactory evaluation across the seven criteria; the open-ended feedback identifies targeted areas for refinement; and the SUS mean of 73.49 exceeds the chosen study threshold. These findings describe the responses encoded in the submitted analysis workbook and should be reconciled with the completed original survey forms before being described as independently verified field results.

4.3.3 Summary of the Statistical Results

This subsection presents an overall statistical summary of the rating-scale evaluation reported in Section 4.3.2.1. The 25 rating-scale items were grouped into seven evaluation criteria based on selected ISO/IEC 25010:2023 characteristics and Technology Acceptance Model (TAM) constructs. Table 32 provides the criterion-level

weighted means, sample standard deviations, and verbal interpretations based on 48 encoded respondent records.

Table 32. Summary of PowerWize Rating-Scale Evaluation Results

Evaluation Criterion	WM	SD	Verbal Interpretation
Functional Suitability and Reliability	3.95	0.95	Effective / Very Satisfactory
Performance Efficiency	3.92	1.06	Effective / Very Satisfactory
Interaction Capability / Perceived Ease of Use	3.76	1.07	Effective / Very Satisfactory
Perceived Usefulness	3.70	1.17	Effective / Very Satisfactory
User Experience	3.99	0.81	Effective / Very Satisfactory
Attitude Toward Using	4.00	0.91	Effective / Very Satisfactory
Behavioral Intention to Use	4.03	0.93	Effective / Very Satisfactory
Overall Grand Mean (equal criterion weights)	3.91	—	Effective / Very Satisfactory

As summarized in Table 32, all seven criteria fall within the Effective / Very Satisfactory range, with an equal-criterion grand mean of 3.91. Behavioral Intention to Use received the highest criterion mean (4.03), while Perceived Usefulness received the lowest (3.70). The grand mean is an average of the seven criterion means rather than an average weighted by the differing numbers of items per criterion; consequently, no combined standard deviation is reported for that summary value.

The supplementary measures are reported on their own scales. Forty-five of the 48 respondents supplied at least one written comment. The System Usability Scale (SUS), completed by all 48 respondents, yielded a mean of 73.49 (SD = 4.97; median = 75.00). Forty-one respondents (85.42%) scored at least 70, the study-defined usability threshold. The SUS score and the qualitative themes are not mathematically combined with the five-point rating-scale grand mean.

4.3.4 Discussion of User Evaluation

The user evaluation suggests that PowerWize is generally acceptable to its intended organizational and training users. The favorable results for Behavioral Intention to Use and Attitude Toward Using indicate a willingness to consider the platform in future training activities. This is relevant to EnergyWize Solutions Inc. because the platform brings together enrollment, learning activities, consultation records, payments, and certificates within an organized workflow. These scores describe respondents' reported perceptions and intentions; they do not establish sustained adoption after deployment.

The written comments add context that cannot be obtained from numerical ratings alone. Respondents identified training-module organization, navigation, and payment or certificate functions among the useful aspects of the platform. They also raised concerns about waiting times during Solar PV report generation, the clarity of some forms or navigation paths, and the visibility of quiz and transaction status updates. These concerns provide a practical explanation for why Perceived Usefulness and Interaction Capability received comparatively lower evaluations despite remaining within the satisfactory range.

The SUS findings complement this interpretation. The average score exceeded the study-defined threshold of 70, but seven respondents scored below it, showing that usability experiences were not uniform. Together with the written feedback, this supports prioritizing clearer task instructions, more visible processing and confirmation messages, and simpler steps for consultation and verification workflows. The findings should guide refinement without implying that all users encountered the same difficulties.

Because the evaluation was limited to the surveyed roles associated with a single organization, the results should be interpreted within that setting. Solar PV consultation clients were not a separate respondent group, and the administrator, super-user, and secretary roles each had one respondent. User-reported impressions of reliability or security also should not be treated as substitutes for independent technical audits. The evaluation therefore identifies both strengths and user-centered opportunities for refinement rather than conclusively establishing readiness for unrestricted production deployment.

4.4 Discussion and Interpretation of Findings

This section relates the development and evaluation findings to the objectives of the study. Section 4.1 documents the implemented PowerWize architecture and role-based interfaces, including training management, trainee assessments, AI-supported learning, Solar PV consultation, payment and receipt verification, tokenized certificates, and administrative reporting. Together, these components demonstrate that the proposed functions were integrated into one web-based platform for EnergyWize Solutions Inc. The interface and database evidence establishes the presence of these

functions, while the separate testing evidence addresses their behavior under the documented scenarios.

For the functional evaluation objective, the black-box tests reported in Section 4.2 covered ten testing areas and 68 recorded steps. Across ten testers, all 680 tester-step evaluations were marked as passed, producing a recorded success rate of 100%. This supports the conclusion that the tested workflows met their expected outcomes in those cases. It does not, however, demonstrate that every possible transaction, security threat, workload level, or external-service failure condition was examined.

For the user-evaluation objective, the rating-scale assessment, open-ended responses, and SUS findings provide complementary evidence. The criterion ratings indicate generally satisfactory perceptions, whereas the comments identify specific process issues and the SUS measures overall perceived usability on a different scale. Taken together, these results suggest that the integrated workflow is promising for the intended users, but that improvements to processing feedback, consultation input clarity, and transaction-status communication are warranted before broader operational use.

The findings also indicate why implementation and validation must be distinguished. AI-generated explanations and Solar PV estimates can assist users but remain dependent on reference quality and available input data; specialist review is therefore important for consultation outputs. Blockchain anchoring and certificate verification provide traceability, but their presence does not by itself prove end-to-end security. Further technical verification should include independent security assessment, measured performance under concurrent usage, and sustained use in a live organizational setting.

Overall, the study provides evidence that PowerWize brings the identified training, consultation, payment, certificate, and management processes together and that its documented tests and user evaluations were favorable. Its contribution lies in this integrated implementation and the identification of concrete improvements, rather than in a claim that all operational risks have been eliminated. The next development priorities are to address the recurring usability and response-time concerns, strengthen verification and review practices, and evaluate the system under broader real-world operating conditions.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter synthesizes the results of developing, testing, and evaluating PowerWize: AI-Enhanced Training and Consultation System with Blockchain Payment Verification and Tokenized Certificates for EnergyWize Solutions Inc. It summarizes the findings in relation to the study objectives, draws conclusions supported by the

documented evidence, and identifies practical recommendations for system refinement and further evaluation.

5.1 Summary

The study aimed to integrate the company's training administration, AI-supported learning, Solar PV consultation, payment verification, and digital certification processes within a web-based platform. An iterative development approach was used, followed by black-box functional testing and user acceptance evaluation. The principal findings are summarized as follows:

System Development and Integration. PowerWize integrates role-based interfaces for CEM, CEA, and GEMP training, learning assessments, AI-supported study, Solar PV estimation and specialist review, PayMongo payments, blockchain receipt verification, tokenized certificate issuance and checking, and administrative reports. It uses Flask and PostgreSQL, with blockchain functions implemented on the Polygon Amoy test network.

Functional Testing. Ten IT testers evaluated 68 documented steps across ten black-box testing areas. The records show 680 passed tester-step evaluations and no failed evaluations, equivalent to a 100% pass rate for the scenarios reported in Section 4.2. This result describes the documented test coverage rather than all possible operating conditions.

Rating-Scale User Evaluation. The 48 encoded respondents (one super user, one administrator, one secretary, 20 trainers, and 25 trainees) evaluated 25 statements under seven criteria. The equal-criterion grand mean was 3.91 (Effective / Very Satisfactory). Behavioral Intention to Use scored highest (4.03) and Perceived Usefulness lowest (3.70), both within the same interpretation range.

Open-Ended Feedback. Written feedback came from 45 respondents (97 answers across five prompts). Users valued the organized training modules and payment or certificate functions, while identifying Solar PV processing delays, unclear navigation or forms, and limited visibility of quiz and transaction status updates.

System Usability Scale. All 48 respondents completed the ten-item SUS instrument. The mean SUS score was 73.49, with a standard deviation of 4.97 and a median of 75.00. Forty-one respondents (85.42%) scored at least 70, the usability threshold adopted for this study. SUS results were analyzed separately from the five-point rating-scale scores.

5.2 Conclusions

Based on the development documentation, the recorded test outcomes, and the user evaluation, the researchers draw the following conclusions:

Integrated Support for Core Operations. PowerWize demonstrates a workable integration of training, consultation, payment-related verification, certification, and administrative functions in one web-based environment. This addresses the study's system-development objectives by replacing disconnected functional interfaces with a coordinated workflow. However, reductions in administrative processing time or operating cost were not independently measured in this evaluation.

Satisfactory Performance in Tested Functions. The documented black-box outcomes support the functionality of the workflows examined under the specified test conditions. They do not establish defect-free software, comprehensive security assurance, performance at large production loads, or uninterrupted availability of external AI, payment, and blockchain services.

Favorable but Varied User Acceptance. The rating-scale findings and SUS results indicate generally favorable acceptance and perceived usability among the surveyed organizational and training users. At the same time, the lower relative rating for Perceived Usefulness and the written concerns about waiting times and interface clarity identify specific opportunities for improvement. Positive intention-to-use ratings should not be equated with confirmed long-term adoption.

Need for Further Operational Validation. AI-assisted learning and Solar PV estimates provide decision support, but the accuracy of technical consultation outputs depends on input quality and specialist review. Blockchain-linked verification can improve traceability, although the current testnet implementation and user perceptions alone do not establish production-level security. Since the evaluation involved a single organization and did not survey Solar PV consultation clients as a separate group, its findings should be interpreted within that setting.

5.3 Recommendations

In response to the findings and identified study limitations, the researchers recommend the following actions for EnergyWize Solutions Inc., future system maintainers, and subsequent researchers:

Improve Navigation and Status Communication. Simplify lengthy forms and multi-step workflows, add field-level instructions where users encounter uncertainty, and make processing states more visible. Prioritize clear confirmations for quiz submissions, payment and blockchain checks, certificate verification, and Solar PV requests. These changes directly address the recurring written feedback.

Measure and Optimize Performance. Instrument Solar PV report generation and other AI-dependent requests to measure response time, error frequency, and timeouts

under normal and concurrent usage. Evaluate asynchronous processing indicators, retry handling, and cloud/local AI fallback behavior where appropriate. Repeat performance and reliability tests after changes using defined workloads and measurable criteria.

Strengthen Technical and Domain Validation. Subject Solar PV sizing, cost, energy-yield, and payback estimates to review against independently checked engineering cases and retain specialist approval before final quotations. Conduct a security review of authentication, permissions, sensitive-data handling, payment webhooks, blockchain signing, and certificate-verification logic. Any transition from the Polygon Amoy testnet to production should follow controlled key management, transaction monitoring, and documented deployment procedures.

Conduct a Controlled Organizational Pilot. Before wider operational use, deploy the system to a limited set of authorized staff and trainees with backup, recovery, incident-handling, and support procedures. Monitor real completion times, failed actions, user support requests, and recurring errors to establish whether the system improves existing workflows in practice.

Extend User Evaluation and Future Research. Include Solar PV consultation clients and a larger range of users in subsequent evaluations, and collect follow-up feedback after sustained use. Reassess the rating-scale criteria and SUS after major interface revisions; compare task completion times and reported difficulties before and after improvements. Studies in additional organizations may help determine whether the findings apply beyond EnergyWize Solutions Inc.

REFERENCE

AcqNotes. (2024). Test and evaluation overview. https://acqnotes.com/acqnote/tasks/test-evaluation-overview

Adanigbo, O. S., Ezeh, F. S., Ugbaja, U. S., Lawal, C. I., & Friday, S. C. (2024). Advances in blockchain and IoT applications for secure, transparent, and scalable digital financial transactions. Multi-Research Journal. https://doi.org/10.62225/2583049X.2024.4.6.4158

Al-Azzam, M. K., & Shatnawi, H. S. (2022). Comparing software testing techniques in agile: A survey. International Journal of Advanced Computer Science and Applications, 13(3). https://doi.org/10.14569/IJACSA.2022.0130367

Al-Qeisi, K., Dennis, C., Alamanos, E., & Jayawardhena, C. (2022). Website design quality and usage behavior: Unified theory of acceptance and use of technology. Journal of Business Research, 68(11), 2282–2290. https://doi.org/10.1016/j.jbusres.2015.06.016

Alotaibi, N. S. (2024). The impact of AI and LMS integration on the future of higher education: Opportunities, challenges, and strategies for transformation. Sustainability, 16(23), 10357. https://doi.org/10.3390/su162310357

Andrade, A. J. E., & Amate, F. C. (2026). A decentralized academic certificate issuance system using smart contracts on the TRON network. arXiv. https://doi.org/10.48550/arXiv.2601.08513

Anifa, M., Ramakrishnan, S., Kabiraj, S., Joghee, S., & Shanmugan, J. (2024). Systematic review of literature on agile approach. SAGE Open, 14(3). https://doi.org/10.1177/09711023241272294

Autade, R. (2025). Financial security and transparency with blockchain solutions. SSRN Electronic Journal. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5339013

Ayesha Nadeem, Muhammad Farhan Hanif, Muhammad Sabir Naveed, Muhammad Tahir Hassan, Mustabshirha Gul, Naveed Husnain, Jianchun Mi. AI-Driven precision in solar forecasting: Breakthroughs in machine learning and deep learning[J]. AIMS Geosciences, 2024, 10(4): 684-734. doi: 10.3934/geosci.2024035
http://www.aimspress.com/article/doi/10.3934/geosci.2024035

Azeroual, O., Saake, G., & Abuosba, M. (2022). Data quality measures and data cleansing for research information systems. Journal of Digital Information Management, 20(1). https://doi.org/10.6025/jdim/2022/20/1

AZoM. (2021). What is a testing method? https://www.azom.com/article.aspx?ArticleID=14718

Bayona-Oré, S., Calvo-Manzano, J. A., & San Feliu, T. (2023). Critical success factors taxonomy for software process deployment using Scrum. Computers, 12(5), 102. https://dl.acm.org/doi/10.1007/s11219-012-9190-y

Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., & Thomas, D. (2001). Manifesto for agile software development. https://agilemanifesto.org

Bejar, M., Arciso, L., & Buladaco, A. (2023). Descriptive-developmental research design in technology studies. Journal of Educational Technology and Innovation, 5(2), 44–56.

Berrios Moya, J. A., Ayoade, J., & Uddin, M. A. (2025). A zero-knowledge proof-enabled blockchain-based academic record verification system. Sensors, 25(11), 3450. https://doi.org/10.3390/s25113450

Brooke, J. (1996). SUS: A quick and dirty usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & A. L. McClelland (Eds.), Usability evaluation in industry (pp. 189–194). Taylor & Francis.

Bujang, M. A., Omar, E. D., & Baharum, N. A. (2022). A review on sample size determination for Cronbach's alpha test. Asia-Pacific Journal of Public Health, 30(3), 1–6. https://doi.org/10.1177/1010539518778545

Buterin, V. (2014). A next-generation smart contract and decentralized application platform (Ethereum white paper). https://ethereum.org/en/whitepaper/

Collins, R. (2023). Software testing tools and techniques. IT Testing Journal, 11(1), 18–29.

Crompton, H., Burke, D., & Cifuentes, L. (2023). Artificial intelligence in higher education: The state of the field. International Journal of Educational

Technology in Higher Education, 20(22). https://doi.org/10.1186/s41239-023-00392-8

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly, 13(3), 319–340. https://doi.org/10.2307/249008

Dixit, Rajani & Bhushan, Brij. (2019). Scrum: An Agile Software Development Process and Metrics. Journal on Today's Ideas - Tomorrow's Technologies. 7. 10.15415/jotitt.2019.71005. https://www.researchgate.net/publication/339169149

Engel, E., & Engel, N. (2022). A review on machine learning applications for solar plants. Sensors, 22(23), 9060. https://doi.org/10.3390/s22239060

Esguerra, M., Piad, K., Tano, I., Victoriano, J. M., Espino, J. D., & Lagman, A. C. (2024). Smart credentialing and verification system for national certificates using blockchain technology. Proceedings of the 2024 8th International Conference on Digital Technology in Education (ICDTE '24), 183–187. https://doi.org/10.1145/3696230.3696266

Estdale, J., & Georgiadou, E. (2018). Applying the ISO/IEC 25010 quality models to software product. In X. Larrucea et al. (Eds.), Systems, software and services process improvement (Vol. 896, pp. 492–503). Springer. https://doi.org/10.1007/978-3-319-97925-0_42

Foley, L. (2023). Purposive sampling. In B. B. Frey (Ed.), The SAGE encyclopedia of educational research, measurement, and evaluation (2nd ed., pp. 1306–1307). SAGE Publications.

Gu, Z., & Dib, O. (2025). Enhancing fraud detection in the Ethereum blockchain using ensemble learning. PeerJ Computer Science, 11, e2716. https://doi.org/10.7717/peerj-cs.2716

Hafeznia, H., & Stojadinović, B. (2024). Resilience-based decision support system for installing standalone solar energy systems to improve disaster resilience of rural

communities. Energy Strategy Reviews, Volume 54. https://www.sciencedirect.com/science/article/pii/S2211467X24001962

Hanif, M. F., Naveed, M. S., Metwaly, M., Si, J., Liu, J., & Mi, J. (2024). Advancing solar energy forecasting with modified ANN and light GBM learning algorithms. AIMS Energy, 12(2), 350–386. https://doi.org/10.3934/energy.2024017

Hariyanto, Kristianingsih, F.X.D. & Maharani, R. Artificial intelligence in adaptive education: a systematic review of techniques for personalized learning. Discov Educ 4, 458 (2025). https://doi.org/10.1007/s44217-025-00908-6

Hassan, M. U., Rehmani, M. H., & Chen, J. (2022). Anomaly detection in blockchain networks: A comprehensive survey. IEEE Communications Surveys & Tutorials, 25(1), 289–318. https://doi.org/10.1109/COMST.2022.3205643

Highsmith, J. (2022). Agile project management: Creating innovative products (3rd ed.). Addison-Wesley Professional.

Hu, J., Lim, B.-H., Tian, X., et al. (2024). A comprehensive review of artificial intelligence applications in photovoltaic systems. CAAI Artificial Intelligence Research, 3, 9150031. https://doi.org/10.26599/AIR.2024.9150031

IMS Global Learning Consortium. (2022). Open Badges specification v3.0. 1EdTech. https://www.imsglobal.org/spec/ob/v3p0

International Organization for Standardization. (2023). ISO/IEC 25010:2023 Systems and software engineering — SQuaRE — Product quality model. ISO. https://www.iso.org/standard/78176.html

Jobayer, M., Shaikat, M. A. H., Rashid, M. N., & Hasan, M. R. (2023). A systematic review on predicting PV system parameters using machine learning. Heliyon, 9(6), e16815. https://doi.org/10.1016/j.heliyon.2023.e16815

Khati, P., Shrestha, A. K., & Vassileva, J. (2023). Student certificate sharing system using blockchain and NFTs. In J. M. Machado et al. (Eds.), Blockchain and applications, 5th international congress (BLOCKCHAIN 2023) (Lecture Notes

in Networks and Systems, Vol. 778, pp. 61–70). Springer. https://doi.org/10.1007/978-3-031-45155-3_7

Khati, P., Shrestha, A. K., & Vassileva, J. (2025). Exploring user acceptance of blockchain-based student certificate sharing system: A study on NFT utilization. In J. Prieto et al. (Eds.), Blockchain and applications, 6th international congress (BLOCKCHAIN 2024) (Vol. 1256, pp. 90–99). Springer. https://link.springer.com/chapter/10.1007/978-3-031-81928-5_9

Kochmar, E., Vu, D.D., Belfer, R. et al. Automated Data-Driven Generation of Personalized Pedagogical Interventions in Intelligent Tutoring Systems. Int J Artif Intell Educ 32, 323–349 (2022). https://doi.org/10.1007/s40593-021-00267-x

Kosie, J. E., & Lew-Williams, C. (2022). Descriptive research in psychology: Bridging the gap between observation and explanation. Advances in Methods and Practices in Psychological Science, 5(3). https://doi.org/10.1177/25152459221099796

Lee, W.-T., & Chen, C.-H. (2023). Agile Software Development and Reuse Approach with Scrum and Software Product Line Engineering. Electronics, 12(15), 3291. https://doi.org/10.3390/electronics12153291

Lemence, A. L. G., & Tamayao, M.-A. M. (2021). Energy consumption profile estimation and benefits of hybrid solar energy system adoption for rural health units in the Philippines. Renewable Energy, 178, 651–668. https://doi.org/10.1016/j.renene.2021.06.090

Lewis, J. R. (2018). The System Usability Scale: Past, present, and future. International Journal of Human–Computer Interaction, 34(7), 577–590. https://doi.org/10.1080/10447318.2018.1455307

Lin, C.-C., Huang, A. Y. Q., & Lu, O. H. T. (2023). Artificial intelligence in intelligent tutoring systems toward sustainable education: A systematic review. Smart Learning Environments, 10, 41. https://doi.org/10.1186/s40561-023-00260-y

Liu, Detian and Zhang, Jianbiao and Wang, Yifan and Shen, Hong and Zhang, Zhaoqian and Ye, Tao (2025). Blockchain smart contract security: Threats and mitigation strategies in a lifecycle perspective. ACM Computing Surveys. https://doi.org/10.1145/3769013

Lotfi Ben Said, Wissem Zghal, Hamdi Hentati, Mohamed Bouaziz, Alaa Chabir, Sattam Alharbi, Mohamed Turki (2026). Integrated machine learning and design-based methodology for technical sizing and performance optimization of photovoltaic systems. Journal of Engineering Research. https://www.sciencedirect.com/science/article/pii/S2307187726000453

Lusiani, & Princes, E. (2024). A user-centric approach using the ISO 25010 quality model. Journal of Logistics, Informatics and Service Science, 11(10), 485–500. https://doi.org/10.33168/JLISS.2024.1028

Mashaly, A., Elmadawy, M., Elgohary, M. et al. Novel and cost-efficient design of stand-alone PV system with simulation using PVsyst and experimental validation. Sci Rep 15, 43383 (2025). https://doi.org/10.1038/s41598-025-28401-y

Masud, S. B., Rana, M. M., Sohag, H. J., Shikder, F., Faraji, M. R., & Hasan, M. M. (2024). Understanding the financial transaction security through blockchain and machine learning for fraud detection. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.5164958

Mazzocca, C., Acar, A., Uluagac, S., Montanari, R., Bellavista, P., & Conti, M. (2025). A survey on decentralized identifiers and verifiable credentials. IEEE Communications Surveys & Tutorials, 27(2), 1456–1488. https://doi.org/10.1109/COMST.2025.3534795

Mickael Fonseca Santos, Ricardo Angelo Filipe, & Joao Carlos Cunha (2025). From traditional to agile methodologies in software project management education: A case study. In Proceedings of the 2024 16th International Conference on Education Technology and Computers (ICETC '24) (pp. 491–497). Association for Computing Machinery. https://doi.org/10.1145/3702163.3702460

Mishra, A., Alzoubi, Y.I. Structured software development versus agile software development: a comparative analysis. Int J Syst Assur Eng Manag 14, 1504–1522 (2023). https://doi.org/10.1007/s13198-023-01958-5

Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. https://bitcoin.org/bitcoin.pdf

Nikolic, S., Matic, S., Capko, D., Vukmirovic, S., & Nedic, N. (2022). Development of a blockchain-based application for digital certificates in education. Proceedings of the 2022 30th Telecommunications Forum (TELFOR 2022). https://doi.org/10.1109/TELFOR56187.2022.9983672

Aaron Paul M. Dela Rosa, Luigi Miguel M. Villanueva, John Mardy R. San Miguel, John Emmanuel B. Quinto (2022). Web-based database courses e-learning application: Evaluation using ISO/IEC 25010. arXiv. https://doi.org/10.48550/arXiv.2212.00104

Okonkwo, C. (2025). Assessing software product quality in DevOps: An ISO 25010:2023 perspective. Proceedings of the 2025 29th International Conference on Evaluation and Assessment in Software Engineering Companion. ACM. https://doi.org/10.1145/3727967.3756847

Okonkwo, C., & Ade-Ibijola, A. (2021). Chatbots applications in education: A systematic review. Computers and Education: Artificial Intelligence, 2, 100033. https://doi.org/10.1016/j.caeai.2021.100033

Palmerola, R. G. (2024). Developmental research methodology in information technology capstone projects. Journal of Computing Education, 12(1), 30–44.

Patel, P., & Desai, R. (2023). Face validity assessment in survey instruments for technology adoption research. International Journal of Research Methodology, 6(2), 15–27.

QA Expertise. (2023). Dichotomous testing: Pass/fail methodology in software quality assurance. https://qaexpertise.com/dichotomous-testing

Qi, Y., et al. (2024). Bringing generative AI to adaptive learning in education. arXiv. https://arxiv.org/abs/2402.14601

Rahman, M. M., et al. (2024). Utilizing blockchain technology for university certificate verification system. International Journal of Applied Information Systems, 12(45), 23–40. https://doi.org/10.5120/ijais2024451977

Republic of the Philippines. (2008). Republic Act No. 9513: Renewable Energy Act of 2008. Official Gazette of the Philippines. https://www.officialgazette.gov.ph/2008/12/16/republic-act-no-9513/

Republic of the Philippines. (2019). Republic Act No. 11285: Energy Efficiency and Conservation Act. Official Gazette of the Philippines.

Rice, R. W. (2022). System testing: Principles, levels, and best practices. Journal of Software Engineering Research and Development, 10(1), 1–14. https://doi.org/10.1186/s40411-022-00110-y

Rojas, H., Renteria, R., Duran, V. M., Gutiérrez, Y. T., Cabrera, M. J. I., & Aminuddin, A. (2025). Mapping the Evolution and Future Directions of ISO/IEC 25010: A Bibliometric and Thematic Analysis. Engineering, Technology & Applied Science Research, 15(5), 27530–27541. https://doi.org/10.48084/etasr.11772

Saima Rafi, Muhammad Azeem Akbar, & Arif Ali Khan (2025). Assessing software product quality in DevOps: An ISO 25010:2023 perspective. In Proceedings of the 2025 29th International Conference on Evaluation and Assessment in Software Engineering Companion (EASE Companion '25) (pp. 10–16). Association for Computing Machinery. https://doi.org/10.1145/3727967.3756847

Schwaber, K., & Sutherland, J. (2020). The Scrum Guide. https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf

Soylu, A. (2022). A comprehensive overview of software testing approaches and trends in the era of agile and DevOps. IEEE Access, 10, 55430–55451. https://doi.org/10.1109/ACCESS.2022.3175551

Taha, A., Makeen, P., & Nazih, N. (2025). Short-term and long-term solar irradiance forecasting with advanced machine learning techniques. Scientific Reports, 15, 39187. https://doi.org/10.1038/s41598-025-24853-4

Tiwari, A., Kanjolia, A., Kumar, B., & Mehra, P. S. (2024). A Survey on Blockchain in Financial Institutions. 2024 11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Noida, India, 2024, pp. 1-6, doi: 10.1109/ICRITO61523.2024.10522341. https://ieeexplore.ieee.org/document/10522341

Tongco, M. D. C. (2022). Purposive sampling as a tool for informant selection. Ethnobotany Research and Applications, 20, 147–158. https://doi.org/10.17348/era.20.0.147-158

Tong, Y., Zhang, X., & Wen, L. (2022). Applying user acceptance testing in software development: A systematic review. IEEE Transactions on Software Engineering, 48(6), 1920–1935. https://doi.org/10.1109/TSE.2021.3068123

Trindade, A., & Cordeiro, L. (2019). Optimal sizing of stand-alone solar PV systems via automated formal synthesis. arXiv preprint arXiv:1909.13139 http://arxiv.org/abs/1909.13139

Trochim, W. M. K. (2025). Research methods knowledge base (3rd ed., online). Conjoint.ly. https://conjointly.com/kb/

Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. Management Science, 46(2), 186–204. https://doi.org/10.1287/mnsc.46.2.186.11926

Wang, X., Huang, R. T., Sommer, M., Pei, B., Shidfar, P., Rehman, M. S., Ritzhaupt, A. D., & Martin, F. (2024). The efficacy of artificial intelligence-enabled adaptive learning systems from 2010 to 2022 on learner outcomes: A meta-analysis. Journal of Educational Computing Research, 62(6), 1568–1603. https://doi.org/10.1177/07356331241240459

Wu, C.-H., & Liu, C.-Y. (2023). Educational applications of non-fungible token (NFT). Sustainability, 15(1), 7. https://doi.org/10.3390/su15010007

Yu, H., Guo, Y., & Yang, S. J. H. (2025). A comprehensive review of AI-based intelligent tutoring systems: Applications and challenges. arXiv. https://doi.org/10.48550/arXiv.2507.18882

Yuensook, T., et al. (2026). AI-driven adaptive learning systems in higher education: A systematic review. Journal of Education and Learning, 15(2). https://www.researchgate.net/publication/398035819_AI-Driven_Adaptive_Learning_Systems_in_Higher_Education_A_Systematic_Review